

CHARACTERISATION AND INVESTIGATION OF THE EXTERNAL FLOW OVER THE STEVE AND CLV LAUNCH VEHICLES

Brent Ross Yelland

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Supervisor: Prof Glen C. Snedden
Co-Supervisor: Prof Michael J. Brooks

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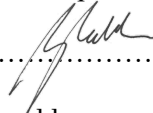
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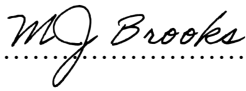
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Declaration 2 - Publications

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Abstract

ASRI (Aerospace Systems Research Institute) is developing a sub-orbital rocket (STeVe) and an orbital rocket (CLV). In this study, the external airflow around these rockets was characterised. Characterisation entailed determining the static aerodynamic coefficients for varying Mach numbers and angles of attack using CFD and semi-empirical methods (RASAero and Missile Datcom).

To ensure the characterisation was accurate, five separate cases were used to validate the CFD model and confirm the accuracy and limitations of the semi-empirical methods. In the validation chapter, both semi-empirical methods and the CFD model were shown to perform well compared to experimental results. In general, for all flow regimes, excluding outliers, the CFD and the semi-empirical methods were within 15% of the experimental wind tunnel results. In most cases, the CFD and the semi-empirical methods were within 5% of the experimental wind tunnel results.

Using the validated model, various aerodynamic coefficients were determined for both launch vehicles. Using these aerodynamic characterisation results, the drag force was determined for STeVe and CLV for a given flight trajectory, comparing the results from CFD and semi-empirical inputs. For STeVe, Missile Datcom predicted a lower peak drag force by 4.8%, while RASAero predicted a higher peak drag force by 7.5%, when compared to CFD. For CLV, Missile Datcom and RASAero both predicted a higher peak drag force by 2.7% and 24.2% respectively, when compared to CFD.

The aerodynamic heating effects on the STeVe nose cone were investigated, in which CFD results were compared with analytical heat transfer calculations. The CFD and analytical temperature results differed by less than 1%. Different nose cone tip radii were simulated to determine the effect on heat transfer, where negligible surface temperature differences were observed. However, the maximum heat transfer coefficient decreased as nose tip radius increased. Missile Datcom was used to modify the STeVe and CLV fuselage geometry, comparing the effects on C_dA and drag force. For STeVe, increasing the diameter by 21% increased the peak drag force by 19%. For CLV, increasing the diameter by 15% increased the peak drag force by 37%.

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Nomenclature

English		
Symbol	Description	Unit
A	Area	m ²
C_d	Drag Coefficient	-
C_l	Lift Coefficient	-
C_p	Heat Capacity at Constant Pressure	J/kg.K
D_{EXIT}	Exit Diameter	m
F_A	Axial Force	N
F_D	Drag Force	N
F_L	Lift Force	N
F_N	Normal Force	N
f_{fl}	First Layer Height Factor	-
f_{tl}	Total Prism Layer Thickness Factor	-
G	Growth Rate	-
h	Convective Heat Transfer Coefficient	W/m ² .K
k	Thermal Conductivity	W/m.K
k	Turbulent Kinetic Energy	J/kg
M	Mach Number	-
N	Number of Prism Layers	-
p	Pressure	Pa
p_0	Stagnation Pressure	Pa
Q	Heat Transfer Rate	W
Q	Rate of Heat Transfer	W
q	Heat Flux	W/m ²
q	Dynamic Pressure	Pa
Re	Reynolds Number	-
r	Recovery Factor	-
S	Sutherland Temperature	K
T	Temperature	K
T_0	Stagnation Temperature	K
T_{aw}	Adiabatic Wall Temperature	K
T_c	Near Wall Cell Temperature	K
T_s	Surface (Wall) Temperature	K
T_∞	Free Stream Temperature	K
T^+	Dimensionless Temperature	-
u	Velocity	m/s
u_∞	Free Stream Velocity	m/s

u^+	Dimensionless Velocity	-
u^*	Friction Velocity (Shear Velocity)	m/s
V	Volume	m ³
Y_{pl}	Total Prism Layer Thickness	m
$Y_{pl,adjusted}$	New Total Prism Layer Thickness	m
y_c	First Cell Layer Distance to Centre of Cell	m
$y_{t,adjusted}$	New First Layer Height	m
y_t	First Layer Height	m
y^+	Dimensionless Distance away from the Wall	-
Greek		
α	Angle of Attack	°
δ_{99}	Boundary Layer Thickness	m
ϵ	Turbulence Dissipation Rate	J/kg.s
μ	Dynamic Viscosity	Pa.s
μ_t	Eddy Viscosity (Turbulent Viscosity)	Pa.s
ν	Kinematic Viscosity	m ² /s
ρ	Density	kg/m ³
ϕ	Roll	°
ω	Specific Turbulence Dissipation Rate	s ⁻¹
Subscripts		
0	Stagnation	
c	Cell	
s	Surface	
∞	Free Stream	

Abbreviations

AEDC	Arnold Engineering and Development Center
AIAA	American Institute of Aeronautics and Astronautics
ASReG	Aerospace Systems Research Group
ASRI	Aerospace Systems Research Institute
CFD	Computational Fluid Dynamics
CLV	Commercial Launch Vehicle
DNS	Direct Numerical Simulation
FEA	Finite Element Analysis
HEEDS	Hierarchical Evolutionary Engineering Design System
HyROPS	Hybrid Rocket Performance Simulator
LES	Large Eddy Simulation
Missile Datcom	Missile Data compendium
NSWC	Naval Surface Warfare Center
NOL	Naval Ordnance Laboratory
OTR	Overberg Test Range
PyROPS	Python Rocket Performance Simulator
RANS	Reynolds-Averaged Navier-Stokes
SAFFIRE	South African First Integrated Rocket Engine
SST	Shear Stress Transport
STeVe	Sub-orbital Test Vehicle
UKZN	University of KwaZulu-Natal

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1. Introduction

The introduction presents a brief background on ASRI, along with an overview of the STeVe and CLV rockets. It also outlines the aims, objectives, and structure of the thesis.

1.1 Background of ASRI

ASRI (Aerospace Systems Research Institute) was founded in 2009 as a research group, formally ASReG (Aerospace Systems Research Group), and is now a research institute based within the University of KwaZulu-Natal's (UKZN) Discipline of Mechanical Engineering. The institute has three aims (ASRI, 2024),

- 1) “to become a global centre of excellence in aerospace propulsion research and development”
- 2) “to support South Africa’s space engineering economy”
- 3) “to develop skilled human capital in aerospace propulsion engineering.”

Since its inception in 2009, ASRI has developed a Phoenix series of rockets, which have achieved multiple successful launches from the Overberg Test Range (OTR). In March 2021, the Phoenix-1B Mk 2-R achieved an apogee of 17.9 km (ASRI, 2021), setting a new African maximum altitude record for a hybrid rocket. The institute has been successful in developing rockets, human capital, test facilities, and launch facilities.

ASRI is currently developing two rockets, namely a Commercial Launch Vehicle (CLV) and a sub-orbital sounding rocket, STeVe (Sub-orbital Test Vehicle). STeVe will test the components and sub-systems that will later be used in CLV. CLV is an orbital rocket that will carry and deploy satellites, representing ASRI’s primary objective. On the path to developing the CLV and STeVe launch vehicles, ASRI is also developing a liquid rocket engine called SAFFIRE (South African First Integrated Rocket Engine) and a permanent test and launch facility at the Overberg Test Range (OTR), among other projects.

1.2 STeVe (Sub-orbital Test Vehicle) and CLV (Commercial Launch Vehicle)

STeVe is a single-stage, sub-orbital sounding rocket, powered by a single SAFFIRE liquid rocket engine. A sounding rocket is a sub-orbital research rocket, usually with the aim of testing equipment and recording measurements. These rockets will carry a payload (such as satellites, or scientific instruments), gaining valuable information during launch and in free fall (near the apogee of flight).

STeVe, shown in Figure 1, aims to enable testing of several systems, including the SAFFIRE rocket engines, propellant tank and aerostructure technologies, and CLV subsystems. It will also build OTR’s space launch capacity, and reach space by achieving an apogee above 100 km. STeVe consists of a Von Kármán nose cone, a cylindrical fuselage, and four hexagonal cross-sectional fins.

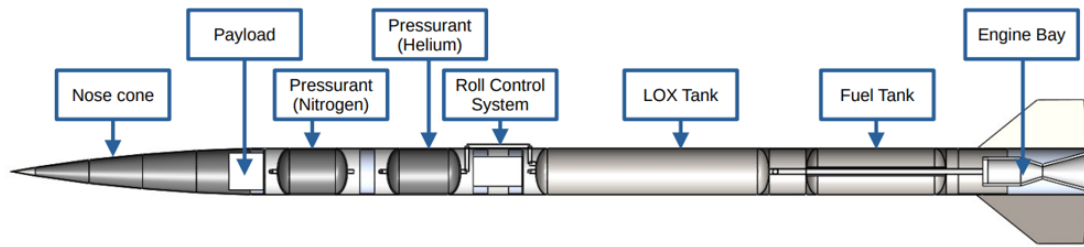


Figure 1. STeVe Sounding Rocket (Mk 1 Configuration)

CLV, shown in Figure 2, is a two-stage rocket, powered by nine SAFFIRE engines in the first stage and a single SAFFIRE engine in the second stage, designed to orbit the Earth for satellite deployment.



Figure 2. Commercial Launch Vehicle

At the start of this study, both STeVe and CLV were still in the design phase and both geometries were not yet finalised. Notably, for STeVe, the fuselage diameter was yet to be confirmed. By July 2024, the initial Computational Fluid Dynamics (CFD) simulations were started for both launch vehicles and the respective geometries were used throughout the thesis.

1.3 Motivation of the Study

Obtaining the aerodynamic coefficients of a rocket is important to predict the flight trajectory and the forces acting on the rocket structure. The aerodynamic coefficients can be accurately determined using CFD software, which was the main goal of this study. An external CFD simulation to obtain the aerodynamic coefficients had not been performed on STeVe or CLV prior to this work.

The aerodynamic coefficients can also be obtained using semi-empirical methods, such as those employed by Missile Datcom or RASAero. The advantage of semi-empirical methods is that they have virtually zero computational cost, however, some accuracy is lost when compared to CFD simulations. A secondary goal of this study was to compare the semi-empirical methods to empirical and CFD results to validate the accuracy and observe the deficiencies of the semi-empirical methods.

Both STeVe and CLV will travel at close to hypersonic speeds, where aerodynamic heating effects will cause substantial increases in rocket surface temperatures, especially at the nose cone. These temperatures must be known and mitigated; therefore, the effects of aerodynamic heating were investigated here.

Since both the launch vehicles were in the design phase during this thesis, the geometries had scope to be modified. Both geometries were changed, analysing the effects on drag and heat transfer.

1.4 Aims and Objectives

The study had the following aims:

- Characterise the external flow over the STeVe sounding rocket and CLV rocket using CFD and semi-empirical methods.
- Compare the accuracy of various semi-empirical methods in determining aerodynamic coefficients.
- Investigate the effects of aerodynamic heating on STeVe.
- Alter the geometry of the launch vehicles to reduce drag and heat transfer.

The study had the following objectives:

- Conduct a literature review on external flow theory and CFD, with a specific focus on meshing and turbulence modelling.
- Compare different CFD software options to determine the optimal software to use for the study.
- Create and validate CFD models using existing wind tunnel experimental data, semi-empirical methods, and other CFD results.
- Use the validation cases to assess the accuracy of the various semi-empirical methods.
- Characterise the external flow over the launch vehicles using the validated CFD model and semi-empirical methods. ‘Characterise’ entails determining the static aerodynamic coefficients for varying Mach numbers and angles of attack.
- Determine the maximum aerodynamic force acting on STeVe and CLV for a given flight trajectory using results from the CFD and semi-empirical results.
- Determine the surface temperature along the nose cone of STeVe using CFD, and compare the results with analytical aerodynamic heating calculations.
- Alter the geometry of the STeVe nose cone, analysing the effects on surface temperature and heat transfer.
- Alter the geometry of the launch vehicles to analyse the effect on drag.

1.5 Methodology and Thesis Layout

The following methodological approach and layout were used for the thesis:

Literature Review and Theory: Research was done to understand useful concepts for the study. This included literature on the entire CFD process (fluid domain creation, meshing, choosing physics models), fluid mechanics, turbulence modelling, aerodynamics, and heat transfer.

Model Validation: A CFD model was created for different rocket geometries, and validated against wind tunnel experimental data, semi-empirical codes, and other CFD results. Various meshing techniques and flow model options were tested in this chapter. The entire method used to create the CFD model became part of the process that was later used for simulating STeVe and CLV.

Computational Aerodynamic Analysis of the Launch Vehicles: The launch vehicles were simulated using CFD, capturing the static aerodynamic coefficients at desired Mach numbers and angles of attack. The entire CFD process is detailed in this chapter.

Aerodynamic Characterisation Results: The STeVe and CLV CFD and semi-empirical results are presented and discussed, including the static aerodynamic coefficients, and the drag forces based on a given trajectory.

Aerodynamic Heating: CFD methods were used to obtain the heat transfer coefficient and surface temperature, which were compared against analytical heat transfer calculations. The impact of changing the STeVe nose cone geometry on heat transfer was then analysed using CFD.

Geometry Modification: Missile Datcom was used to observe trends in drag, when the fuselage geometries were varied.

Conclusion: The main findings and recommendations are discussed, followed by a list of potential future work related to this master's topic.

2. Literature Review and Theory

This chapter discusses important literature and theory regarding the topics of turbulence modelling, aerodynamics, aerodynamic heating, meshing, and semi-empirical aeroprediction codes.

2.1 Turbulence Modelling

Due to the high-speed nature of rocket flight, there will be turbulent flow. This chapter discusses when turbulence will occur and how to model turbulent flow using CFD.

2.1.1 Reynolds Number

The Reynolds number, Re , is a dimensionless number, defined by the ratio of the inertial forces over the viscous forces,

$$Re_x = \frac{\rho \cdot u \cdot x}{\mu} \quad 2.1$$

where ρ is the density (kg/m^3),

u is the fluid velocity (m/s),

x is the distance along the plate (m), and

μ is the dynamic viscosity ($\text{Pa}\cdot\text{s}$).

The Reynolds number is used to determine whether a flow is turbulent or not. For external fluid flow over a flat plate, after some distance, laminar flow transitions to turbulent flow. This is known as the transitional zone, and begins at a Reynolds number of $\sim 5 \times 10^5$ for flow over a flat plate. After a further distance, the flow leaves the transitional zone and becomes fully turbulent. Figure 3 shows the laminar flow transitioning to turbulent flow, indicating the boundary layer thickness (δ) and velocity profile.

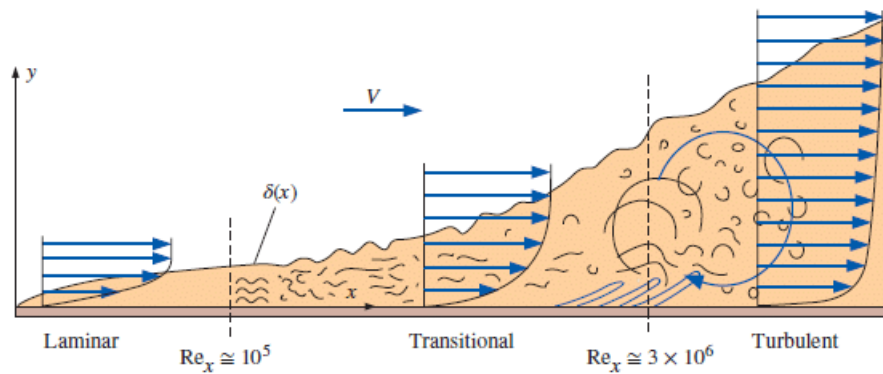


Figure 3. Laminar to Turbulent Transition Over a Flat Plate (Cengel and Cimbala, 2014)

2.1.2 Boundary Layer

The no-slip condition is a boundary condition in fluid dynamics which states that at a solid boundary (a wall), the fluid velocity relative to that wall is zero. At some distance away from the wall, the velocity will reach a free stream value (denoted by subscript ∞). The distance away from the wall at which the velocity is 99% of the maximum velocity, is called the boundary layer thickness, δ_{99} . In other words, when

$$y = \delta_{99} \quad 2.2$$

the fluid velocity, u , is

$$u = 0.99 \cdot u_{\infty} \quad 2.3$$

where u_{∞} is the free stream velocity.

Therefore, it can be concluded that there exists a velocity gradient in the distance from the surface to the free stream, indicated by the blue line in Figure 4. This region, from the surface to the free stream, is defined as the boundary layer. It is important to note that there are large changes in velocity, temperature, and other flow properties in the boundary layer. The changes in flow properties are larger, closer to the surface. These large changes must be accurately captured using CFD, which is described in Chapter 2.1.4.

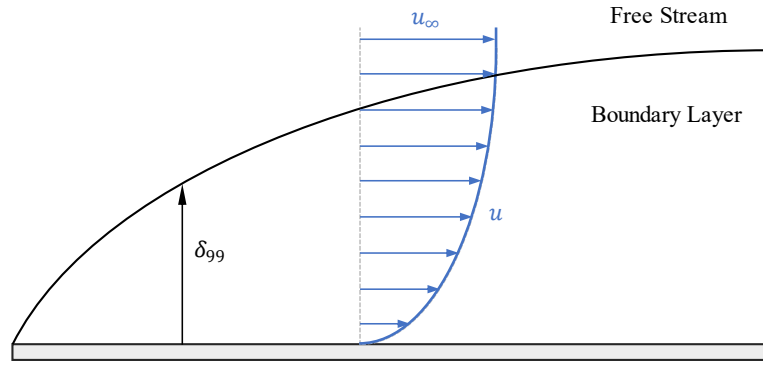


Figure 4. Boundary Layer Formation

The total boundary layer thickness can be estimated for external fluid flow using equations for flow over a flat plate. The boundary layer has a different thickness for laminar and turbulent flow, and that is why it is important to know what type of flow is present. The total boundary layer thickness for laminar flow, $Re < 5 \times 10^5$, can be calculated using the Blasius solution (Schlichting, 1979),

$$\delta_{99} = 5.0 \cdot \frac{x}{\sqrt{Re}} \quad 2.4$$

The total boundary layer thickness for turbulent flow, $Re > 5 \times 10^5$, can be calculated using the following equation (Cengel and Cimbala, 2014),

$$\delta_{99} = \frac{0.38 \cdot x}{Re^{\frac{1}{5}}} \quad 2.5$$

2.1.3 y^+ and Wall Functions

y^+ is a dimensionless number that represents the perpendicular distance away from the wall used to characterise the regions of flow through the boundary layer. It exists in both reality and in CFD, with slightly differing definitions in each context. In reality, y^+ is defined as

$$y^+ = \frac{u^* \cdot \rho \cdot y}{\mu} \quad 2.6$$

where u^* (m/s) is the friction velocity, and

y (m) is a perpendicular distance away from the wall.

In CFD, the y^+ definition uses the distance from the wall to the centre of the first cell as shown in Equation 2.7,

$$y^+ = \frac{u^* \cdot \rho \cdot y_c}{\mu} \quad 2.7$$

where y_c (m) is the distance from the wall surface to the cell centre, shown later in Figure 8.

It must be noted that y^+ is an output value from a CFD simulation. An initial guess can be made for the first layer height, to achieve a desired y^+ value. However, it is crucial to remember that this is just an initial guess, and the y^+ values from the simulation must be monitored, so the first layer height can be updated.

Before discussing the importance of y^+ , the dimensionless velocity, u^+ , should be defined,

$$u^+ = \frac{u}{u^*} \quad 2.8$$

A graph of u^+ vs. y^+ , highlighting the different wall functions is shown in Figure 5.

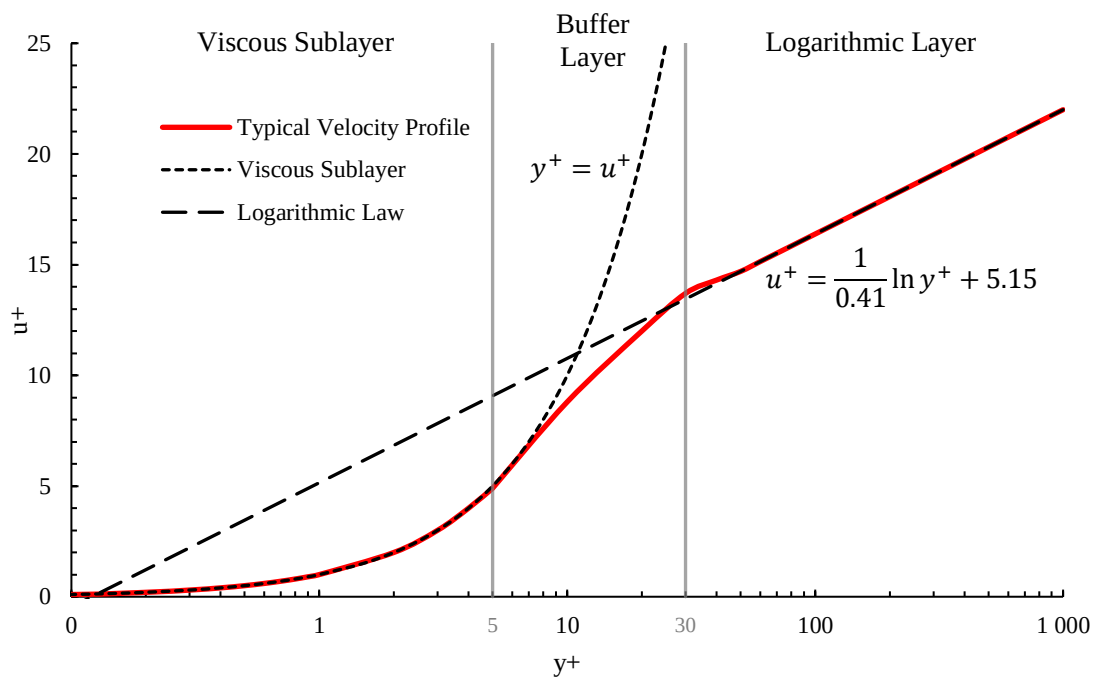


Figure 5. Graph of Different Wall Functions

In Figure 5, u^+ is the non-dimensional velocity tangential to the surface, and y^+ is the non-dimensional perpendicular distance away from the surface. The red line indicates a typical velocity profile that would occur in reality. The dashed lines indicate the different relationships that aim to fit the typical velocity profile, and are known as ‘wall functions’.

For y^+ values less than 5, the y_c value is reduced to resolve the boundary layer without using empirical equations. This region is known as the viscous sublayer and the wall function is the linear function defined by

$$u^+ = y^+ \quad 2.9$$

For y^+ values greater than 30, an empirical equation can be used to fit the velocity profile. This region is known as the logarithmic layer and the empirical equation, known as the ‘logarithmic law’, is a logarithmic function defined by

$$u^+ = \frac{1}{K} \ln y^+ + C \quad 2.10$$

where K and C are dimensionless constants.

For y^+ values above 5 and below 30, the two discussed wall functions do not fit the velocity profile. This region is known as the buffer region. Other wall functions, proprietary to CFD solvers, can be used to fit the velocity profile in this region, as well as in the viscous sublayer and logarithmic layer. One such unified wall function was developed by Spalding (1961) and is defined as follows

$$y^+ = u^+ + e^{-Kc} \left(e^{Ku^+} - 1 - Ku^+ - \frac{1}{2}(Ku^+)^2 - \frac{1}{6}(Ku^+)^3 \right) \quad 2.11$$

CFD solvers can automatically switch between wall functions, depending on the y^+ value. In Star-CCM+ the ‘all y^+ ’ option can be selected, which automatically chooses the correct wall function to apply depending on the y^+ . Star-CCM+ will change the y^+ as the solution develops to a steady state.

The difference in how the wall functions resolve the velocity gradient can be shown graphically, in Figure 6. The blue points represent a flow property (such as velocity) at the centre of each cell.

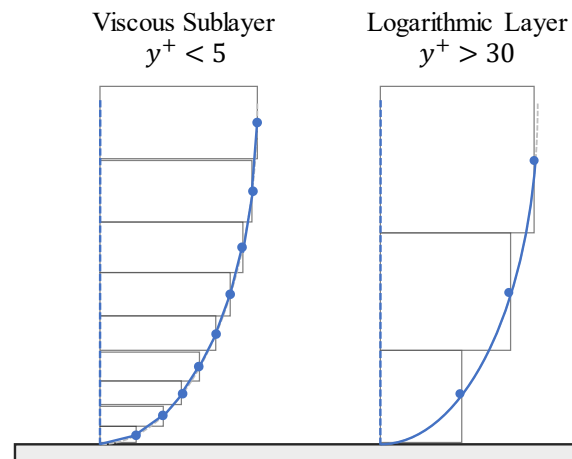


Figure 6. Resolving the Boundary Layer

In the viscous sublayer, the boundary layer is resolved by linearly interpolating values between neighbouring cells. The cells must be small enough so that the linear connections between cells appear smooth, and that is why the y^+ is limited to 5 in this region. For y^+ values greater than 30, the ‘logarithmic law’ resolves the boundary layer by using a non-linear empirical equation to interpolate between cells.

The empirical equations for the y^+ regions were developed based on data from flow over a flat plate. Therefore, inaccuracies are likely to occur in flows involving adverse pressure gradients, flow separation, and curvature. It can be concluded that it is most accurate to aim for y^+ values less than 5, as empirical equations are not used in this region. However, based on trial and error

from industry, a y^+ value of ~ 1 is targeted, to accurately resolve the boundary layer. This is because the transition from the viscous sublayer to the buffer zone varies for different cases (if there is flow separation for example). It is therefore a good idea to aim for a y^+ further away from the transition point to the buffer zone at $\sim y^+$ of 5. It is also important to not have a y^+ value that is too small, as this wastes computational resources and may cause problems when trying to model turbulent structures near the wall.

2.1.4 Prism Layers (Inflation Layers)

Prism layers (sometimes called inflation layers) are small cells near the surface of objects that aim to capture the boundary layer formation as described in Chapter 2.1.2. Due to the large velocity and temperature gradients near the surface, the cells must be small to capture these changes in flow properties. Figure 7 shows a diagram of the prism layers (grey cells) and boundary layer thickness (red line) and is useful for the following explanation.

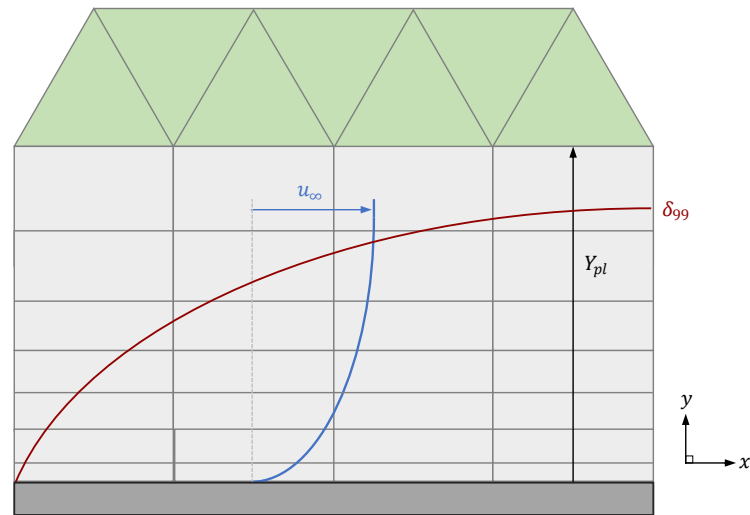


Figure 7. Prism Layer Formulation with the Boundary Layer Thickness Overlayed

There is a crucial difference between the total prism layer thickness and the boundary layer thickness. The total prism layer thickness, Y_{pl} , is a result of the mesh and only exists in CFD. In contrast, the boundary layer thickness, δ_{99} , is a flow phenomenon and exists in reality.

Figure 8 shows an exaggerated view of the first few prism cells, and is useful for the following explanation.

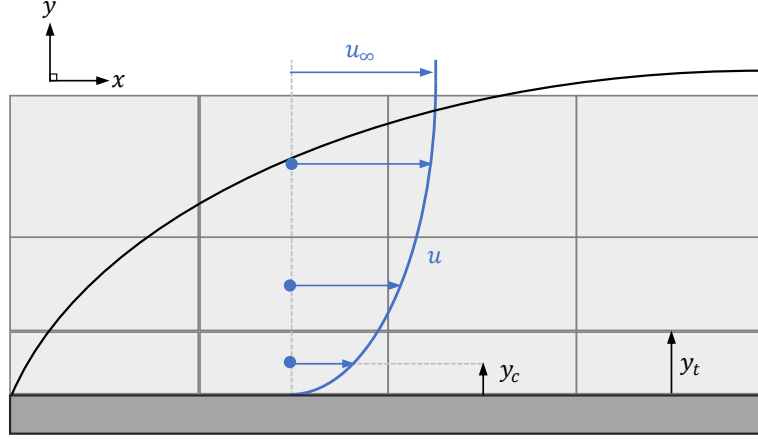


Figure 8. Boundary Layer Formation Over Prism Layers

The prism layer thickness is directly related to y^+ . Equation 2.7 can be rearranged to calculate the distance from the surface to the centre of the first cell, y_c , as shown below.

$$y_c = \frac{y^+ \cdot \mu}{\rho \cdot u^*} \quad 2.12$$

CFD software specifies the input of the first layer as the total thickness of the first cell. To obtain the total first cell height (y_t), y_c is doubled, as shown in Equation 2.13:

$$y_t = 2 \cdot y_c \quad 2.13$$

The next parameter related to prism layers is the total prism layer thickness, Y_{pl} . The total height of all the prism layers should be at least the height of the boundary layer thickness (Figure 7), to adequately capture the boundary layer formation. To calculate the total prism layer thickness, the growth rate, G , must be defined. The growth rate is the ratio of a larger cell's thickness, to its neighbouring smaller cell's thickness and is constant for all prism layers. Growth rate is defined according to Equation 2.14, in terms of the thickness of the n^{th} prism layer.

$$G = \frac{y_{t,n}}{y_{t,n-1}} \quad 2.14$$

The total prism layer thickness, Y_{pl} , can be calculated by summing the thickness of each prism layer, resulting in a geometric series, as shown in Equation 2.15:

$$\begin{aligned} Y_{pl} &= y_{t,1} + y_{t,2} + y_{t,3} + \dots + y_{t,N-1} + y_{t,N} \\ &= y_{t,1} + G \cdot y_{t,1} + G \cdot y_{t,2} + \dots + G \cdot y_{t,N-2} + G \cdot y_{t,N-1} \\ &= y_{t,1} + G \cdot y_{t,1} + G^2 \cdot y_{t,1} + \dots + G^{N-2} \cdot y_{t,1} + G^{N-1} \cdot y_{t,1} \\ &= \sum_{k=0}^{N-1} y_t G^k \\ &= y_t \frac{1 - G^N}{1 - G} \end{aligned} \quad 2.15$$

where N is the number of prism layers,
 y_t is the first layer height, and

the numbered subscripts denote the n^{th} prism layer.

The total prism layer thickness can be equated to the total boundary layer thickness, forming an equation to calculate G and N , as shown in Equation 2.16:

$$\begin{aligned} Y_{pl} &= \delta_{99}(L) \\ Y_{pl} - \delta_{99}(L) &= 0 \\ y_t \frac{1 - G^N}{1 - G} - \frac{0.38 \cdot x}{Re^{\frac{1}{5}}} &= 0 \end{aligned} \quad 2.16$$

Since G and N are the only unknowns in Equation 2.16, N can be varied while G automatically changes to satisfy Equation 2.16.

All the calculations for first layer height, number of prism layers, and growth rate are based on theory from a flat plate and are used as an initial guess. Therefore, it is likely that all those parameters should be adjusted for following simulation runs, to achieve the desired distance from the wall to the centre of the first cell, or in CFD parlance, y^+ . This is achieved using two new factors: the first layer height factor, f_{fl} , and total prism layer thickness factor, f_{tl} . The first layer height factor adjusts the theoretical calculation for the first layer height as shown in Equation 2.17:

$$y_{t,adjusted} = f_{fl} \cdot 2 \cdot y_c \quad 2.17$$

where $y_{t,adjusted}$ is the new first prism layer height.

If f_{fl} is 1, there is no adjustment. If f_{fl} is less than one, then the first cell thickness would be smaller than the theoretical value. If f_{fl} is greater than one, then the first cell thickness would be larger than the theoretical value. The first layer factor can be increased to increase y^+ , and decreased to decrease y^+ .

The total prism layer calculation is affected by the first layer factor as shown in Equation 2.18:

$$Y_{pl,adjusted} = y_{t,adjusted} \frac{1 - G^N}{1 - G} \quad 2.18$$

where $Y_{pl,adjusted}$ is the new total prism layer thickness based on the first layer height factor.

The total prism layer thickness factor adjusts the theoretical calculation for the total boundary layer thickness as shown in Equation 2.19:

$$\delta_{99,adjusted}(L) = f_{tl} \frac{0.38 \cdot L}{Re^{\frac{1}{5}}} \quad 2.19$$

where $\delta_{99,adjusted}(L)$ is the new total boundary layer thickness, based on the total layer height factor.

This adjusted total boundary layer thickness and adjusted boundary layer thickness can be equated to form Equation 2.20:

$$\begin{aligned} Y_{pl,adjusted} &= \delta_{99,adjusted}(L) \\ Y_{pl,adjusted} - \delta_{99,adjusted}(L) &= 0 \\ y_{t,adjusted} \frac{1 - G^N}{1 - G} - f_{tl} \frac{0.38 \cdot L}{Re^{\frac{1}{5}}} &= 0 \end{aligned} \quad 2.20$$

As shown in Equation 2.18, the total prism layer thickness is a function of G and N . An Excel solver was created to satisfy Equation 2.20, while adjusting G , for a user input of N . If f_{tl} is less than 1, then $Y_{pl,adjusted}$ decreases and G decreases, and vice versa. If N is decreased, then G increases, and vice versa.

In summary, the user can input the number of prism layers, the first layer height factor, and the total layer height factor. The changes in growth rate, first layer height, and total prism layer thickness can then be observed. Based on the y^+ results from CFD, the user inputs can then be adjusted to achieve the desired y^+ value.

2.1.5 Turbulence Modelling

The Navier-Stokes equations describe all fluid flow, including turbulence and instantaneous velocity fluctuations. However, the equations have no analytical solution due to their complexity. There are two methods to apply the Navier-Stokes equations in CFD: either solving them numerically or reducing and simplifying them before solving. This chapter examines these methods to simulate turbulent flow.

Due to the high Reynolds number flow that occurs in external airflow over rockets, there will be turbulent flow, which CFD turbulence models aim to capture. Three main approaches are used to model turbulence: Direct Numerical Simulation (DNS), Large Eddy Simulation (LES), and Reynolds Averaged Navier-Stokes (RANS).

DNS: It is possible to model turbulent flow by solving the Navier-Stokes equations numerically, and not rely on turbulence modelling. This is the principle behind DNS, which requires large computational resources. No information is available regarding the exact computational cost of DNS, except that the computational cost is orders of magnitude larger than RANS. Therefore, DNS was not chosen for this study.

LES: LES is a simpler version of DNS, as it separates the flow into large and small-scale turbulent structures. LES aims to capture the larger turbulent structures, while ignoring the smaller turbulent structures. This reduces mesh size, since the smaller cells required to model smaller turbulent structures are ignored. However, solution accuracy is not sacrificed, since most of the turbulent kinetic energy is contained in the large turbulent structures. Even though computational resources are saved using this method, LES is still far too computationally demanding for this study.

RANS: RANS differs from DNS and LES, as it aims to model the entire flow, and not directly resolve the turbulent structures. In reality, for steady, turbulent flow, the velocity profile in a specific area at a point in time is never going to be exactly the same as the velocity profile in that same area at a different point in time. The velocity profile at a point in time is defined as the instantaneous velocity profile. In other words, the instantaneous velocity profile varies with time in steady flow, as shown for steady flow over a flat plate at two time points, t_1 and t_2 , in Figure 9.

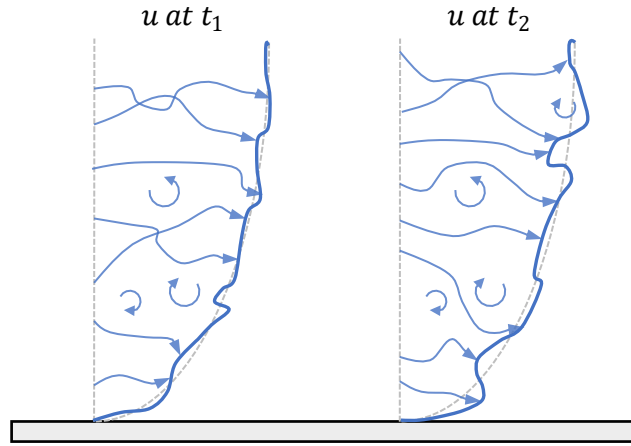


Figure 9. Instantaneous Velocity Profile at Two Different Timepoints in Steady Flow

When analysing fluid flow, the instantaneous velocity profile is not useful; rather, it is the average velocity profile that is used for analysis. The average velocity profile is obtained by averaging all instantaneous velocity profiles over time. This can be determined using Reynolds decomposition, which splits the instantaneous velocity into its average and fluctuating components,

$$u = \bar{u} + u' \quad 2.21$$

where u is the instantaneous velocity,

$\bar{u}$ is the average velocity component, and

u' is the fluctuating velocity component.

The instantaneous velocity can be rewritten in vector form,

$$\vec{u} = \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} \bar{u} + u' \\ \bar{v} + v' \\ \bar{w} + w' \end{bmatrix} \quad 2.22$$

The Reynolds decomposition can be represented graphically as shown in Figure 10.

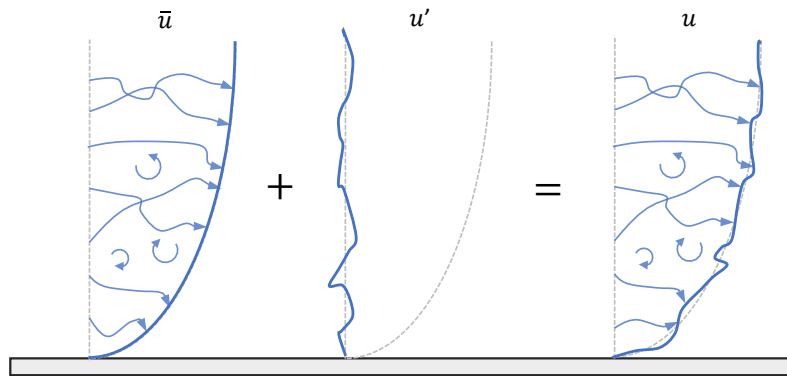


Figure 10. Reynolds Decomposition of Instantaneous Velocity for Flow Over a Flat Plate

For this explanation of Reynolds decomposition, velocity was used; however, other flow variables (pressure, density, or temperature) can also be used. Reynolds decomposition of the instantaneous velocity can be applied to the Navier-Stokes Equations to obtain the Reynolds Averaged Navier-Stokes (RANS) set of equations, which are listed below:

The conservation of mass (continuity equation):

$$\begin{aligned}\nabla \cdot \vec{\bar{u}} &= 0 \\ \frac{\partial \bar{u}}{\partial x} + \frac{\partial \bar{v}}{\partial y} + \frac{\partial \bar{w}}{\partial z} &= 0\end{aligned}\tag{2.23}$$

$$\begin{aligned}\nabla \cdot \vec{u'} &= 0 \\ \frac{\partial u'}{\partial x} + \frac{\partial v'}{\partial y} + \frac{\partial w'}{\partial z} &= 0\end{aligned}\tag{2.24}$$

The conservation of momentum (momentum equation) in the x direction only:

$$\begin{aligned}\rho \left(\frac{\partial \bar{u}}{\partial t} + \bar{u} \frac{\partial \bar{u}}{\partial x} + \bar{v} \frac{\partial \bar{u}}{\partial y} + \bar{w} \frac{\partial \bar{u}}{\partial z} + \frac{\partial \overline{u'u'}}{\partial x} + \frac{\partial \overline{u'v'}}{\partial y} + \frac{\partial \overline{u'w'}}{\partial z} \right) \\ = -\frac{\partial \bar{p}}{\partial x} + \mu \left(\frac{\partial^2 \bar{u}}{\partial x^2} + \frac{\partial^2 \bar{u}}{\partial y^2} + \frac{\partial^2 \bar{u}}{\partial z^2} \right) + \rho a_x\end{aligned}\tag{2.25}$$

where $\vec{\bar{u}}$ is the flow velocity vector, defined in terms of the cartesian velocity components, and

$$\vec{\bar{u}} = u_x \hat{i} + u_y \hat{j} + u_z \hat{k}\tag{2.26}$$

∇ is the standard derivative of a vector,

$$\nabla = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}\tag{2.27}$$

The momentum equation contains three terms,

$$\frac{\partial \overline{u'u'}}{\partial x} + \frac{\partial \overline{u'v'}}{\partial y} + \frac{\partial \overline{u'w'}}{\partial z}\tag{2.28}$$

which are referred to as the ‘Reynolds stresses’. Unlike the other terms in the momentum equation, the Reynolds stresses depend on the turbulent velocity fluctuations. The process of approximating the Reynolds stresses is referred to as ‘closing the turbulence model’, which is described in the following chapter.

2.1.6 Closing the Turbulence Model

The turbulence model in the RANS momentum equation must be closed. To do this, the Reynolds stresses are approximated using one of three broad models: the Eddy-Viscosity Model, the Non-Linear Eddy-Viscosity Model, or the Reynolds Stress Model.

Eddy-Viscosity Model: The Eddy Viscosity Model introduces a term called the eddy viscosity, μ_t , (also known as the turbulent viscosity) to determine the Reynolds stresses. Boussinesq, 1877, pioneered this method, and showed that the Reynolds stresses can be determined using the following equation,

$$-\rho \overline{u'u'} = \mu_t \left[(\nabla \vec{u}) + (\nabla \vec{u})^T - \frac{1}{3} (\nabla \cdot \vec{u}) I \right] - \frac{2}{3} \rho k I\tag{2.29}$$

where k is the turbulent kinetic energy in J/kg, and

I is the identity matrix.

To calculate the Reynolds stresses, the eddy viscosity term must first be calculated. There are various methods to calculate the eddy viscosity term, which are discussed later in Chapter 2.1.7.

Reynolds Stress Model (RSM): The RSM (also known as the Reynolds Stress Transport Model) does not assume a linear or non-linear relationship to calculate the Reynolds stresses, unlike the eddy viscosity models. The RSM uses a separate transport equation to solve for each Reynolds stress term. The RSM is physics based and can be highly accurate; however, the RSM is resource intensive.

Non-Linear Eddy Viscosity Model: The non-linear eddy viscosity model uses the same principles as the eddy viscosity model. However, the equation used to determine the Reynolds stresses uses a non-linear function. The standard Eddy Viscosity Model uses a linear function, as shown in Equation 2.29.

Due to the higher computational resources required to use Reynolds stress transport models and non-linear eddy viscosity models; eddy viscosity models were chosen for use in this study.

2.1.7 Eddy Viscosity Models

Eddy viscosity models are a class of turbulence model that approximate the eddy viscosity term, to determine the Reynolds stress terms. There are three broad categories of eddy viscosity models: Algebraic models, one equation models, and two equation models.

Algebraic Models (Zero Equation Models): Solutions to additional equations are not required. The eddy viscosity term can be directly calculated using flow variables. These models are too simple for most flow cases. Two common algebraic models are the ‘Baldwin-Lomax’ model and the ‘Cebeci-Smith’ model.

One Equation Models: One additional transport equation is introduced. The most common one equation model is ‘Spalart-Allmaras’, which solves for the turbulent kinematic viscosity, ν_t . Spalart Allmaras was developed for the aerospace industry involving wall-bounded and aerodynamic flows, and can also perform well with adverse pressure gradients. It is recommended to aim for a y^+ of ~ 1 when using Spalart Allmaras.

Two Equation Models: Two additional transport equations are introduced. The three most popular two equation models (which have become the industry standard turbulence models) are the $k - \epsilon$, $k - \omega$, and the $k - \omega$ SST (Shear Stress Transport) models.

$k - \epsilon$ is most accurate in modelling flow in the free stream; however, it is inaccurate when modelling adverse pressure gradients and shock waves. Two variables, k and ϵ , must be determined to calculate the eddy viscosity,

$$\mu_t = C_\mu \frac{\rho k^2}{\epsilon} \quad 2.30$$

where C_μ is a model coefficient, and

ϵ is the turbulence dissipation rate (J/kg.s).

$k - \omega$ is most accurate in modelling flow near the wall, separation, and adverse pressure gradients. ω (s^{-1}) is the specific turbulence dissipation rate and is related to ϵ by

$$\omega = \frac{\epsilon}{C_\mu k} \quad 2.31$$

Here, ϵ and ω describe the same flow principle, the turbulence dissipation rate. The two transport equations for $k - \omega$ and $k - \epsilon$ are very similar. However, for the second transport equation, $k - \omega$ solves for ω , and $k - \epsilon$ solves for ϵ . The main difference between the two

models is that $k - \epsilon$ requires the use of damping functions in the viscous sublayer, and these damping functions do not perform well for adverse pressure gradients and flow separation.

$k - \omega$ SST is a two-equation model that is the most widely used turbulence model in industry. $k - \omega$ SST is a combination of the $k - \omega$ and $k - \epsilon$ models, where $k - \epsilon$ is used to model flow in the free stream and $k - \omega$ is used to model flow near the wall. To transition between $k - \omega$ near the wall and $k - \epsilon$ in the free stream, a blending function is introduced. The blending function, F , has a value of 0 near the wall, and a value of 1 in the free stream, and (in Star-CCM+) is defined as follows,

$$F = \tanh(arg) \quad 2.32$$

where F is the blending function, and

arg is a function that is analogous to the distance away from the wall, and is not discussed in this thesis.

The operation of the blending function is shown in Figure 11. For values close to the wall, $k - \omega$ is used, and for values in the free stream, $k - \epsilon$ is used.

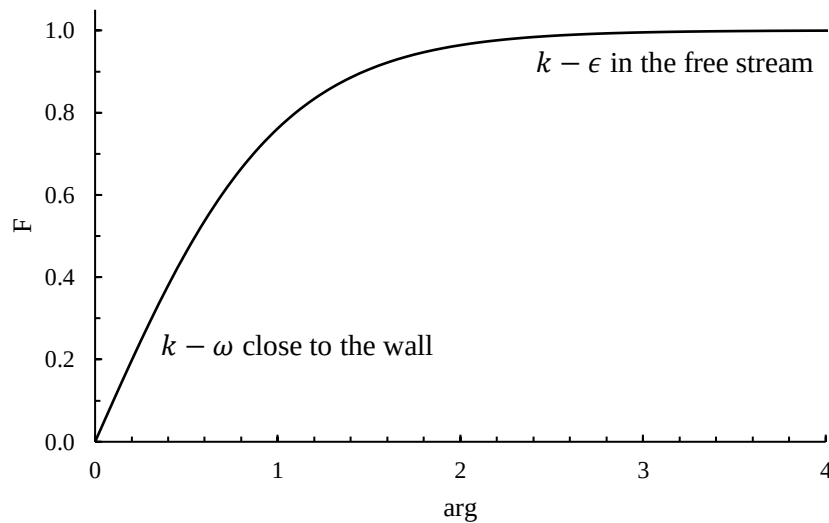


Figure 11. $k - \omega$ Blending Function

2.1.8 Transition Zone Modelling

Transition zone models aim to improve the accuracy of simulations in the transition region ($Re \cong 5 \times 10^5$), where the flow is neither fully turbulent nor laminar. These models are most useful for flow that transitions at a point further from the ends of a body. For example, flow at Mach 2 will enter the transition zone after ~ 11 mm. In this case, the flow will become turbulent so quickly that it becomes unnecessary to accurately model the transition zone, as the flow will be turbulent for over 99% of a 5 m long rocket. Therefore, for high-speed external flows, transition zone models are not required.

2.1.9 Coupled and Segregated Flow Solvers

Flow solvers in CFD affect how the governing (transport) equations are solved, which in turn affects solution speed, accuracy, and the resources required. Two types of flow solvers can be used: segregated and coupled solvers. Segregated solvers (pressure-based solvers) solve the

transport equations sequentially, while coupled solvers (density-based solvers) solve the transport equations simultaneously. The differences between coupled and segregated solvers are summarised in Table 1.

Table 1. Flow Solver Differences

Segregated (Pressure-Based) Solver	Coupled (Density-Based) Solver
Less memory	More memory
Incompressible flows	Compressible flows
Low-speed flows	High-speed flows
Simple geometries	Complex geometries
Less robust: Can encounter instabilities as the problem's complexity increases.	More robust: Unlikely to encounter instabilities as the problem's complexity increases.
Solves faster, in general. Unless the problem is complex	Solves slower, in general. Unless the problem is complex
More iterations are required to achieve convergence	Fewer iterations are required to achieve convergence
Does not solve any equation simultaneously	Solves the continuity, momentum, and energy equations simultaneously (coupled)
	Solves the turbulence equations sequentially (segregated)

A coupled solver was used for this study as coupled solvers are suited for high-speed external aerodynamics. If there was difficulty in achieving convergence for an initial run, then a segregated solver was used to obtain an initial solution, before switching to a coupled solver to obtain the final results.

2.2 Aerodynamics

Aerodynamics is the study of the interaction of air flowing over bodies. The following chapters describe important concepts in aerodynamics, relevant to this study.

2.2.1 Drag

Drag is the force that acts parallel and opposite to the relative motion of the fluid around an object, through the centre of pressure of the object. Three factors affect drag: the object geometry, the relative velocity of the object in the fluid, and the fluid properties. Multiple kinds of drag force arise when analysing rocket flight, which are described below. Firstly, the total drag can be decomposed into skin friction drag and pressure drag.

Skin friction drag: this is a result of the friction between the fluid and the object moving through the fluid. Due to the differences in velocity between the fluid layers (in the boundary layer), there are viscous shear forces between the layers, resulting in friction drag.

Pressure drag (form drag): This is caused by the pressure difference between the leading and trailing sections of an object. There is a stagnation point at the leading edge of an object, where the dynamic pressure is converted into static pressure, resulting in a high pressure zone at the

leading edge. Often there are sharp angles at the trailing section of objects, where the fluid separates from the object's surface. This creates a wake (recirculation zone) at the trailing section of the object, resulting in a low static pressure. Due to the pressure difference, the object is 'pushed' in the direction of high-to-low pressure, which opposes the motion of the object, as shown in Figure 12. Pressure drag can also be decomposed further into wave drag and base drag.

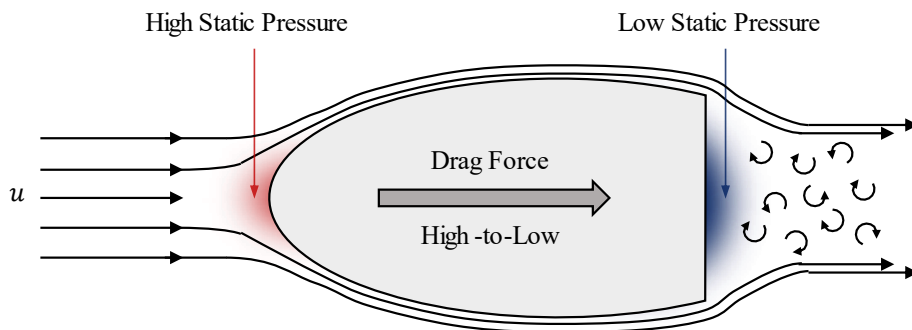


Figure 12. Pressure Drag

Wave drag: This is a type of pressure drag caused by shock wave formation in transonic and supersonic flight. Wave drag is a sudden and large increase in drag when shock waves are present (typically around Mach 0.8), and the onset of wave drag can easily be seen in drag coefficient vs. Mach number plots. Wave drag is directly proportional the cross-sectional area of the object (Whitcomb, 1960) and is independent of viscous effects in the fluid. To reduce wave drag, the object's cross-sectional area should ideally be as small as possible and have smooth changes in cross-sectional area along the length of the object.

Base drag: This is a type of pressure drag that is caused by sharp angles at the trailing edges of objects. When objects have blunt trailing edges, a wake forms behind the trailing face of the object, creating an area of low static pressure. This pressure difference typically exerts a force in a direction opposite to that of the object's travel. Base drag can be reduced by creating a 'boat tail' shape at the trailing edge of an object to reduce flow separation.

There are two additional kinds of drag: lift induced drag and interference drag. These kinds of drag both form part of pressure and skin friction drag; however, neither fits entirely into a single category.

Lift induced drag: This is a result of lift and occurs at the rocket fins at a non-zero angle of attack. Due to the angled fins, on one side of the fin there is a high-pressure zone, and a low-pressure zone on the other side of the fin. This creates a flow from the high-pressure side, along the fin, towards the tip, around the tip, and along the low-pressure side of the fin. This results in vortices at the fin tip, which forces the fluid near the fin tip from the high-pressure side to the low-pressure side of the fin. Lift induced drag in rockets is only a concern if the fins are canted or the rocket is at a non-zero angle of attack.

Interference drag: This is when fluid flowing across one section of the aircraft mixes with fluid flowing from another section of the aircraft. This occurs when there are junctions between different geometries, such as the junction of the fins and the rocket body in rockets.

2.2.2 Lift

Lift acts perpendicular to the relative motion of the fluid around an object, through the centre of pressure. In rocketry, lift will only occur at non-zero angles of attack.

2.2.3 Angle of Attack

The angle of attack, α , is the difference in the angle between the relative fluid velocity and the axial line through the rocket body (or chord line). A positive angle of attack is when the aircraft is pitched up, relative to the fluid velocity, and a negative angle of attack is when the aircraft is pitched down, relative to the fluid velocity. Figure 13 shows a positive angle of attack for a rocket.

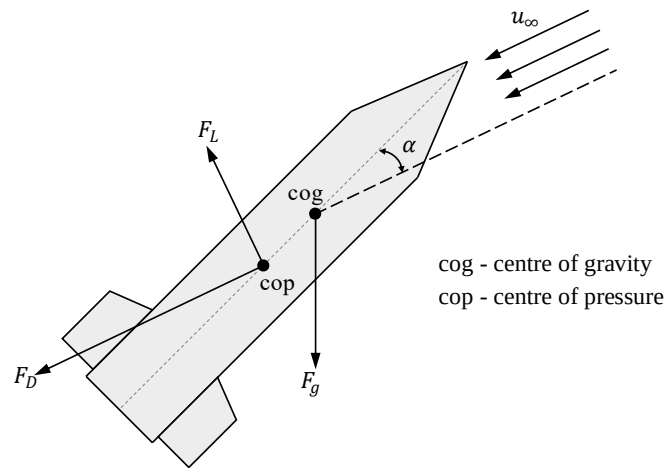


Figure 13. Positive Angle of Attack

2.2.4 Axial and Normal Force

The drag force acts parallel with the relative motion of the fluid around an object, through the centre of pressure. In contrast, the axial force acts parallel with the length of the object (or chord line), through the centre of pressure. The lift force acts perpendicular to the relative motion of the fluid around an object, through the centre of pressure. In contrast, the normal force acts perpendicular to the length of the object (or chord line), through the centre of pressure.

At 0° angle of attack, the axial force is equal to the drag force, and the normal force is equal to the lift force. When the angle of attack changes, the lift, drag, normal, and axial forces are related by the following set of equations:

$$F_L = F_N \cos(\alpha) - F_A \sin(\alpha) \quad 2.33$$

$$F_D = F_N \sin(\alpha) + F_A \cos(\alpha) \quad 2.34$$

$$F_N = F_L \cos(\alpha) + F_D \sin(\alpha) \quad 2.35$$

$$F_A = -F_L \sin(\alpha) + F_D \cos(\alpha) \quad 2.36$$

where F_L , F_D , F_N , and F_A are the lift, drag, normal, and axial force respectively.

Figure 14 shows a labelled aerofoil, indicating the normal, axial, drag and lift forces when the aerofoil is pitched to a certain angle of attack.

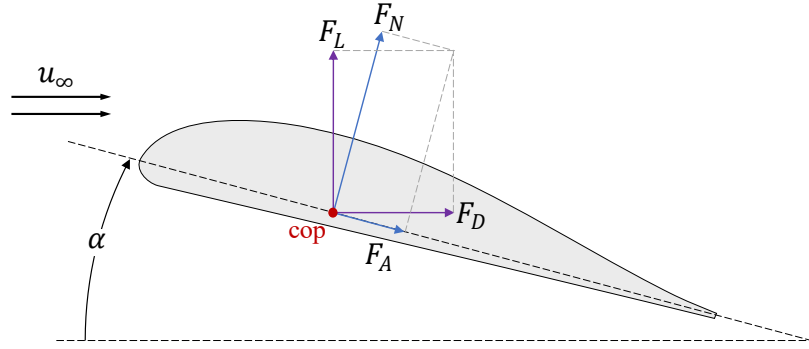


Figure 14. Aerofoil at a Non-Zero Angle of Attack

2.2.5 Aerodynamic Coefficients

Aerodynamic coefficients can be categorised into either static or dynamic coefficients. Static coefficients characterise the aircraft aerodynamics at constant and steady flow properties. These coefficients can be used for the initial response of the aircraft, or if the aircraft only makes gradual changes in its orientation during its flight. Dynamic coefficients characterise the aircraft aerodynamics in unsteady flow conditions and relate to aircraft stability.

If an aircraft is performing a rapid manoeuvre, the next iteration depends not only on the current flow properties, but also on the flow properties of the previous iteration. This is where dynamic coefficients, such as the roll/pitch/yaw damping moment coefficients are used. Dynamic coefficients were outside of the scope of this study, therefore only static coefficients were analysed.

The drag, lift, normal, and axial force coefficients are non-dimensional, static coefficients used to characterise aircraft performance and are calculated using the same equation,

$$C_x = \frac{2 \cdot F_x}{\rho \cdot A \cdot u^2} \quad 2.37$$

where F_x (N) is the chosen force to analyse,

C_x is the corresponding coefficient, and

A (m²) is the object's reference area (often the largest fuselage section, in rocketry).

2.2.6 Dynamic Pressure

Dynamic pressure is the pressure as a result of a fluid's velocity, and is defined by

$$q = \frac{1}{2} \cdot \rho \cdot u^2 \quad 2.38$$

The dynamic pressure is a non-negative quantity, as fluid density and velocity are non-negative quantities. At take-off, the rocket velocity is at zero, so according to Equation 2.38, the dynamic pressure is zero. Once the rocket reaches space, the fluid density is zero, so according to Equation 2.38, the dynamic pressure is zero. Therefore, there exists a point in the rocket flight where there is a maximum dynamic pressure. This point is known as 'max q'.

Max q is the limiting factor for the structural design of the rocket, because the maximum mechanical stress occurs at max q. Determining max q and the maximum stress on the rocket is crucial for designing its structure. If the maximum stress is under-predicted, then the rocket could undergo catastrophic failure during flight. If the maximum stress is over-predicted, then

the rocket structure would be over-engineered, resulting in a lower payload capacity than what could have been achieved. CFD can be used to determine the forces that the rocket would undergo during flight.

2.2.7 Shock Waves

Qualitative analysis can be very useful when analysing the flow phenomena in high-speed external aerodynamics. In supersonic flight, the flow phenomenon that is largely responsible for drag is shock waves, which are unavoidable when travelling at supersonic speeds. A shock wave can be characterised by a discontinuous and abrupt increase in the Mach number, pressure, temperature, and density shown in Figure 15.

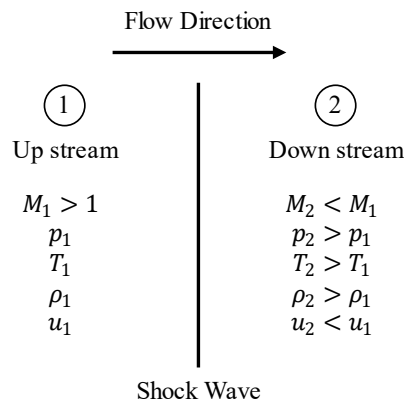


Figure 15. Shock Wave Flow Properties

Shock waves can be categorised into three types, by the shape formed: Normal, oblique, and bow shocks, as pictured in Figure 16. Normal shocks propagate perpendicular to the flow direction. Oblique shocks propagate at some angle ($0^\circ < \theta < 90^\circ$) to the flow direction, and form at the nose tip, leading edges, or any sharp geometric change on a rocket body. Bow shocks (detached shocks) form at a distance from the leading edge of an object, due to the large rounded leading edge of the object, and unlike normal and oblique shocks, bow shocks are curved.

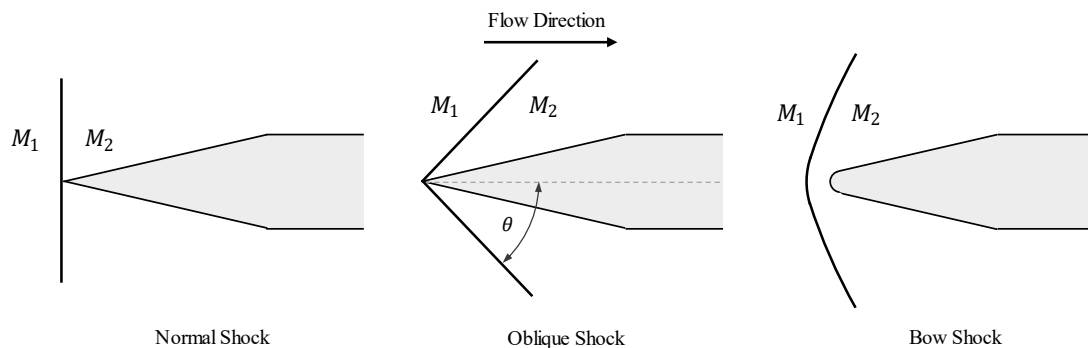


Figure 16. Types of Shocks

The rocket geometry can be adjusted to allow for the formation of either oblique or bow shocks. Bow shocks produce greater drag than oblique shocks, and this can be useful when slowing down an object, such as for capsule re-entry. However, this is extremely detrimental for STeVe and CLV.

2.3 Aerodynamic Heating

Aerodynamic heating is when the temperature of the air surrounding an object increases due to stagnation and viscous effects.

- Stagnation effects: When a fluid is slowed down, the fluid's kinetic energy is transferred into thermal energy (heat) into the surrounding fluid.
- Viscous effects: In the boundary layer, the flow increases in velocity as the distance away from the wall increases. Due to this velocity difference, there is friction between neighbouring fluid layers, which causes the surrounding fluid to be heated.

The impact of aerodynamic heating becomes more of a problem as the velocity increases, because stagnation and viscous heating effects both increase with velocity. The areas on a rocket where aerodynamic heating would be greatest are the parts that first contact the oncoming flow, such as the nose cone and fins. This is because stagnation heating contributes more to the maximum temperature than viscous heating.

As a part of this master's study, the heat transfer and surface temperature distribution were analysed for the nose cone of STeVe. The effects of aerodynamic heating were analysed using CFD, with the required theory detailed in this chapter.

2.3.1 Thermal Boundary Layer

Previous mention of boundary layer theory in Chapter 2.1.2 only discussed the velocity profile. However, similarly to the velocity boundary layer, there also exists a thermal boundary layer, and temperature profile, as shown in Figure 17.

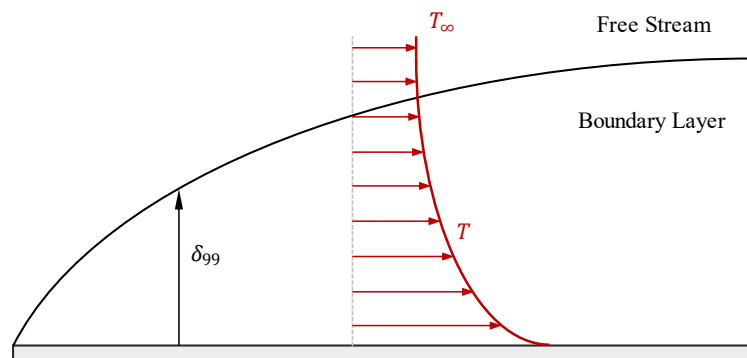


Figure 17. Temperature Profile and Thermal Boundary Layer for a Flat Plate

Unlike the velocity profile, which has a zero velocity at the wall, the temperature is at a maximum at the wall. For clarification, the temperature at the wall refers to the temperature of the first cell, nearest the wall, which is different to the wall temperature, which is the temperature of the solid wall. In reality, there is a temperature gradient across the wall. However, for this analysis, the temperature across the wall will be considered uniform.

2.3.2 Thermal Boundary Condition

In reality, heat is transferred via convection from the fluid to the rocket body, and heat is transferred via conduction through the rocket wall. A transient conjugate heat transfer analysis should be performed to simulate this accurately. This would require meshing the rocket shell along with the fluid domain. The rocket surface temperature, as well as the heat flux into the

rocket surface, could then be determined. However, performing such a simulation would be computationally intensive, and would still require an assumption of what the inner rocket temperature would be during flight. Therefore, only the fluid was simulated, with a thermal boundary condition on the rocket surface. The following thermal boundary conditions, shown in Table 2, are available for selection in Star-CCM+:

Table 2. Star-CCM+ Thermal Boundary Conditions

Thermal Boundary Condition	Specification
Heat flux	The amount of energy flowing across the boundary in W/m ² .
Heat source	The heat source in W.
Temperature	The boundary temperature in K.
Adiabatic	There is no energy flowing across the boundary. Therefore, there is no heat flux.
Convection	The convection flux across the boundary in W/m ² . The ambient temperature, heat transfer coefficient, and thermal resistance are required as input.

2.3.3 Heat Transfer Coefficient

There are three modes of heat transfer: convection, conduction, and radiation. Due to the high velocity flow present is aerodynamic heating, forced convection is the dominant heat transfer mode. The convective heat transfer coefficient can be determined using CFD as follows.

Newton's law of cooling, for convection, provides the equation for the rate of heat transfer,

$$Q = h \cdot A \cdot (T_s - T_{ref}) \quad 2.39$$

where Q is the rate of heat transfer in W,

h is the convective heat transfer coefficient in W/m².K,

A is the cross-sectional area where heat transfer occurs in m²,

T_{ref} is the reference temperature in K, and

T_s is the surface (wall) temperature in K.

A more useful quantity, called the heat flux, can be defined to represent the rate of heat transfer per unit area,

$$q = \frac{Q}{A} = h \cdot (T_s - T_{ref}) \quad 2.40$$

where q is the heat flux in W/m².

In Star-CCM+, the convective heat transfer is modelled locally based on the following equation,

$$q = \frac{\rho \cdot C_p \cdot u^*}{T^+} \cdot (T_s - T_c) \quad 2.41$$

where T_c is the near wall cell temperature in K, and

T^+ is the dimensionless temperature. T^+ is a function of y^+ and the Prandtl number, and is calculated differently, depending on the chosen wall function and y^+ value.

In Star-CCM+, the heat transfer coefficient can be obtained using four methods, each described below.

1) Local heat transfer coefficient:

Equation 2.40 and 2.41 can be equated to solve for h .

$$h_1 = \frac{\rho \cdot C_p \cdot u^*}{T^+} \quad 2.42$$

Equation 2.40 can be used to solve for the reference (free stream) temperature,

$$T_{ref,1} = T_s - \frac{q}{h_1} \quad 2.43$$

T^+ is dependent on the near wall y^+ for this method, and is therefore sensitive to any mesh changes.

2) Specified y^+ heat transfer coefficient:

This method calculates h and T_{ref} using the same as method one. However, the y^+ value is user inputted. This gives the user more control when setting a reference temperature. For completeness, the equations for h_2 and $T_{ref,2}$ are provided below

$$h_2 = \frac{\rho \cdot C_p \cdot u^*}{T^+} \quad 2.44$$

$$T_{ref,2} = T_s - \frac{q}{h_2} \quad 2.45$$

According to the Star-CCM+ user's manual, this method is less dependent on the mesh than method one, and the user specified y^+ value should be less than 150 (with a y^+ of 100 shown to work well).

3) Heat transfer coefficient:

This method calculates the local heat flux using Equation 2.41. Equation 2.40 can then be used to calculate the heat transfer coefficient based on a user specified reference temperature, $T_{ref,3}$

$$h_3 = \frac{q}{T_s - T_{ref,3}} \quad 2.46$$

Since q is calculated first, h will be zero when using an adiabatic thermal boundary condition with this method. This is because $T_s = T_c$ for an adiabatic thermal boundary condition, and therefore $q = 0$. According to the Star-CCM+ user's manual, this method is "primarily used for comparing the heat transfer coefficients to experimental data and or other previous work".

4) Virtual local heat transfer coefficient:

The energy equation is not solved, and only flow field values are used in calculating the heat transfer coefficient

$$h_4 = \frac{\rho \cdot C_{p,user} \cdot u^*}{T^+} \quad 2.47$$

The user inputs the fluid specific heat ratio and the Prandtl number. According to the Star-CCM+ user's manual, this method should not be used for laminar flows.

There are two important facts to remember when choosing a heat transfer coefficient and a reference temperature. Firstly, “there is an infinite number of heat transfer coefficient and reference temperature combinations that give the same surface heat flux upon convergence” (Siemens, 2023b). Secondly, the choice of heat transfer coefficient will not affect the rest of the simulation.

Since there is an infinite number of heat transfer coefficient and reference temperature pairs, a user specified h or T_{ref} can be used to find a ‘complete pair’ based on an existing pair.

$$q_1 = q_2$$

$$h_1(T_s - T_{ref,1}) = h_2(T_s - T_{ref,2}) \quad 2.48$$

This also means that comparing heat transfer coefficients between different methods is not useful, unless the heat transfer coefficient and reference temperature pairs are known, and there is heat transfer occurring ($q \neq 0$).

2.3.4 Recovery Factor

When free stream flow contacts a surface, the fluid is slowed down. The kinetic energy of the fluid is converted into thermal energy (heat), and the temperature near the surface increases. The stagnation temperature, T_0 , is defined as the temperature the fluid would be if it is slowed down to zero velocity. In this case, all the kinetic energy from the fluid is converted into thermal energy.

$$T_0 = T_\infty \left(1 + \frac{\gamma - 1}{2} \cdot M_\infty^2 \right) \quad 2.49$$

In reality, it is not possible for all the kinetic energy from the fluid to be converted into heat. In other words, the stagnation temperature will never be reached. The actual temperature at an adiabatic wall is defined as the adiabatic wall temperature, T_{aw} .

Based on the previous definitions, a parameter called the recovery factor, r , can be defined as

$$r = \frac{T_{aw} - T_\infty}{T_0 - T_\infty} \quad 2.50$$

The recovery factor can be thought of as how close the actual temperature is to the stagnation temperature or how much of the kinetic energy has been recovered as thermal energy. Using the recovery factor and stagnation temperature definitions, the adiabatic wall temperature can be written in terms of M_∞ and T_∞ ,

$$T_{aw} = r \cdot (T_0 - T_\infty) + T_\infty$$

$$T_{aw} = r \cdot \left(T_\infty \left(1 + \frac{\gamma - 1}{2} \cdot M_\infty^2 \right) - T_\infty \right) + T_\infty$$

$$T_{aw} = r \cdot T_\infty \cdot \left(1 + \frac{\gamma - 1}{2} \cdot M_\infty^2 - 1 \right) + T_\infty$$

$$T_{aw} = r \cdot T_\infty \cdot M_\infty^2 \cdot \frac{\gamma - 1}{2} + T_\infty$$

$$T_{aw} = T_\infty \cdot \left(r \cdot M_\infty^2 \cdot \frac{\gamma - 1}{2} + 1 \right) \quad 2.51$$

The recovery factor from CFD can be used to approximate the wall temperature at flight conditions, based only on the free stream temperature at which that Mach number occurs. The application of this method is shown in Chapter 6.

2.4 Meshing

Creating a mesh is one step, if not the most crucial in the CFD process. CFD functions by dividing the fluid volume into a finite number of small elements, called a mesh. The mesh will influence convergence, accuracy, precision, and the speed of the simulation. Due to the importance of mesh selection, this chapter of the literature review presents and analyses research on different elements, methods, and software that can be used in the meshing process.

2.4.1 Types of Mesh Elements

The most common three-dimensional meshing elements available are hexahedral, tetrahedral, and polyhedral elements, as shown in Figure 18 (Sosnowski *et al.*, 2019). In this chapter, literature examining the effectiveness of the different meshing elements is provided, beginning with a brief description of the elements:

- 1) Hexahedral: Elements with 6 quadrilateral faces, 8 vertices, and 12 edges.
- 2) Tetrahedral: Elements with 4 triangular faces, 4 vertices, and 6 edges.
- 3) Polyhedral: Elements with more than 4 polygonal faces.

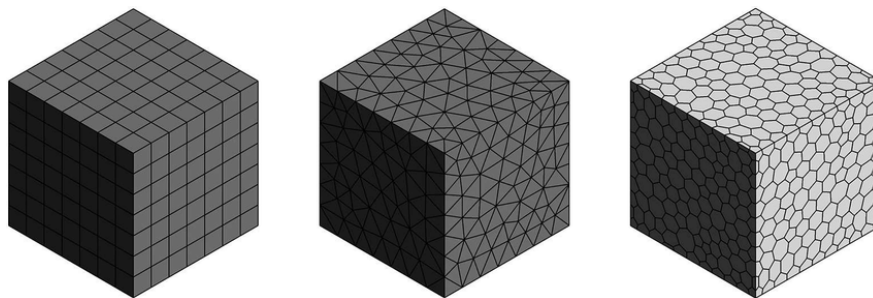


Figure 18. Hexahedral (left), tetrahedral (middle), and polyhedral (right) mesh elements (Sosnowski *et al.*, 2019)

Balafas (2014) analysed hexahedral, tetrahedral, and polyhedral mesh generation for CFD applications, to determine the efficiency and quality of polyhedral elements. He used two cases to compare the different meshes: laminar flow through a cube, and laminar flow around a cylinder. The number of iterations required to achieve a certain level of convergence was determined for the flow through the cube, and is shown in Table 3. The tetrahedral mesh took roughly 9.5 times the number of iterations to achieve a solution, compared to the hexahedral mesh. The tetrahedral mesh also had roughly 4.6 times the number of elements, compared to the hexahedral mesh. The hexahedral mesh had ~0.4% fewer elements and took ~17% fewer iterations to converge, compared to the polyhedral mesh.

Table 3. Iterations required to achieve the same level of convergence (Balafas, 2014)

Mesh type	Number of Cells	Iterations to Convergence
Tetrahedral	6168	229
Polyhedral	1336	28
Hexahedral	1331	24

A residual is the leftover value that arises when solving the governing equations in CFD. It is a measure of the convergence of the solution, where a low residual indicates that the solution has

converged and there is a low convergence error. In a simulation, each cell contains a residual value for each governing equation; however, the residuals for each governing equation are averaged (using various methods) over the entire fluid domain to easily assess convergence. Figures 19 and 20 show the residuals for the flow through the cube and around the cylinder respectively. The residuals for the polyhedral and hexahedral meshes were both similar, while the tetrahedral mesh residuals were much higher.

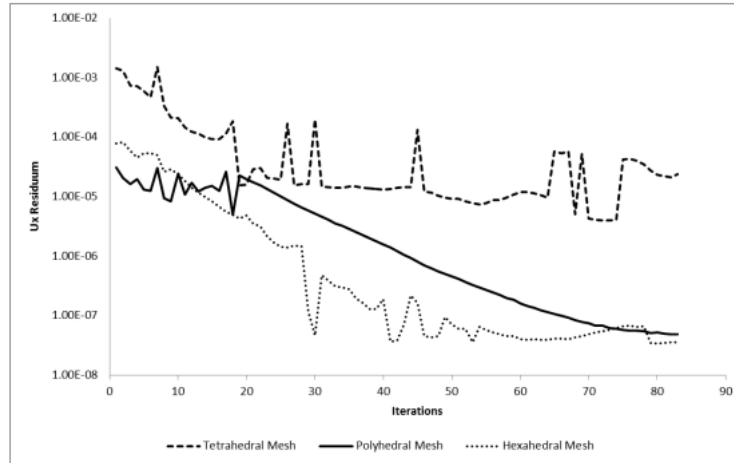


Figure 19. Flow through a Cube Residuals (Balafas, 2014)

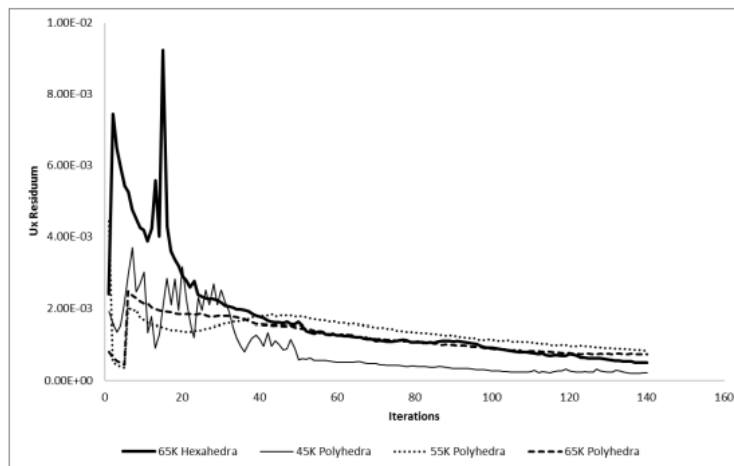


Figure 20. Flow around a cylinder, residuals for 45k – 65k elements (Balafas, 2014)

Based on the results from Balafas, tetrahedral elements are unsuitable for CFD meshing and should not be used. It can also be concluded that polyhedral elements can be a suitable replacement for hexahedral elements, as the mesh sizes and convergence were similar for both. This is useful for complex geometries, where hexahedral mesh elements often cannot be used.

Ibraheem (2021) evaluated the efficiency of polyhedral elements, compared to tetrahedral elements, for flow around a ship's rudder. He found that the polyhedral mesh converged almost three times quicker than the tetrahedral mesh. The polyhedral and tetrahedral mesh had a percentage error of 2.4% and 4.7% respectively for the lift coefficient, and 5.6% and 15% respectively for the drag coefficient. This confirms the increased accuracy and quicker solution time of the polyhedral mesh, when compared to the tetrahedral mesh.

Table 4 summarises the advantages and disadvantages of the different elements, based on the literature.

Table 4. Advantages and Disadvantages of the Different Mesh Elements

Element Type	Advantages	Disadvantages
Hexahedral	• Fewer elements are required than tetrahedral. • Better convergence than tetrahedral. • Regular and structured elements provide accurate and fast solutions.	• Difficult to generate mesh for complex geometries.
Tetrahedral	• Flexible, will generate a mesh for almost any geometry.	• Greater number of elements required than hexahedral and polyhedral. • May be difficult to reach convergence. • Higher probability of low-quality elements, which could cause poor convergence.
Polyhedral	• Flexible, will generate a mesh for almost any geometry. • Fewer elements are required than tetrahedral. • Improve the quality in the transition zone between different elements. • Each element has many neighbouring elements, resulting in greater accuracy. • Less sensitive to stretching than tetrahedral elements.	• Could lead to higher computational cost, due to increased complexity in solving the equations (due to more neighbouring elements). • Chance to generate low-quality elements in complex geometries.

2.4.2 Difference in Software Approach

Each CFD software approaches meshing differently. In this study, Ansys Fluent meshing and Star-CCM+ were chosen for comparison, with the optimal meshing software chosen to use for the rest of the study.

Ansys Fluent meshing uses a proprietary meshing technique called Mosaic meshing. According to Ansys, Mosaic meshing “accelerates the meshing process with a reduced face count, higher quality cells and efficient parallel scalability” (Ansys, no date). Mosaic meshing uses polyhedral elements to connect different element types to each other. Due to the complex nature of the rocket geometry, with thin fins and a sharp nose cone; a purely hexahedral element type is not possible. Ansys Fluent meshing can be used to create a poly-hexcore mesh (as shown in Figure 21), which comprises of three different mesh types:

- 1) Poly-prism elements in the boundary layer, near the object surface.
- 2) Structured hexahedral elements in the main fluid domain.
- 3) Polyhedral elements connecting the prism layer elements to the main fluid domain.

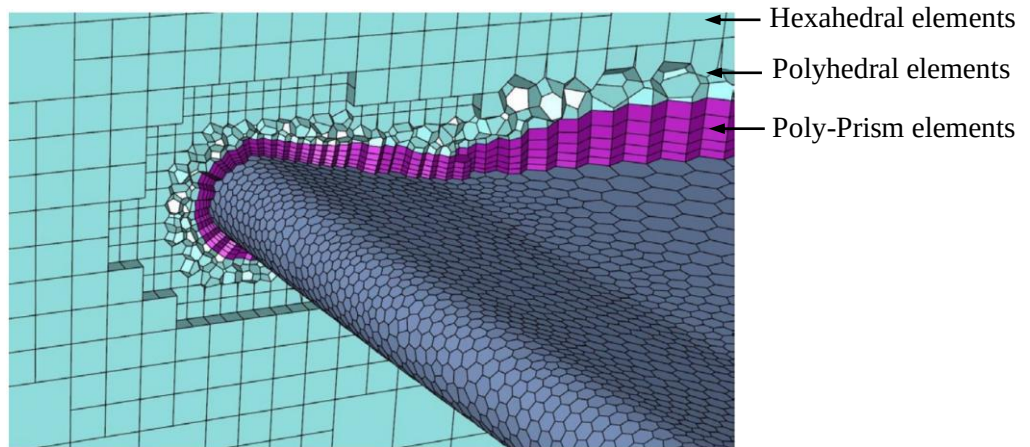


Figure 21. Ansys Poly-hexcore mesh (Ansys, 2020)

Ansys Fluent meshing has four element options for meshing the volume. These options are explained below as background to the following sections:

- 1) Polyhedral: The entire fluid domain is filled with polyhedral elements.
- 2) Poly-hexcore: The bulk of the fluid domain is filled with high-quality hexahedral elements with poly-prism elements in the boundary layer. In the transition regions between the boundary layer and the bulk mesh, polyhedral elements are used.
- 3) Hexcore: The entire fluid domain is filled with hexahedral elements. Often this is not possible due to complex geometry, so tetrahedral elements are used in combination with the hexahedral elements.
- 4) Tetrahedral: The entire fluid domain is filled with tetrahedral elements.

Zore et al. (2019) analysed the effectiveness of a poly-hexcore mesh, compared to a hexcore mesh, for a 100-passenger jet airliner. It was found that (in general) using a poly-hexcore mesh reduces the total element count by 20% to 50%, when compared to a hexcore mesh. This in turn speeds up the Fluent solver by 10% to 50%. When comparing poly-hex with hexcore in terms of the jet airliner: the total element count was reduced by 48%, and the computational time was reduced by 41%. For the lift coefficient at an 18.58° angle of attack, the poly-hexcore and hexcore mesh were, respectively, $\sim 1.5\%$ and $\sim 0.4\%$ higher than wind tunnel data, as shown in Figure 22. This confirms the accuracy of both meshes, with the hexcore mesh being slightly closer to the experimental results, than the poly-hexcore mesh.

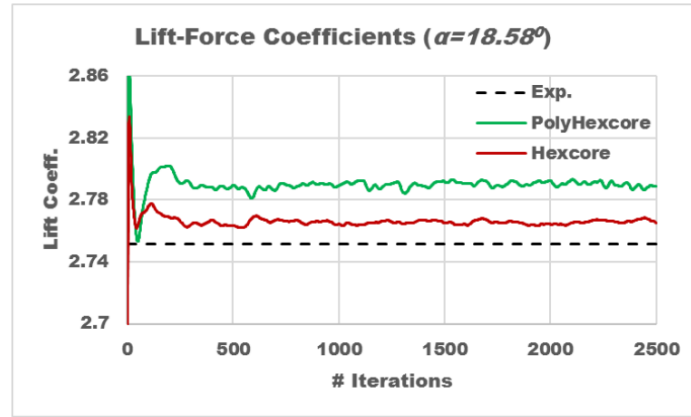


Figure 22. Graph of Lift Coefficient vs. Number of Iterations (Zore *et al.*, 2019)

Figure 23 shows the graph of the drag coefficient vs. angle of attack. The drag coefficient results from the poly-hexcore, hexcore, and experimental data are similar in value.

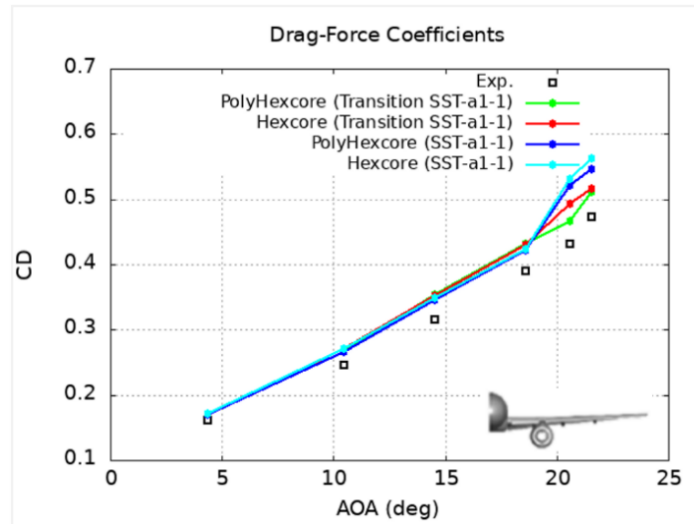


Figure 23. Graph of Drag Coefficient vs. Angle of Attack (Zore *et al.*, 2019)

It must be noted that Zore (2019) works as a Senior R&D Engineer at Ansys, and more research is required to determine the downsides of poly-hexcore meshes. The results from Zore's report are summarised in Table 5.

Table 5. Summarised Results from Zore *et al.* (2019)

Variable	Poly-hexcore	Hexcore
Total element count	Reduced by 48%	-
Computational time	Reduced by 41%	-
Lift coefficient	~1.5% greater than wind tunnel data	~0.4% greater than wind tunnel data

The following information regarding meshing is from an Ansys white paper report (Ansys, 2020). Ansys developed Mosaic meshing technology to connect one type of mesh accurately and efficiently to another type of mesh. This technology forms the basis for the poly-hexcore mesh, as the prism layers are connected to the hexcore elements, using polyhedral elements.

Ansys (2020) states that simulations using poly-hexcore meshes are “faster with greater solution accuracy while using less RAM”. Ansys simulated a Formula One wing using poly-hexcore and standard tet-hexcore meshing. The poly-hexcore mesh was 46% smaller in element size and had greater quality than the tet-hexcore mesh. Ansys then compared a poly-hexcore mesh with an all-polyhedral mesh on a triangular prism bluff body flame holder. The poly-hexcore mesh had 19% fewer elements, was 47% faster, and used 34% less RAM when compared to the all-polyhedral mesh.

The other CFD software option was Star-CCM+, which has three meshing options for the volume mesh:

- 1) Trimmed cell: A trimmed cell mesh is a structured mesh that uses entirely hexahedral elements in the volume. However, in some cases with complex geometries, in the transition between the prism layers and the volume, some tetrahedral and polyhedral elements are generated. Trimmed cell meshes are fast and efficient to generate.
- 2) Polyhedral: An unstructured mesh where only polyhedral elements are used. Polyhedral meshes are suitable for complex geometries.
- 3) Tetrahedral: An unstructured mesh where only tetrahedral elements are used.

Each of these volume mesh options can be combined with one of the following prism layer meshing options:

- 1) Prism layer mesher: This generates a volume around the surface based on the total specified prism layer thickness. Each prism layer is then generated until that volume is filled.
- 2) Advancing layer mesher: This generates each entire layer successively, starting with the prism layer closest to the surface. Each prism layer is generated until the total number of prism layers specified is reached.

Both the prism layer meshing options seem similar; however, there are some key differences. The advancing layer mesher keeps the total prism layer thickness and the first layer thickness constant over the entire model geometry. This can result in some poor prism cells over curves, as there might be a large volume change between neighbouring cells in the prism layer. The advancing layer mesher also takes a substantially longer time to generate a mesh, compared to the prism layer mesher. In contrast, the prism layer mesher has many input parameters that can be used to adapt sections of the prism layer to capture complex features in the model. The prism layers can be adapted to fit certain geometric features or be removed entirely for areas where prism layers are not required, which is not possible when using the advancing layer mesher.

Star-CCM+ and Ansys Fluent both use different approaches to meshing. As discussed above, there is positive literature regarding the accuracy and efficiency of Ansys poly-hexcore meshes. However, due to the limited literature available comparing Star-CCM+ and Ansys Fluent meshing, it was decided to use the first validation case to compare the two meshing options. The optimal meshing option was then used for the rest of the study.

2.4.3 Mesh Domain Construction

There are no exact guidelines on mesh domain construction for high-speed external airflow. Therefore, the mesh domains from previous CFD studies on external rocket aerodynamics are presented and discussed in this chapter of the literature review, to build a catalogue and create

guidelines to start a mesh. In total, 14 papers were researched, with the literature and mesh details summarised in Tables 6 and 7:

Table 6. Literature Details

Number	Author/s	Year of Publish	Rocket Geometry	Software
1	Moore, McInville and Robinson	1999	Multi-fin weapon	-
2	Fujimoto and Fujii	2005	Apollo configuration	-
3	Gülay et al.	2011	Modified finner	Ansys Fluent
4	Allen and Ghoreyshi	2018	Basic finner	-
5	Dahalan et al.	2017	Curved fin rocket	Ansys Fluent
6	Ocokoljić, Rašuo and Bengin	2017	Guided missile	Ansys Fluent
7	Runduo and Xiaobing	2018	Guided rocket	Ansys Fluent
8	Shelton, Martin and Silva	2018	Basic finner	NASA FUN3D
9	Hammargren	2018	Maxus sounding rocket	Ansys CFX
10	Sahbon et al.	2022	Sounding rocket	Ansys Fluent
11	Sankalp et al.	2022	Sounding rocket	Ansys Fluent
12	Milne et al.	2023	Sounding rocket	Ansys Fluent
13	Bunescu et al.	2023	Basic finner	-
14	Żurawka et al.	2023	Grot sounding rocket	Ansys Fluent

Table 7. Summary of Mesh Details from the Chosen Literature

Number	Element type	Domain Shape	Domain Dimensions	Number of Prism Layers	Number of Elements (10 ⁶)
1	Hexahedral	Bullet (with a full symmetry plane)	25 <i>D</i> diameter 0.59 <i>L</i> behind <i>L</i> ahead Radial: 51 Circumferential: 54 Axial: 165	-	-
2	Hexahedral	Sphere (with a full symmetry plane)	21 <i>D</i> diameter Radial: 117 Circumferential: 85 Axial: 91	-	-
3	Tetrahedral	-	15 <i>L</i> ahead 24 <i>L</i> behind	24	-
4	Tetrahedral	-	-	-	20.7
5	Tetrahedral	Cylinder	28.5 <i>D</i> diameter 2.86 <i>L</i> length	-	-
6	Tetrahedral	Ellipsoid	3 <i>L</i> major axis 16 <i>D</i> minor axis 4.34 <i>L</i> length 15 <i>D</i> ahead 17 <i>D</i> behind	13	2.6
7	Hexahedral	Cylinder	20 <i>D</i> diameter 10 <i>L</i> length	-	5
8	Tetrahedral	Cube (with a full symmetry plane)	120 <i>D</i> lengths 12 <i>L</i> lengths	20	65
9	Hexahedral and tetrahedral	Cylinder	-	-	-
10	Tetrahedral	Bullet	30 <i>D</i> diameter 4 <i>L</i> length from base to centre of hemisphere	10	-
11	Tetrahedral	Cuboid	-	-	4.2
12	Tetrahedral	-	-	15	0.43
13	Polyhedral	Cylinder	1667 <i>D</i> diameter	-	9
14	Tetrahedral and hexcore	Cylinder	4 <i>L</i> (102 <i>D</i>) diameter 5 <i>L</i> behind 2 <i>L</i> ahead	30 to 300	2.5

The following observations were made based on the researched literature:

Software: The most common CFD software used was Ansys Fluent, with 57% of the literature using it.

Domain shape: The most common domain shape was the cylinder. The second most common domain shape was the bullet (cylinder with a hemisphere at the inlet) geometries. A bullet shape was chosen, as it has fewer elements when compared to a cylindrical shape of the same length.

Number of elements: There was a large variation in the number of elements, ranging from 400 000 elements to 65 million elements.

Element type: Tetrahedral and hexahedral elements were the most common element type, with only the most recent report (2023) using polyhedral elements.

Prism layers: When the y-plus was mentioned, all reports aimed to have a y-plus of around 1. The number of boundary layers ranged from 10 to 24, when mentioned.

Domain dimensions: The total length of the domains ranged from 2.86 to 16 model lengths. The domain diameters ranged from 21 to 30 model diameters, in most cases. One paper mentioned the distance in front of the model to be 2 model lengths, and the distance behind the model to be 5 model lengths.

2.5 Semi-Empirical Aeroprediction Codes

Semi-empirical aeroprediction codes determine the aerodynamic coefficients of rocket geometries using a combination of analytical calculations and experimental data. Two semi-empirical codes, RASAero and Missile Datcom, were used in this research project as another data source to compare with the CFD model and to investigate the accuracy of the semi-empirical codes.

Engineers at ASRI use RASAero and Missile Datcom to determine the aerodynamic coefficients of rockets being developed, within an in-house six-degree-of-freedom code called PyROPS. The coefficients are supplied to the in-house software to determine the rocket flight path, apogee, and impact points. ASRI has not validated the aerodynamic coefficients from the semi-empirical methods against CFD data. It would be useful to know the accuracy of various aerodynamic coefficients at different Mach numbers and angle of attacks for future use.

2.5.1 RASAero

RASAero (Rogers AeroScience Aero) is a free software application that can perform aerodynamic analyses and flight simulations for model rockets to sounding rockets. It uses empirical wind tunnel and free flight data combined with analytical calculations to determine the aerodynamic coefficients. A large database of interference effects between different bodies is used to obtain results for different geometry combinations. RASAero has been calibrated against “NACA and NASA wind tunnel model data, free-flight model data and sounding rocket data, missile aerodynamic data, and professional engineering method missile aerodynamic analysis programs, against which in comparison tests RASAero has demonstrated equivalent numerical accuracy” (Rogers and Cooper, 2019b). The latest RASAero software (version 1.0.2.0) was used in this study.

The following paragraphs explain some of the features and limitations of the RASAero software, referencing the RASAero user manual (Rogers and Cooper, 2019a).

Geometry Creation: RASAero has five input parts for geometry creation, which can all be combined in different orders: Nose cone, body tube (fins), transition, fin canister, and boattail.

There are seven nose cone inputs: Conical, tangent ogive, Von Kármán ogive, power law, LV-Haack, parabolic, and elliptical. All nose cone profiles, bar the power law, are tangential to the body tube, which is a major limitation of the RASAero software. For the spinner validation geometry, the nose cone is an arc with a constant radius that is not tangential to the body tube (secant ogive). This nose cone shape cannot be created directly using RASAero. To bypass this limitation, the closest power law nose cone shape was fitted to the spinner nose cone. The 0.75 power law profile fits the spinner nose cone geometry closely, as shown in Figure 24, and was used for validation.

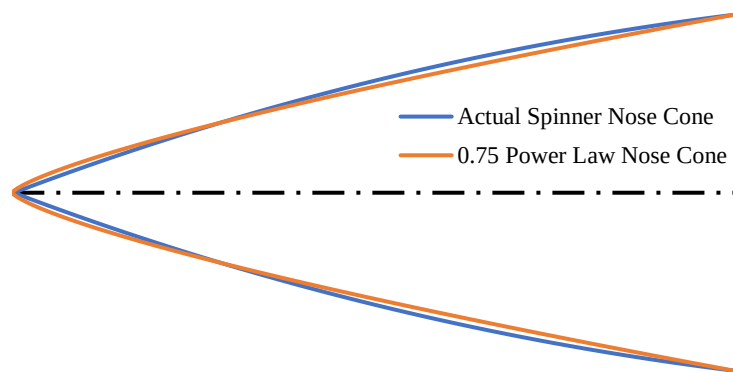


Figure 24. Spinner and 0.75 Power Law Nose Cone Geometry

The body tube is a simple cylinder with the diameter being equal to the diameter at the end of the previous geometry item. Fins can be added to the base of the tube. The following fin options are available for use: hexagonal, NACA, double-wedge, biconvex, hexagonal blunt-base, single wedge, rounded, and square aerofoils. RASAero covers all fin geometries. However, there is no option to add fin cant in RASAero. This is a limitation, as STeVe contains fins with cant, so there will be some inaccuracy present.

The transition allows for a length and an end diameter to be inputted, to create a truncated cylinder profile. This part can be used to create a step if a small length (0.0001 in) is inputted. The fin canister part is effectively a transition part and a body tube part combined. The boattail part is effectively a transition part, where the end diameter must be less than the start diameter.

Modelling Options: RASAero has three main modelling options that can be selected for various reasons: surface finish, Rogers modified Barrowman, and all turbulent flow.

Table 8 shows the following surface roughness options that can be selected.

Table 8. RASAero Surface Roughness Options

Surface Finish Selection	Equivalent Sand Roughness (in)
Smooth	0
Polished	0.00005
Sheet metal	0.00016
Smooth paint	0.00025
Camouflage paint	0.0004
Rough camouflage paint	0.0012
Galvanized metal	0.006
Cast iron (very rough)	0.01

Either the Barrowman method (Barrowman, no date) or the Rogers Modified Barrowman method (Rogers and Cooper, 2019a) can be used for calculating the subsonic centre of pressure. According to the RASAero documentation, the Rogers Modified Barrowman method includes several factors left out by the original Barrowman method. These factors improve the accuracy of the centre of pressure calculations.

The equations to determine the skin friction coefficient depend on whether the flow is laminar or turbulent. Therefore, the skin friction coefficient depends on the Reynolds number. RASAero calculates the Reynolds number of the flow along the rocket body. If the Reynolds number is less than 5×10^5 , the laminar skin friction calculations are performed and if the Reynolds number is greater than 5×10^5 , the turbulent skin friction calculations are performed. If the ‘all turbulent’ option is selected, RASAero immediately transitions to turbulent flow for the skin friction calculations.

Aerodynamic Coefficients: The aerodynamic coefficients can be generated by observing the ‘aero plots’. RASAero uses the largest cross-sectional area of the geometry when calculating the aerodynamic coefficients. This does not include the cross-sectional area of the fins. The drag, lift, normal force, and axial force coefficients, along with the centre of pressure can be obtained from Mach 0.01 to 25 in 0.01 increments. The angle of attack can also be varied; however, this is limited. The angle of attack can only be varied up to 16° for Mach 0.1, 0.5, 1.1, 2, and 5. For Mach numbers not in this selection, the aerodynamic coefficients can only be determined at 0° , 2° , and 4° angles of attack.

2.5.2 Missile Datcom

Missile Datcom is a semi-empirical aeroprediction code, similar to RASAero, that was developed in Fortran by the United States Air Force. Initially released in 1985, it is continuously updated to this day. The information in this chapter is sourced from the Missile Datcom 1998 user’s manual (Air Force Research Laboratory, 1998).

Geometry Creation: Missile Datcom does not have some of the geometry limitations that RASAero has. For example, fin cant and roll angles can be added. Additionally custom nose

cone geometries can be created based on specified points. However, like RASAero, Missile Datcom can not generate a secant ogive nose cone as part of the default nose cones. In the user's manual, it states that if “a secant ogive cannot be defined, it is recommended that the nose be approximated with a POWER nose.” In this study, a power law nose cone approximation was used, consistent with the approach taken in RASAero. Figure 25 shows a summary of geometry input in Missile Datcom. D_{EXIT} was always set to 0 to account for the full effects of base drag.

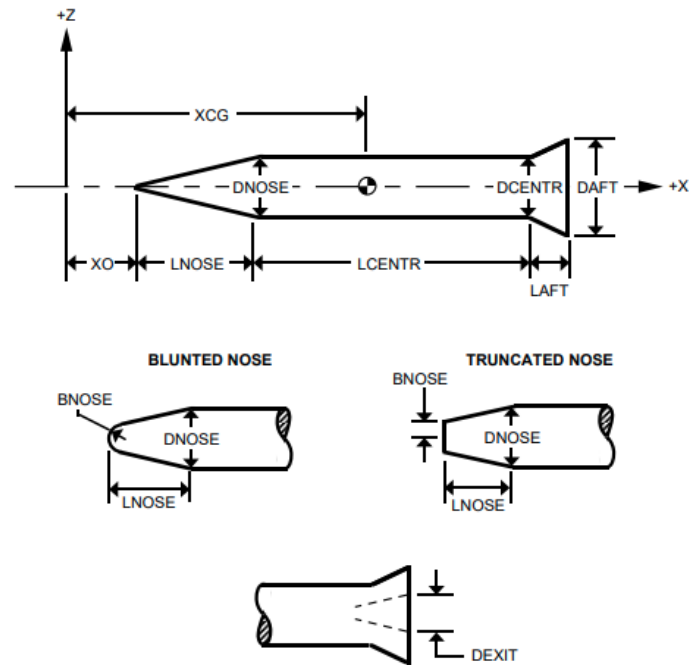


Figure 25. Missile Datcom Geometry Input (Air Force Research Laboratory, 1998)

Fins are constructed by specifying chord cross sections at varying spanwise locations. Figures 26 and 27 show how to specify Missile Datcom fin geometry.

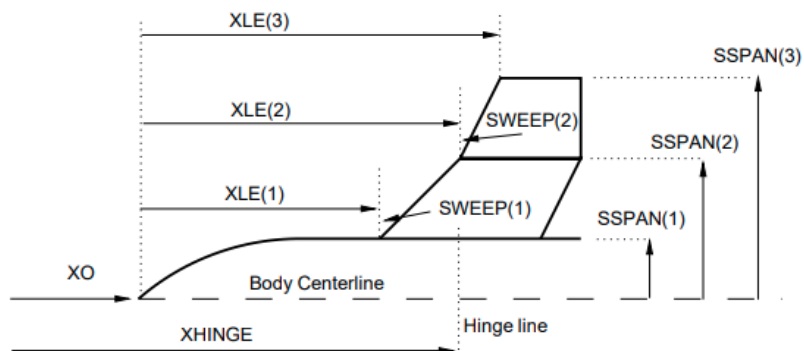


Figure 26. Fin Placement (Air Force Research Laboratory, 1998)

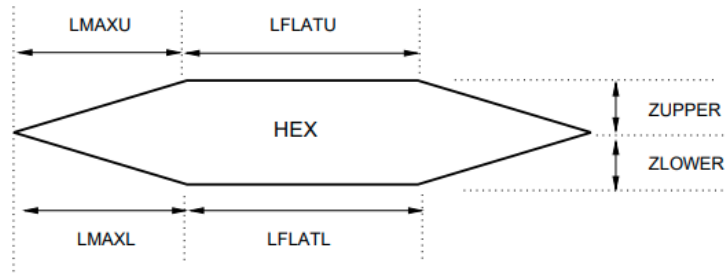


Figure 27. Fin Cross Section (Air Force Research Laboratory, 1998)

Modelling Options: The various roughness options that can be selected in Missile Datcom are shown in Table 9, with the recommended RHR values for different materials shown in Table 10. ‘RHR’ is the value written in the Missile Datcom input file.

Table 9. Missile Datcom Surface Roughness Options

Surface Finish Selection	Equivalent Sand Roughness (in)	RHR
Aerodynamically Smooth	0	0
Polished Metal or Wood	0.00002 to 0.00008	6 to 26
Natural Sheet metal	0.00016	53
Smooth matte paint, carefully applied	0.00025	83
Camouflage paint, average application	0.0004	133
Camouflage paint, mass production spray	0.0012	400
Dip galvanized metal surface	0.006	2000
Natural surface or cast iron	0.01	3333

Table 10. Missile Datcom Recommended RHR Values

Application	RHR
Steel structural parts	250
Aluminium and titanium structural parts	125
Close tolerance surfaces	63
Seals	32

Aerodynamic Coefficients: Missile Datcom has a larger number of outputs, and fewer restrictions, when compared to RASAero. It can output all static and dynamic aerodynamic coefficients at any combination of Mach number, roll angle, and angle of attack.

3. Model Validation

When designing any aerial vehicle, it is important to predict the aerodynamic performance under varying conditions, such as Mach number, altitude, and angle of attack. To achieve this, the aerodynamic characteristics must be determined. Aerodynamic characteristics refer to the properties of an aircraft that influence the surrounding air flow. These characteristics can be determined experimentally (wind tunnel testing), using CFD simulations, semi-empirical aeroprediction codes, or analytical calculations. CFD simulations are a relatively inexpensive and quick method to obtain these aerodynamic coefficients. However, the results from the CFD simulations must be validated to ensure accurate and reliable results.

Chapter 3.1 explains fully the process of validation, and how it differs from verification. The rest of this chapter then addresses the entire validation process done in this study, from defining the geometry to creating the flow models, and finally discusses the results.

3.1 Validation and Verification

Since validation is a fundamental part of this study, this chapter clarifies the subject of validation and how it differs from verification.

When discussing validation and verification, the concepts of error, accuracy, and precision must first be mentioned. Error is defined as “a recognisable deficiency in any phase or activity of modelling and simulation that is not due to lack of knowledge.” (AIAA G-077-1998). Accuracy and precision are two measures used to observe and analyse errors in results. Accuracy is a measure of the difference between the obtained results and the actual results, while precision is a measure of the difference between just the obtained results. Since semi-empirical methods and most CFD codes are deterministic in nature, precision is not of importance in this study. However, measuring accuracy must be discussed further.

Validation and verification are both parameters used to assess the accuracy of a solution. Verification is “The process of determining that a model implementation accurately represents the developer's conceptual description of the model and the solution to the model.” (AIAA G-077-1998). Since a CFD code is not being created or modified in this study, verification was not required. Validation is “the process of determining the degree to which a model is an accurate representation of the real world from the perspective of the intended uses of the model.” (AIAA G-077-1998). Validation compares results with experimental data, and in the case of this study, CFD results were validated by comparing them with experimental wind tunnel data.

To confirm that the results from the CFD model are accurate to reality, the results must be validated against experimental data from wind tunnel testing as wind tunnels capture flow phenomena in reality. Comparing against other CFD simulations, semi-empirical aeroprediction codes, or analytical calculations is also important and was done in this study; however, technically speaking, this would not be considered validation.

In this study, five different validation cases were used to test various parameters that would affect the accuracy of the CFD model. Some of the parameters were the domain size, meshing techniques, mesh independence, flow models, geometry complexity, different CFD applications, and the accuracy of various semi-empirical methods, among other options. The

different effects of these parameters can then be observed, and this information can be used to validate the CFD models for the STeVe and CLV rockets.

3.2 Geometry Description

The geometry, as specified by the author, for each validation case is shown below. Often the dimensions provided by the authors were incomplete, so assumptions were provided where necessary. Appendix A shows the labelled orthographic and isometric drawings for each validation geometry and fluid domain.

3.2.1 Dunn Spinner

Figure 28 shows the geometry of the Army Navy spinner, as provided by Dunn (1989) with the dimensions are given in terms of calibre units.

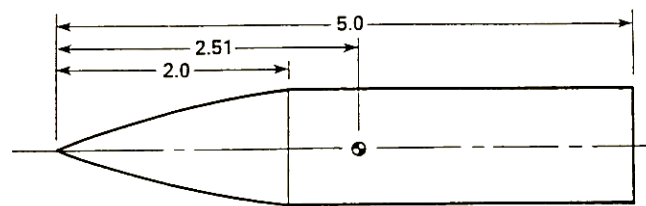


Figure 28. Dunn Spinner, Dimensions in Calibres (Dunn, 1989)

The calibre diameter or Reynolds number used for the wind tunnel tests were not provided by Dunn. The ogive nose shape was also not described in enough detail to accurately replicate. A paper by Weinacht and Danberg, 2005, used the Army Navy spinner, as shown in Figure 29, with a calibre of 20 mm for CFD simulations. The calibre and ogive shape used by Weinacht, were used for the Dunn spinner validation. A small radius of $0.001D$ was added to the spinner nose, as sharp points cause problems when meshing and could create poor-quality cells. The reference area for the aerodynamic coefficients was the largest cross-sectional area, 314.16 mm^2 .

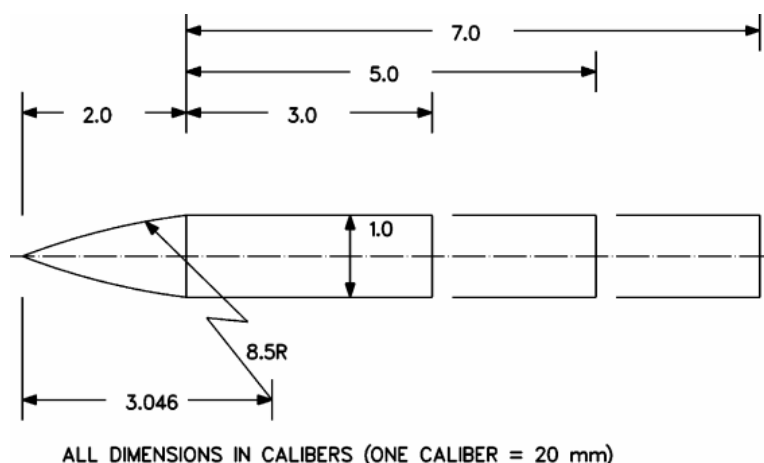


Figure 29. Weinacht and Danberg Spinner (Weinacht and Danberg, 2005)

Due to reasons discussed in Chapter 2.5, a secant ogive nose cone cannot be created in RASAero or Missile Datcom. Therefore, for the Dunn Spinner, a 0.75 power law nose cone geometry was used as it fits the Dunn Spinner's ogive nose cone well, as shown in Figure 24.

3.2.2 Dunn Finner

The basic finner geometry used by Dunn (1989) is shown in Figure 30, with the dimensions given in calibres. A small radius of $0.001D$ and $0.0005D$ was applied to the nose cone tip and fin leading edges respectively. The Reynolds number and calibre were not given in the report, so a diameter of 20 mm was chosen, to maintain consistency with the Dunn spinner. The reference area for the aerodynamic coefficients was the largest fuselage cross-sectional area, 314.16 mm^2 .

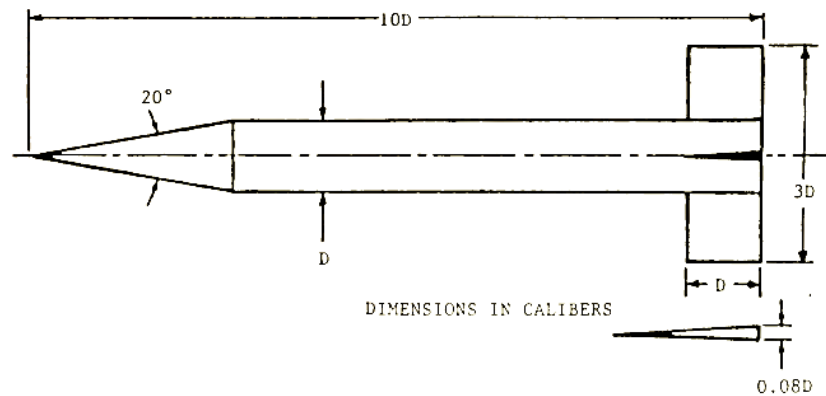


Figure 30. Dunn Finner (Dunn, 1989)

3.2.3 Dunn Modified Finner

The geometry Dunn used for the modified finner (air slew demonstrator) is shown in Figure 31 (Dunn, 1989), with dimensions given in inches. Dunn mentioned that a “tangent ogive body” was used for the nose shape. Not enough information was given for the fin geometry to replicate accurately. A calibre size of 426.72 mm (16.8 in) was used. The reference area for the aerodynamic coefficients was the largest fuselage cross-sectional area, $1.43 \times 10^5 \text{ mm}^2$.

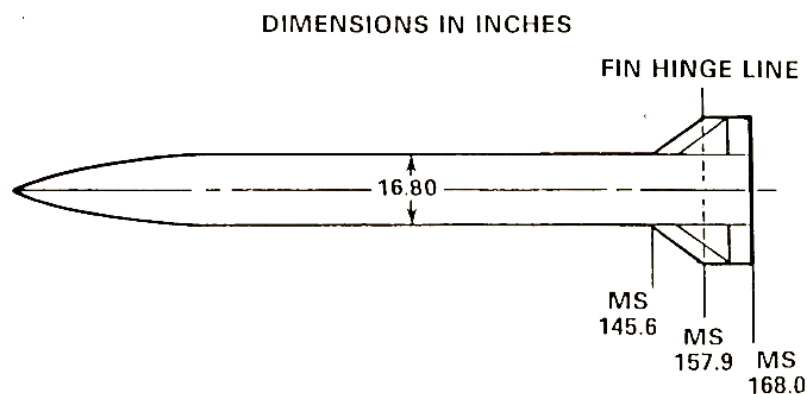


Figure 31. Dunn Modified Finner (Dunn, 1989)

A small radius of $0.001D$ and $0.0005D$ was applied to the nose cone tip and fin leading and trailing edges respectively. The fin geometry from a modified finner used in a report by Bhagwandin (2016) is shown in Figure 32, and was used to supplement the geometry given by Dunn.

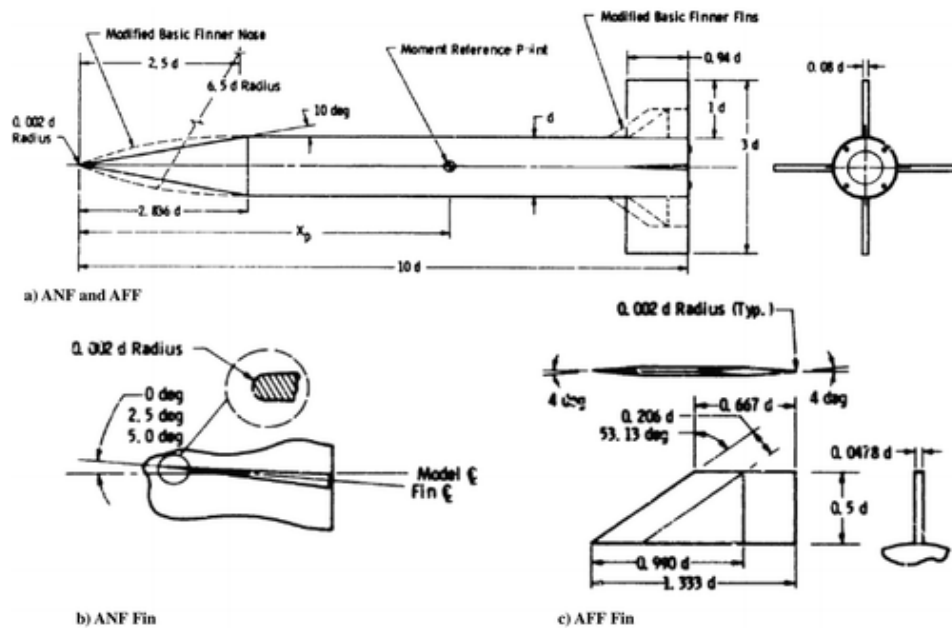


Figure 32. Bhagwandin Modified Finner (Bhagwandin, 2016)

3.2.4 Shantz Finner

The finner geometry provided by Shantz is shown in Figure 33 (Shantz and Groves, 1960), with some minor modifications made to improve legibility. This geometry is identical to the finner used by Dunn. Therefore, the same geometry as described in Chapter 3.2.2 was used for validation, with the only difference being the calibre. The calibre was specified to be 25.4 mm (1 in) by Shantz. The reference area for the aerodynamic coefficients was the largest cross-sectional area, 506.71 mm².

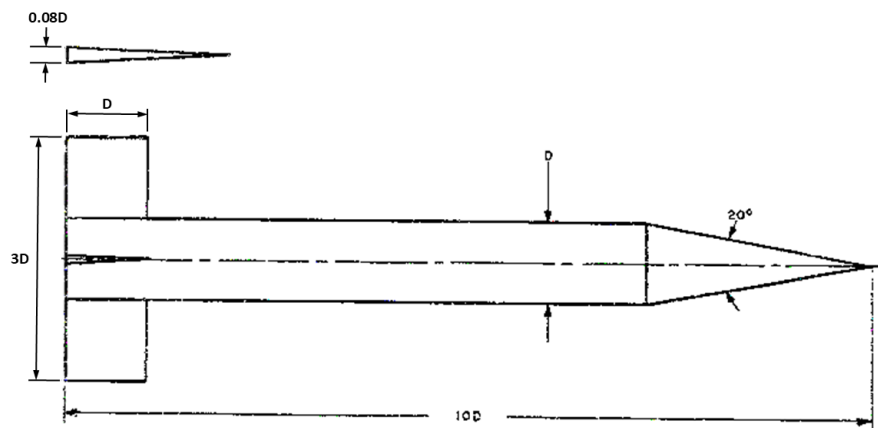


Figure 33. Shantz Finner (Shantz and Groves, 1960)

3.2.5 Hammargren Maxus

The final validation case used was a previously launched sounding rocket, the Maxus. The overall geometry is shown in Figure 34, and is an extract from a report by Elvemar (1990) at Saab.

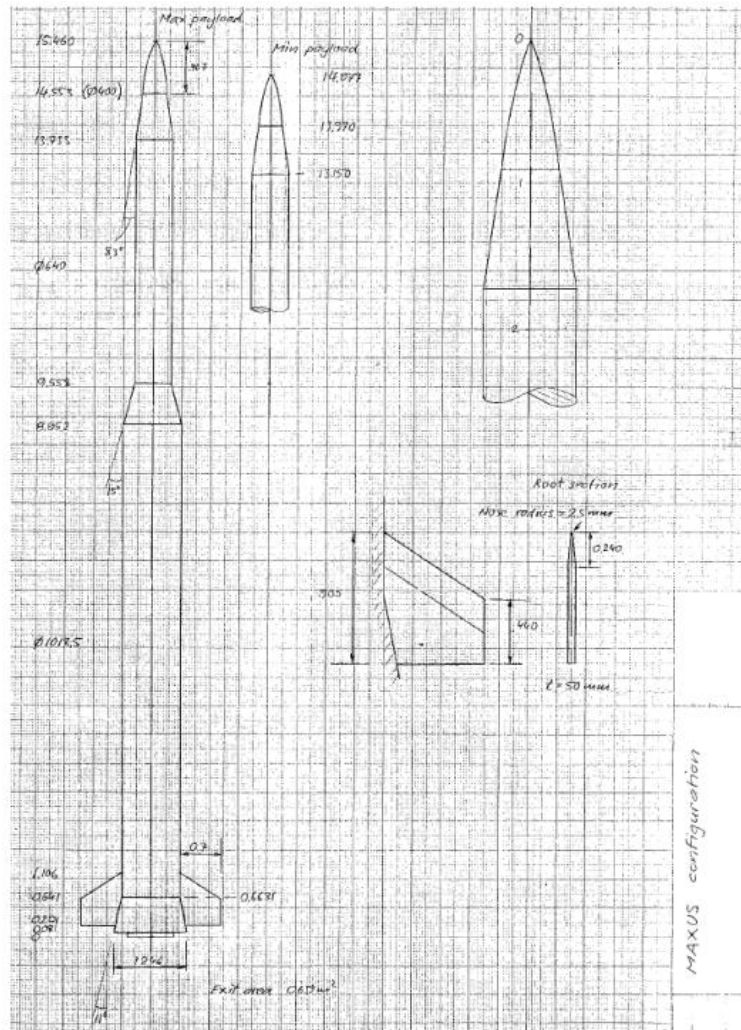


Figure 34. Elvemar Maxus Sketch (Elvemar, 1990)

The nose cone geometry as shown in Figure 35 is from a report by Andersson (1990) and is a secant ogive shape.

Dimensionen på OGIVEN är som följer :

STATION (mm) (TDR)	DIAMETER (mm)
0	0
100	67
200	128
300	183
400	234
500	277
600	314
907	400

Figure 35. Anderson Maxis Nose Geometry (Andersson, 1990)

A small radius of $0.001D$ was applied to the nose cone tip. The reference area for the aerodynamic coefficients was the largest cross-sectional area (the base area), 1.219 m^2 .

Due to reasons discussed in Chapter 2.5, a non-tangent, ogive nose cone cannot be created in RASAero or Missile Datcom. Therefore, for the Hammargren Maxus, a 0.73 power law nose cone geometry was used as it fits the Hammargren Maxus's ogive nose cone well, as shown in Figure 36.

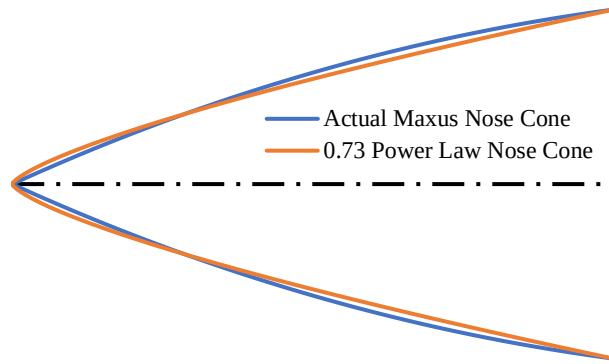


Figure 36. Maxus RASAero Nose Cone

3.3 Physics Modelling

This chapter summarises the fluid properties, turbulence models, and boundary conditions chosen for each validation case. An ideal gas (air) was used with the following properties:

Table 11. Properties of Air for the Validation Cases

Description	Variable	Value
Dynamic viscosity*	μ	$8.1332 \times 10^{-5} \text{ Pa.s}$
Temperature	T	293.15 K
Heat capacity at constant pressure	C_p	1003.62 J/kg.K
Molar Mass	M	28.97 kg/kmol

* The dynamic viscosity was calculated using Sutherland's Law. If a gas is pure and non-reacting, the viscosity is only dependent on temperature (Anderson, 2006). Since air is roughly 99% nitrogen and oxygen (Walker, 1977), which have similar properties, air can be considered a pure and non-reacting gas. Sutherland's law is an approximation that can be used to calculate the dynamic viscosity based on the temperature for pure and non-reacting gases. The equation for Sutherland's law is

$$\mu = \mu_{ref} \left(\frac{T}{T_{ref}} \right)^{\frac{3}{2}} \cdot \frac{T_{ref} + S_{\mu}}{T + S_{\mu}} \quad 3.1$$

where the Sutherland temperature, $S_{\mu} = 110.4 \text{ K}$ (for air),
the reference temperature, $T_{ref} = 273.15 \text{ K}$, and
the reference viscosity, $\mu_{ref} = 1.716 \times 10^{-5} \text{ Pa.s}$.

The reference values for air were obtained from White (1992). The viscosity values calculated using Sutherland's law had a 2% error for a temperature range of 170 K to 1900 K, according to White (1992).

Table 12 shows the types of eddy viscosity and transition zone models used for each validation case.

Table 12. Turbulence Models for the Validation Cases

Validation Case	Eddy Viscosity Model	Transition Zone Model
Dunn Spinner	$k - \omega$ SST	None
Dunn Finner	$k - \omega$ SST	$\gamma - Re_\theta$
Dunn Modified Finner	$k - \omega$ SST	$\gamma - Re_\theta$
Shantz Finner Mach 1.58	$k - \omega$ SST	$\gamma - Re_\theta$
Shantz Finner Mach 3.86	Spalart Allmaras	None

A bullet domain was chosen, as discussed in Chapter 2.4.3. A simplified cross-sectional view of the bullet domain is shown in Figure 37, with the labelled boundary conditions listed in Table 13.

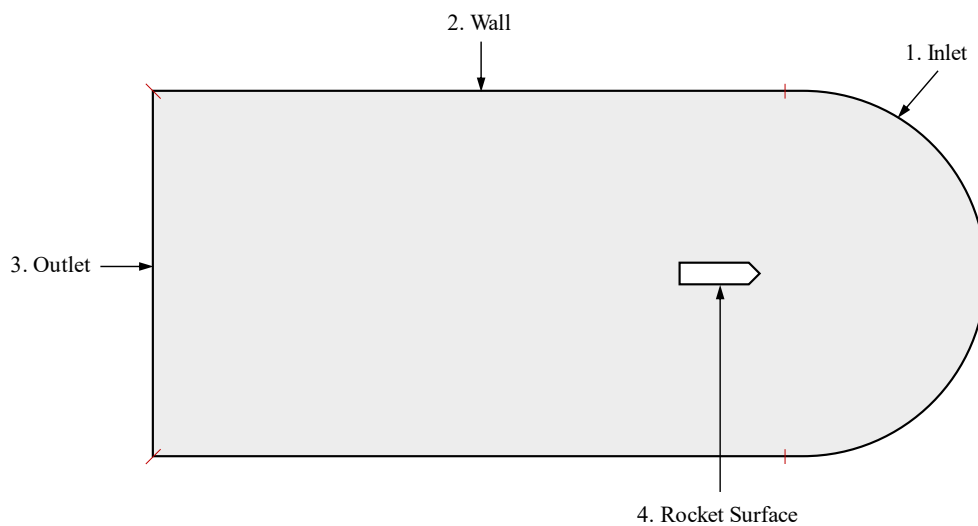


Figure 37. Boundary Conditions for Bullet Fluid Domain

Table 13. Boundary Conditions

Number	Description	Boundary Condition	Variable	Value
1	Inlet	Free Stream	Temperature	293.15 K
2	Wall		Mach Number	-
3	Outlet		Static Pressure (Gauge)	0 Pa
4	Rocket Surface (Broken up further for complex geometries)	Wall (No Slip)	-	-

3.4 Results

The results obtained for each validation case are summarised in Table 14, where the Mach numbers and angles of attack are inputs for the discrete CFD results. The results are then described in the following chapters, where the experimental data are understood to represent the benchmark against which all modelled and CFD results are compared. The RASAero and Missile Datcom input parameters for the validation cases are shown in Appendix B.

Table 14. Table of the Validation Results

Model Geometry	Author	Methods (from Author)	Mach Number/s	Angle/s of Attack	Output Parameter
Spinner	Dunn (1989)	NSWC code, Missile Datcom, wind tunnel	0.6, 0.8, 0.95, 1.05, 1.2, 1.5, 2, 2.5, 3, 3.5, 4	0°	Drag coefficient
Finner			0.5, 0.75, 0.9, 1, 1.1, 1.25, 1.5, 2, 2.5, 3	0°	Drag coefficient
Modified finner			1.3	0°, 10°, 20°, 30°, 40°, 50°	Normal force coefficient
Finner *	Shantz (1960)	Wind tunnel	1.58, 3.86	0°, 4°, 8°, 12°, 16°, 20°	Normal force coefficient
Maxus sounding rocket	Hammargren (2018)	Analytical methods, CFD	0.5, 0.8, 1, 1.2, 1.5, 2, 3, 4, 5	0°	Drag coefficient

* The Shantz finner case varied the roll angle between 0° and 45°.

3.4.1 Dunn Spinner (Drag Coefficient vs. Mach Number)

Figure 38 shows the drag coefficient vs. Mach Number for the Dunn spinner validation case. Dunn (1989) provided the experimental wind tunnel results and results from the Naval Surface Warfare Center (NSWC) code. The Star-CCM+, Ansys Fluent, Missile Datcom, and RASAero results were generated as part of this study.

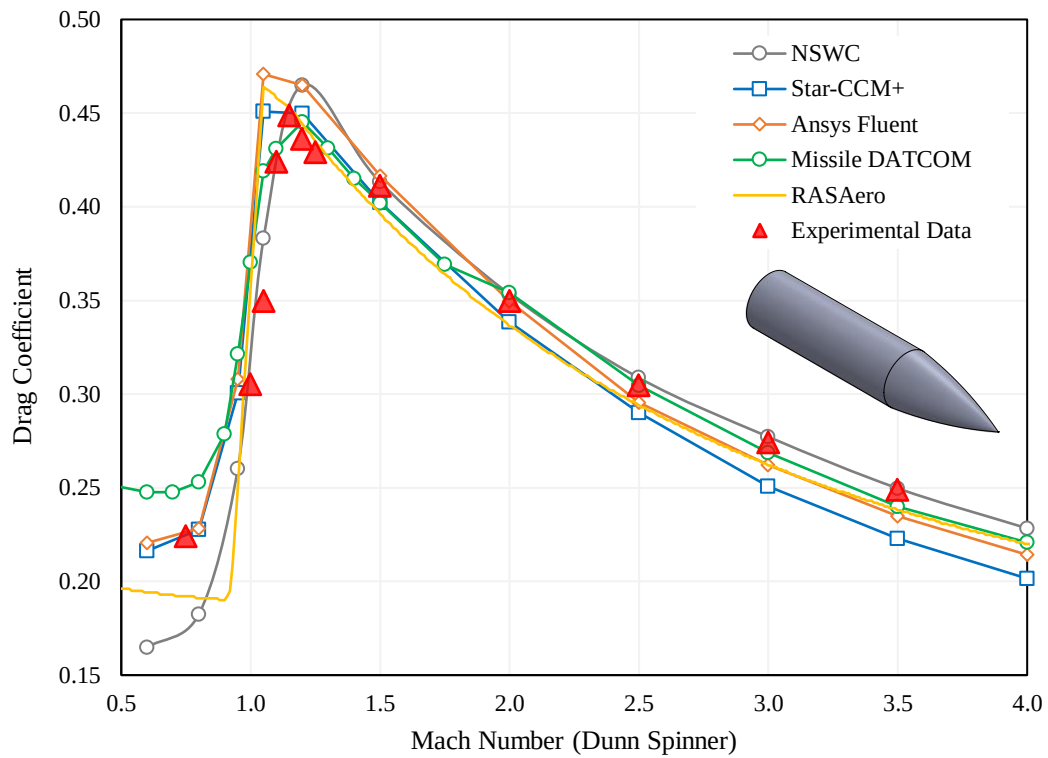


Figure 38. Graph of Drag Coefficient vs. Mach Number for the Dunn Spinner Validation

Table 15 shows the mesh comparison between Ansys Fluent and Star-CCM+ for the Dunn Spinner at Mach 0.6.

Table 15. Mesh Comparison Table for the Mach 0.6 Mesh

	Ansys Fluent	Star-CCM+
Number of Elements	3 837 556	3 876 367
Minimum Orthogonality	0.1646	-
Average Orthogonality	0.979	-
Average Cell Aspect Ratio	-	0.9385
Minimum Cell Aspect Ratio	-	0.01175
Maximum Cell Aspect Ratio	259.76	1
Average Surface Skewness	0.002277	24.43°
Minimum Surface Skewness	4.59×10^{-8}	0°
Maximum Surface Skewness	0.6991	85.18°
Average Volume Change	2.448	0.5982
Minimum Volume Change	0.010	0.005486
Maximum Volume Change	776.67	1
Average Cell Quality	-	0.7082
Face Validity	-	1
Chevron Quality Indicator	-	0

* Even though some of the mesh metrics have the same name between different applications, the metrics are defined differently depending on the software.

Images of the Star-CCM+ and Ansys Fluent spinner mesh at Mach 0.6 are shown in Figures 39 and 40.

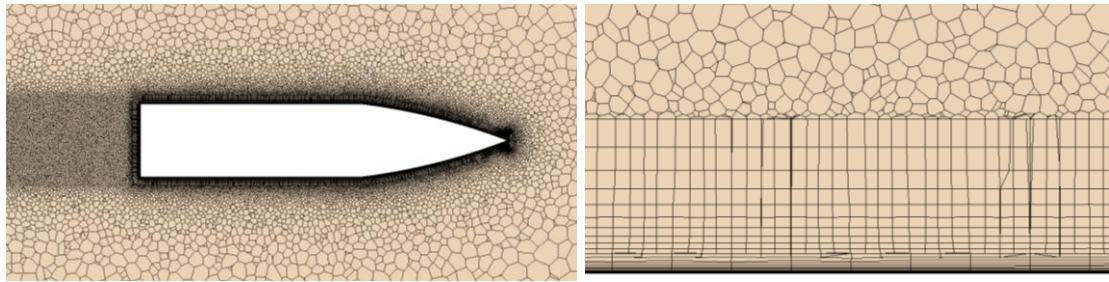


Figure 39. Star-CCM+ Spinner Mesh at Mach 0.6

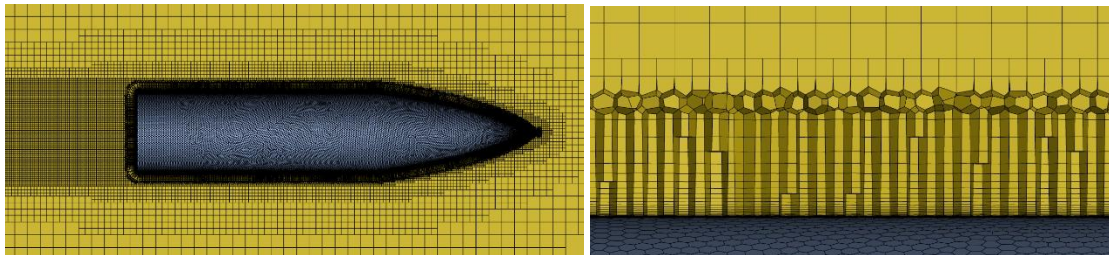


Figure 40. Ansys Fluent Spinner Mesh at Mach 0.6

Finally, a qualitative comparison was done between Star-CCM+ and Ansys Fluent. The Mach number and static pressure plots are shown at Mach 0.6 and Mach 4, in Figures 41 and 42.

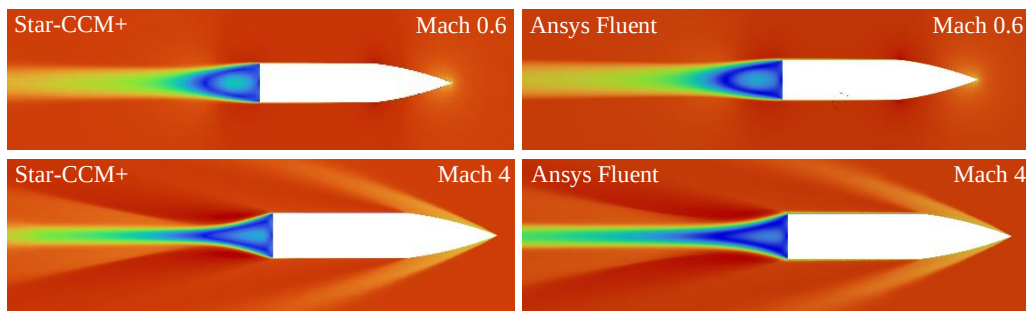


Figure 41. Mach Number Plots

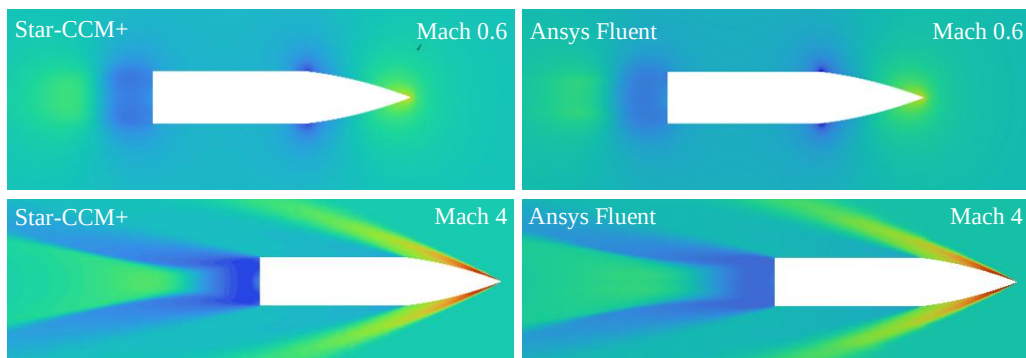


Figure 42. Static Pressure Plots

A domain size independence study was conducted by changing the size of the domain ahead, behind, and around the spinner by a constant factor, obtaining the results shown in Figure 43.

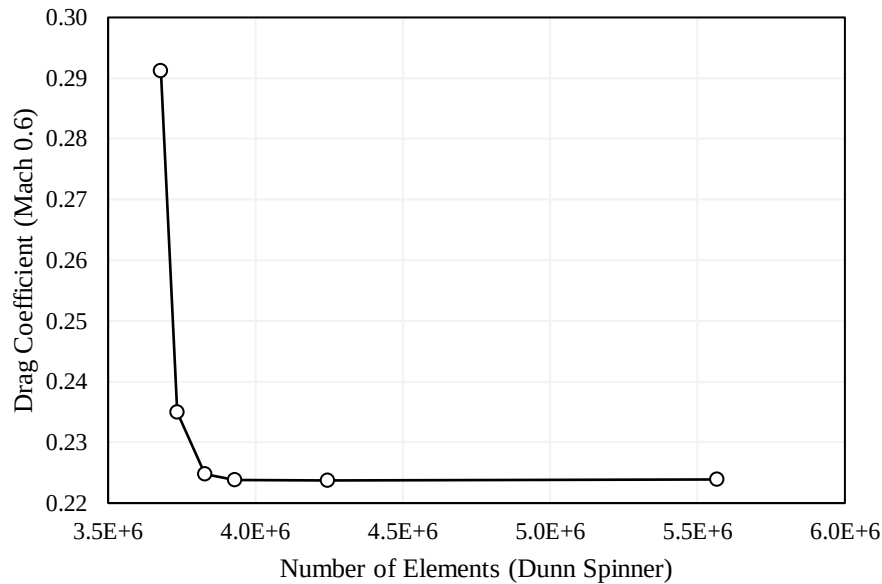


Figure 43. Graph of Dunn Spinner Domain Independence at Mach 0.6

A mesh independence study was done at Mach 0.6, as shown in Figure 44. The prism layer mesh properties were not changed during the independence study.

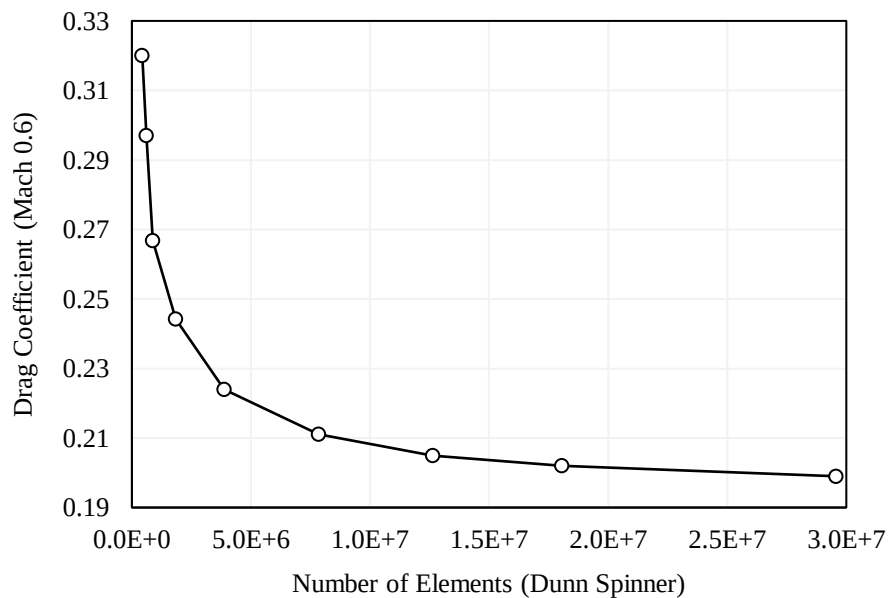


Figure 44. Graph of Dunn Spinner Mesh Independence at Mach 0.6

Firstly, the Star-CCM+ drag results are compared to the Ansys Fluent results. In the subsonic region, the Star-CCM+ and Ansys drag coefficients were nearly identical (2.3% difference at most). There was only one experimental data point in the subsonic region, which lay on the Ansys and Star-CCM+ curves. In the transonic region, Ansys overpredicted the peak drag coefficient by 4.9%, while Star-CCM+ overpredicted the peak drag coefficient by 0.5%. Accurately predicting the peak drag is crucial for rocket design, as peak drag directly relates to the maximum force and is crucial in designing the rocket structure. In the supersonic region,

Ansys and Star-CCM+ underpredicted the drag coefficient by 5.9% and 10.7% at most, respectively. When comparing the data from Star-CCM+ and Ansys with the NSWC code, the CFD data was more accurate in the subsonic and transonic regions. In contrast, the NSWC code was more accurate in the supersonic region. Overall, the difference between Star-CCM+ and Ansys was negligible. However, Star-CCM+ predicted the drag coefficient more accurately when compared to Ansys, which is important for determining the maximum aerodynamic forces on the rocket.

The two semi-empirical methods were close to the experimental data. RASAero underpredicted the drag in the subsonic region by 15.2%, overpredicted the drag in the transonic region by 3.4%, and underpredicted the drag in the supersonic region by 4.4%. Missile Datcom underpredicted the drag in the subsonic region by 10.5%, underpredicted the drag in the transonic region by 2.1%, and overpredicted the drag in the supersonic region by 3.6%. This confirmed the accuracy of RASAero and Missile Datcom, especially in the transonic and supersonic regions, and coupled with the small computational cost, semi-empirical methods are a useful tool for obtaining aerodynamic coefficients. At Mach 2, there is a kink in the Missile Datcom graph. This could be due to the software changing methods to solve in the supersonic region, not the transonic region.

It is not straightforward to compare two meshes from different software. This is because different software uses different methods to calculate mesh metrics. Even though the mesh metrics might have the same name, they are calculated using different formulae. Therefore, to compare the meshes, the recommended mesh metrics for the Ansys mesh and the Star-CCM+ mesh are analysed below (for the Mach 0.6 mesh). The recommended values provided are not ‘hard’ limits, and it can be justified if certain cells are outside recommended limits by small amounts.

Ansys mesh:

- The minimum orthogonal quality was 0.1646, which is considered ‘bad’ according to the Ansys guidelines. The average orthogonal quality was 0.98, which is well within the ‘excellent’ category according to Ansys guidelines (Ansys, 2023).
- The maximum aspect ratio was 259.76, with the high aspect ratio cells in the boundary layer. In the boundary layer, the cells are longer in the direction of fluid flow (small to no velocity change), and shorter in the direction perpendicular to fluid flow (large velocity change). Therefore, having large aspect ratio cells in the boundary layer should not cause errors.
- The maximum skewness was 0.6991, which is in the ‘fair’ category specified by Ansys. The average skewness was 0.002277, which is well into the ‘excellent’ category (Ansys, 2023).
- The maximum volume change was 776.67 and the average 2.448. These values are quite large. The large average value could be due to the hexcore section of the mesh. Due to the definition of hexcore, when a small hexahedral element transitions to a larger hexahedral element, the side length doubles, and the volume increases by a factor of 8.

Star-CCM+ mesh:

- The average and minimum aspect ratios were 0.9385 and 0.01175 respectively. No guidelines were provided by Star-CCM+ for the cell aspect ratio, except that values closer to 1 are preferred, which was achieved in this mesh.

- The maximum skewness angle was 85.18° , which is 0.18% greater than the Star-CCM+ recommended maximum skewness angle of 85° (Siemens, 2023a).
- The minimum volume change was 0.005486, which is 45% below the Star-CCM+ minimum recommended value of 0.01 (Siemens, 2023a).
- The minimum cell quality was 2×10^{-5} , which is double the Star-CCM+ minimum recommended value of 1×10^{-5} (Siemens, 2023a).
- The face validity was 1, which confirms that all the faces were valid.
- The chevron quality indicator was 0, which confirms that there were no chevron cells.

The meshes both satisfied some criteria and failed other criteria. Both meshes had well above acceptable cell quality on average; however, there were regions with high volume change. It can be concluded that both meshes were of acceptable quality and had no substantial quantitative differences in mesh metrics between each other. However, qualitatively, there were differences between the meshes, as seen in Figures 39 and 40. Ansys Fluent poly-hexcore meshing aims to connect a structured hexagonal mesh in the main fluid domain to the prism layers, using polyhedral elements. This resulted in some poor-quality cells in the region between the prism layers and the fluid domain. Star-CCM+ meshing created orthogonal prism layers and filled the rest of the fluid domain with polyhedral elements. Ansys Fluent does have the option to have a completely polyhedral domain. However, a poly-hex mesh was used to test a different meshing technique compared to the standard polyhedral mesh in Star-CCM+. Some of the key differences between Star-CCM+ and Ansys Fluent are shown in Table 16.

Table 16. Key Differences Between Star-CCM+ and Ansys Fluent

Star-CCM+	Ansys Fluent
Has built in macro functionality using JavaScript programming.	Can download a Pyfluent extension to create journal files using the Python programming language.
Has the ability to view updated scalar and vector plots while the solution is progressing.	Has proprietary meshing technology, called Mosaic meshing, which can generate polyhedral elements between the prism layers and the rest of the mesh.
One application for geometry importing, modifying, mesh generation, and CFD modelling. This improves workflow by reducing the time to switch between applications.	Easy integration with other Ansys applications. For example, a Finite Element Analysis (FEA) can be conducted on the same geometry block that is already being used for Fluent CFD.

A qualitative comparison was done between Star-CCM+ and Ansys Fluent, plotting the Mach number and static pressure at Mach 0.6 and Mach 4, as shown in Figures 41 to 42. The shock waves formed at the same angle and the wake had the same shape between the CFD codes. At Mach 4, there was a slightly lower static pressure (shown by the dark blue zone) for Star-CCM+, than compared to Ansys Fluent. In summary, the plots were practically identical between different software options.

A domain size independence study was then conducted at Mach 0.6, as shown in Figure 43. Initially, a large enough domain size was chosen based on research from the literature, as

discussed in Chapter 2.4.3. The domain was defined based on three dimensions: Distance in front of the spinner, distance behind the spinner, and the domain diameter. Each of these dimensions was changed by a constant factor and the drag coefficient was monitored. It was clear that there was a point where the domain size did not affect the results. This point was roughly when the domain was 20 model diameters in front of the spinner, 35 model diameters behind the model, and 25 model diameters in diameter.

A mesh independence study was conducted at Mach 0.6, as shown in Figure 44. When performing a mesh independence study, the prism layer specifications were not changed. This is because correctly sizing the y^+ and boundary layer thickness are independent of other mesh independence studies. It was observed that the drag coefficient only started converging at around 20 million elements, which was larger than initially predicted. There was an 11.2% difference between the fully converged value and the mesh at ~4 million elements. For all the validation cases, mesh sizes of roughly 4 to 10 million elements were used. Validation aimed to see the effects of changes made to the CFD model, by comparing CFD results to existing data. It was not an effective use of computational resources to do a complete mesh independence study for each validation case, when the main aim of the study was to create an accurate CFD model for STeVe and CLV.

3.4.2 Dunn Finner (Drag Coefficient vs. Mach Number)

Figure 45 shows the drag coefficient vs. Mach Number for the Dunn finner validation case. Dunn (1989) provided the experimental wind tunnel results and results from the NSW code. The Star-CCM+, Missile Datcom, and RASAero results were generated as part of this study.

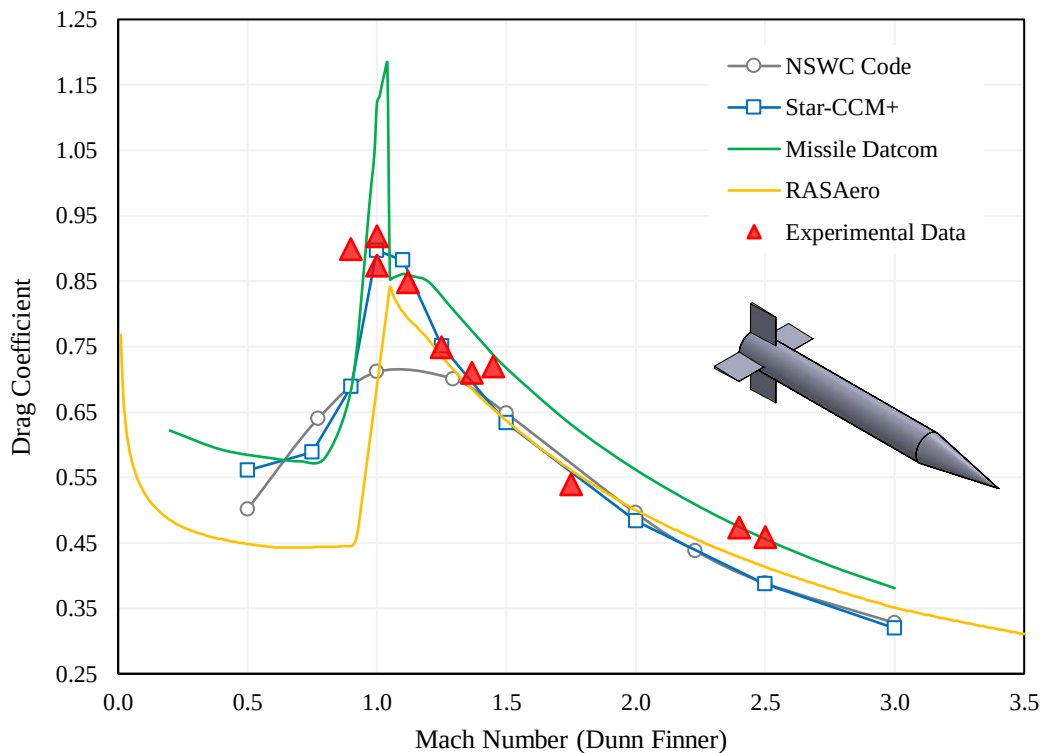


Figure 45. Drag Coefficient vs. Mach Number for the Dunn Finner

The drag coefficient results from the CFD simulation, shown in Figure 45, are discussed below. The peak drag coefficient predicted by Star-CCM+ was within the range of peak drag from the

experimental data. At Mach 1.25, Star-CCM+ overpredicted the drag coefficient by 0.34%. At Mach 2.5, Star-CCM+ underpredicted the drag coefficient by 15.63%. There was no experimental data for the subsonic region; however, in the CFD results were within 12% of the NSW code, in this region. In the supersonic region, the CFD results were within 2.19% of the NSW code.

RASAero underpredicted the drag coefficient in the subsonic region by 20% at most, compared to CFD results. In the transonic and supersonic region, the RASAero data was accurate, compared to the experimental data. RASAero underpredicted the peak drag coefficient by 6%, and underpredicted the drag coefficient by 6% at Mach 2.5.

In the subsonic region, Missile Datcom underpredicted the drag by 3.4% at most, when compared to the CFD data. The data from Missile Datcom overpredicted the peak drag by 28.9%, which can be seen as a substantial spike in the drag plot. At Mach 2.5, the experimental data lay on the Missile Datcom curve.

3.4.3 Dunn Modified Finner (Normal Force Coefficient vs. Angle of Attack)

Figure 46 shows the normal force coefficient vs. angle of attack for the Dunn modified finner validation case. Dunn (1989) provided the experimental wind tunnel results, results from the NSW code, and Missile Datcom. The Star-CCM+, Missile Datcom, and RASAero results were generated as part of this study.

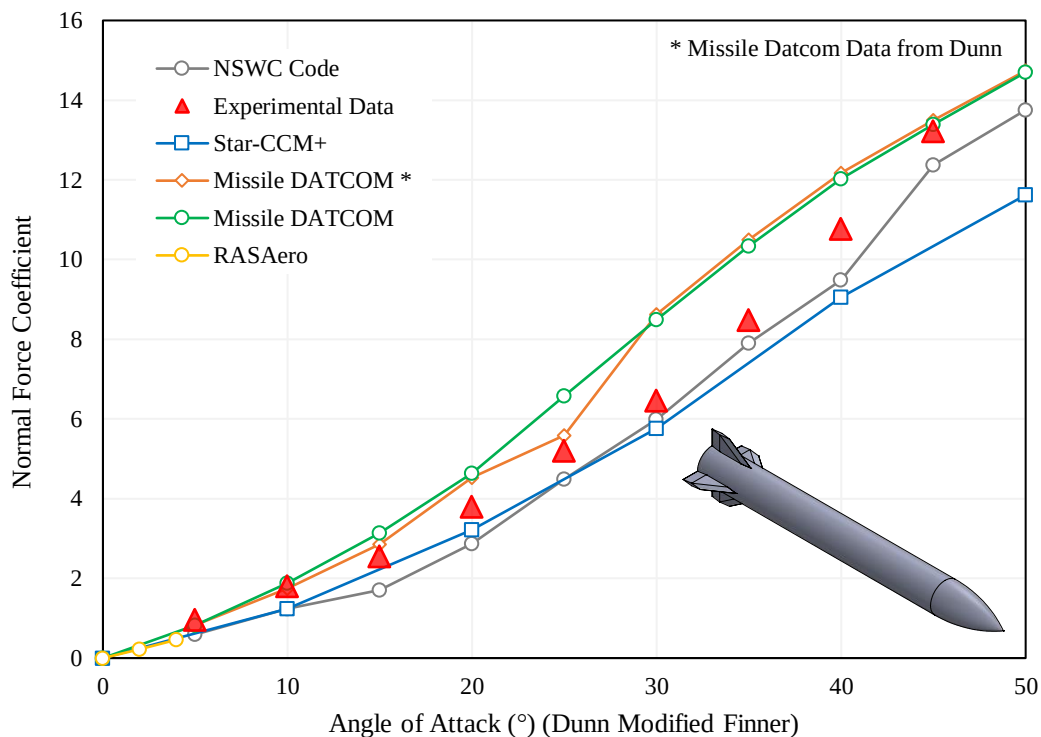


Figure 46. Normal Force Coefficient vs. Angle of Attack for the Dunn Modified Finner

The normal force coefficient was plotted against angle of attack, as shown in Figure 46. In terms of the experimental data, Star-CCM+ underpredicted the normal force coefficient by 31%, 15%, 11%, and 16% for 10°, 20°, 30°, and 40° angles of attack respectively. In terms of the NSW code, Star-CCM+ varied at most by 15.5% for angles of attack up to 50°.

In terms of the experimental data, Missile Datcom overpredicted the normal force coefficient by 4%, 22%, 31%, and 12% for 10°, 20°, 30°, and 40° angles of attack respectively. Due to the limitations discussed previously regarding RASAero, the normal force coefficient could only be plotted up to an angle of attack of 4°. The Star-CCM+ and RASAero data agreed well up to this point.

It can be observed that Star-CCM+ predicted the normal force coefficient well, considering the high angles of attack that were simulated. At high angles of attack, there is extreme flow separation, which should be difficult for the CFD software to capture accurately.

3.4.4 Shantz Finner (Normal Force Coefficient vs. Angle of Attack)

The normal force coefficient was plotted against the angle of attack at Mach 1.58 and Mach 3.86, for 0° and 45° roll angles for the Shantz finner validation case. Shantz (1960) provided the experimental data; while the Star-CCM+, Missile Datcom, and RASAero data were generated as part of this study. The results are shown in Figures 47 to 49 and discussed below.

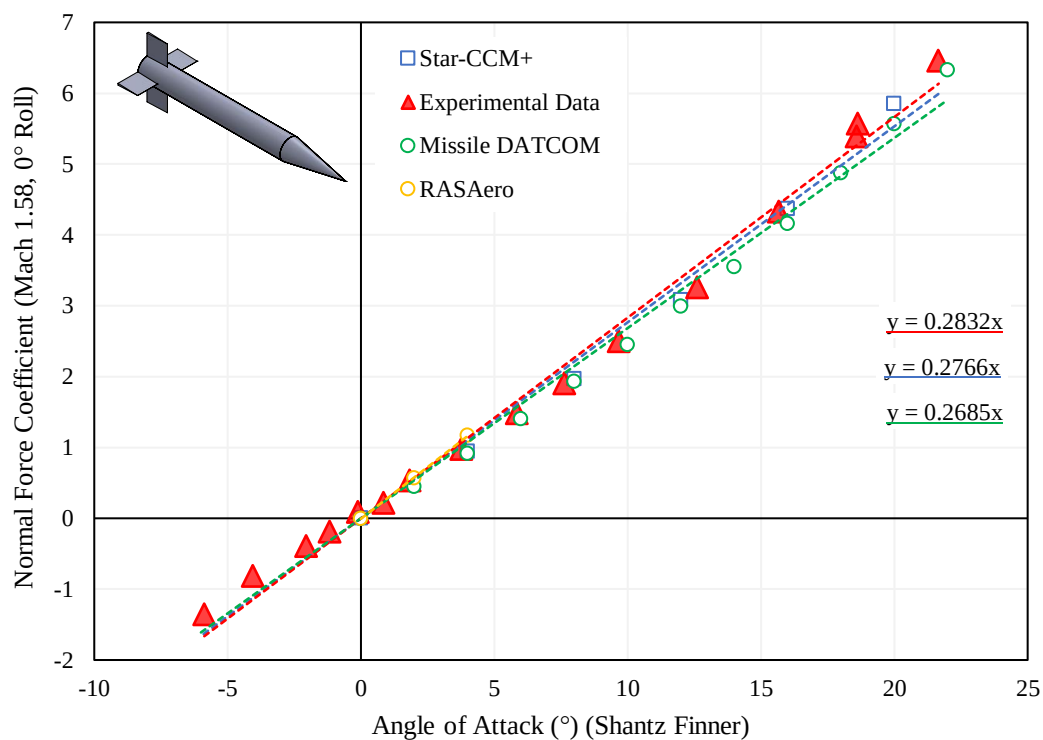


Figure 47. Graph of Normal Force Coefficient vs. Angle of Attack at Mach 1.58 and 0° Roll

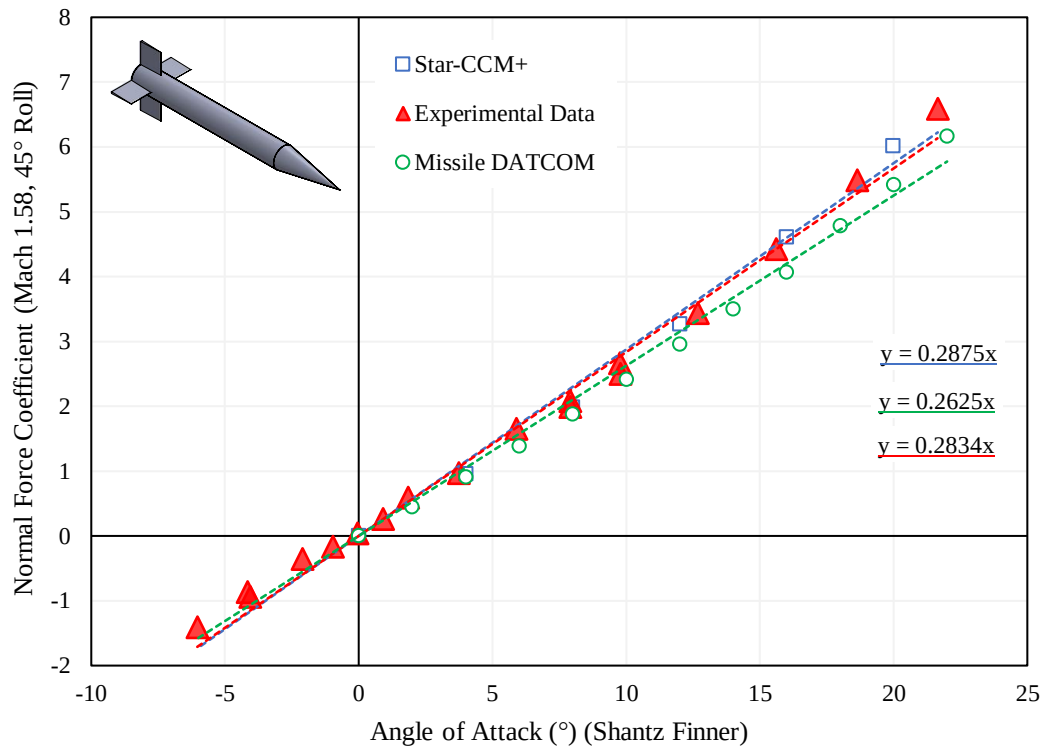


Figure 48. Graph of Normal Force Coefficient vs. Angle of Attack at Mach 1.58 and 45° Roll

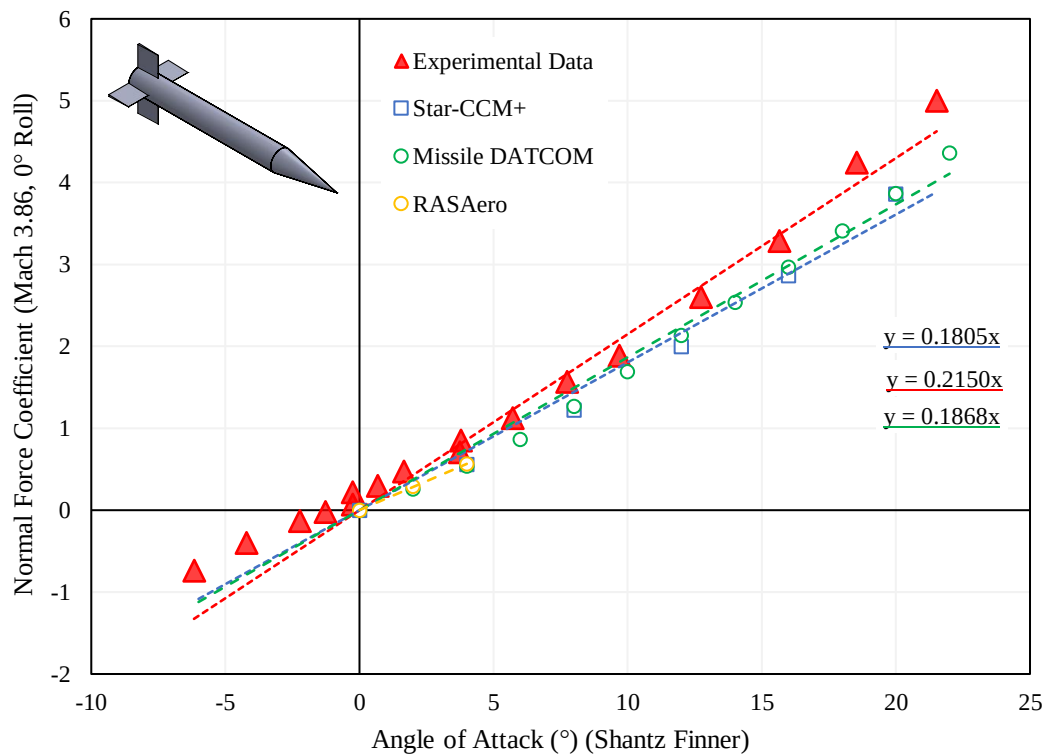


Figure 49. Graph of Normal Force Coefficient vs. Angle of Attack at Mach 3.86 and 0° Roll

Due to the low angles of attacks analysed, a linear trendline can be plotted. The gradients of the trendlines are used to obtain an average output for each data source, as shown in Table 17.

Table 17. Shantz Finner Validation Data

Mach	Roll	Data Source	Gradient	% Difference
1.58	0°	Experimental Data	0.2832	-
		CFD	0.2766	-2.33%
		Missile Datcom	0.2685	-5.19%
1.58	45°	Experimental Data	0.2834	-
		CFD	0.2875	+1.45%
		Missile Datcom	0.2625	-7.37%
3.86	0°	Experimental Data	0.2150	-
		CFD	0.1805	-16.05%
		Missile Datcom	0.1868	-13.12%

At the lower Mach number (1.58), the CFD results are extremely close to the experimental results, with the Missile Datcom results less accurate. At the higher Mach number (3.86), the CFD results become more inaccurate. At Mach 3.86, the Spalart Allmaras turbulence model was used as there were convergence problems with the $k - \omega$ SST model at this high Mach number. This could be why the values at Mach 3.86 were more inaccurate. The finner at 45° roll and Mach 3.86 was not analysed as the solution did not converge fully. The RASAero data were only available for angles of attack up to 4°, where it was accurate.

3.4.5 Hammargren Maxus (Drag Coefficient vs. Mach Number)

Figure 50 shows the drag coefficient vs. Mach number for the Hammargren Maxus validation case. Hammargren provided the pre-existing data, and CFD simulation data. The Star-CCM+ and RASAero results were generated as part of this study.

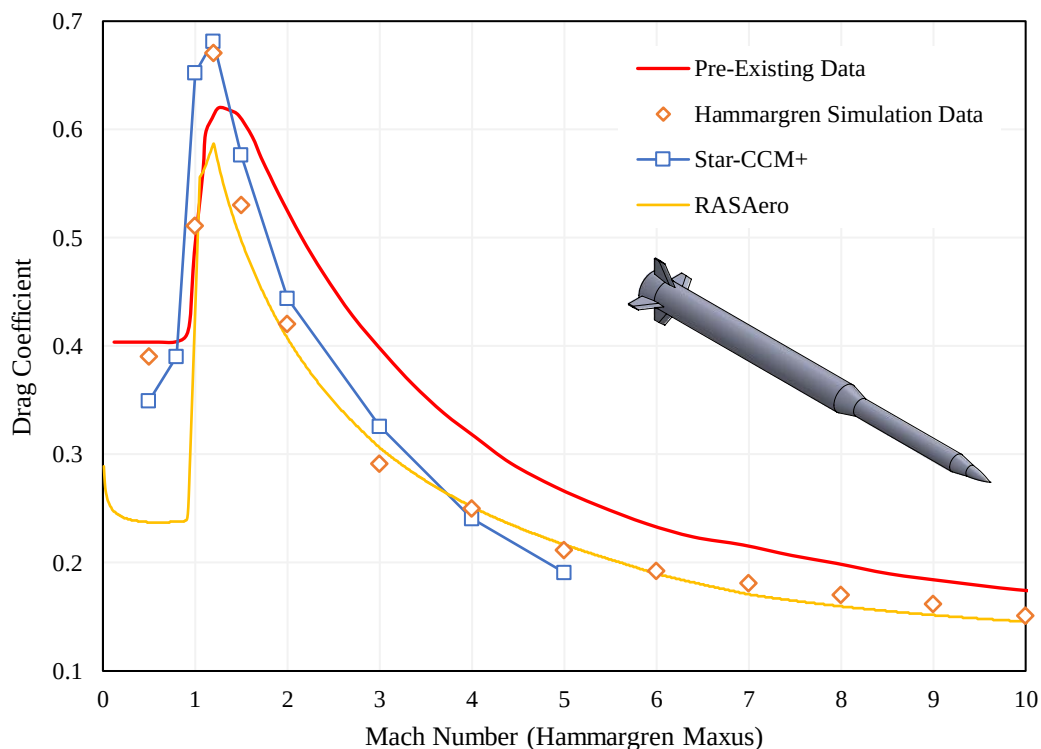


Figure 50. Graph of Drag Coefficient vs. Mach Number

The drag coefficient was plotted against Mach numbers up to Mach 10, as shown in Figure 50. For the Star-CCM+ model, the drag coefficient was only plotted up to Mach 5, as above Mach 5, the flow becomes hypersonic. This means a purely aerodynamic analysis cannot be performed, but rather an aerothermodynamic analysis must be performed in which the air's chemical composition and effects such as dissociation and ionisation are taken into account. This was not in the scope of this research, so the CFD results were not obtained above Mach 5. The reference area used in calculating the drag coefficient was the larger 'skirt' diameter at the base of the rocket.

In terms of the pre-existing data, Star-CCM+ overpredicted the peak drag coefficient by 9.9%. In the supersonic region, Star-CCM+ underpredicted the drag coefficient by at most 25.7%. In the subsonic region, Star-CCM+ underpredicted the drag coefficient by at most 13.4%. When compared to Hammargren's CFD data, the Star-CCM+ data were very similar. The peak drag coefficient differed by 1.6% between the two.

The RASAero data showed an underprediction in the peak drag by 5.3%. In the subsonic region, RASAero underpredicted the drag by 41.3%. This underprediction in the subsonic region is consistent with previous validation results. In the supersonic region, RASAero underpredicted the drag by 18.5%. The Missile Datcom data were so inaccurate (pronounced overprediction) it was left out of the analysis. In Missile Datcom, the Maxus geometry had to be created using a custom set of points, which was not done previously and could have caused the inaccurate results.

3.5 Conclusion

In summary, validation began by researching different rocket geometries that had wind tunnel data available. Five different validation cases were used, of varying geometry complexities: the spinner, finner, modified finner, and Maxus rocket.

The Dunn spinner case was used to compare the differences between Star-CCM+ and Ansys Fluent. It was decided to use Star-CCM+ for the rest of the study due to the better workflow with JavaScript macros and slightly more accurate peak drag prediction.

A mesh independence and domain independence study were conducted at Mach 0.6 for the Dunn Spinner validation. The number of elements it took to achieve convergence was larger than expected. It was confirmed that the domain size chosen was already in the converged region.

For low angles of attack, the drag and normal force coefficients were predicted accurately for all validation cases. The only substantial inaccuracies occurred when simulating at large angles of attack ($>15^\circ$) at high Mach numbers (>3.86). However, these large angles of attack and Mach number combinations are unrealistic for real rocket flight.

Missile Datcom and RASAero were close to experimental data in terms of predicting aerodynamic coefficients. Both methods were more accurate in the transonic and supersonic regions, than the subsonic region. RASAero consistently underpredicted drag in the subsonic region. A noticeable problem with Missile Datcom is that there are often significant unrealistic jumps in values between adjacent Mach numbers, as observed in the Dunn Spinner and Dunn Finner validation cases. This is likely due to the code changing correlations from the transonic to supersonic region. In the Missile Datcom user's manual (Air Force Research Laboratory,

1998) it is stated that the body alone aerodynamic methodology switches from subsonic/transonic correlations to supersonic correlations at Mach 1.2. The fin alone aerodynamic methodology switches from transonic correlations to supersonic correlations at Mach 1.4. It therefore makes sense, that in some instances there might be substantial changes around this transition point between correlations. Missile Datcom also overpredicted the drag for the Maxus rocket by a substantial amount.

In summary, the CFD model created in Star-CCM+ was close to experimental data for existing rocket geometries. This was true for varying geometry complexities, Mach numbers, and angles of attack, when measuring the drag and normal force coefficients. Both semi-empirical methods (RASAero and Missile Datcom) were accurate, besides for occasional flaws in the Missile Datcom results. The same methods used to create the validation simulations were used for the STeVe and CLV simulations, to maintain accuracy.

4. Computational Aerodynamic Analysis of the Launch Vehicles

The following chapter describes the implementation of computational analyses for the determination of the aerodynamic performance of the STeVe and CLV launch vehicles. This includes the creation of the vehicle geometries and meshes, the selection of the appropriate physics models, and the generation of a set of data in which ASRI engineers can have confidence given the validation of the process in Chapter 3. The results of this chapter will serve as crucial inputs to the design cycle of both vehicles as their development progresses.

4.1 Geometry

STeVe was still in the design phase at the start of this thesis; therefore, in June 2024, the latest geometry was chosen to be used throughout the thesis. The detailed geometry is shown in Appendix A, with a rendered image shown below in Figure 51.



Figure 51. STeVe Geometry, as of June 2024

STeVe has four hexagonal cross-section fins (canted at 0.7°), a boat tail, and a Von Kármán nose cone. The geometry used for the CFD had to be simplified, resulting in a constant diameter cylinder for the body tube section. The boat tail was also adjusted to start at the same point as the fins. The nose cone tip was given a 0.4 mm radius round, and the leading and trailing edges of the fins were both given 0.75 mm radius rounds to represent the as-machined components and to assist with meshing at sharp edges.

CLV was still in the design phase at the start of this study; therefore, in June 2024, the latest geometry was chosen to be used throughout the study. The detailed geometry is shown in Appendix A, with a rendered image shown below in Figure 52.



Figure 52. Commercial Launch Vehicle

CLV is a two-stage rocket, in which only the first phase configuration (two stages attached) was analysed in this study. To simplify the CFD, the launch lugs were removed from the rocket body, and the rocket was smooth with a constant diameter.

4.2 Fluid Domain

Based on the research conducted in the literature review, Chapter 2.4.3, a bullet shaped mesh domain was used for both launch vehicle simulations. The domain dimensions for STeVe and CLV are shown in Figures 53 and 54, as well as in Appendix A.

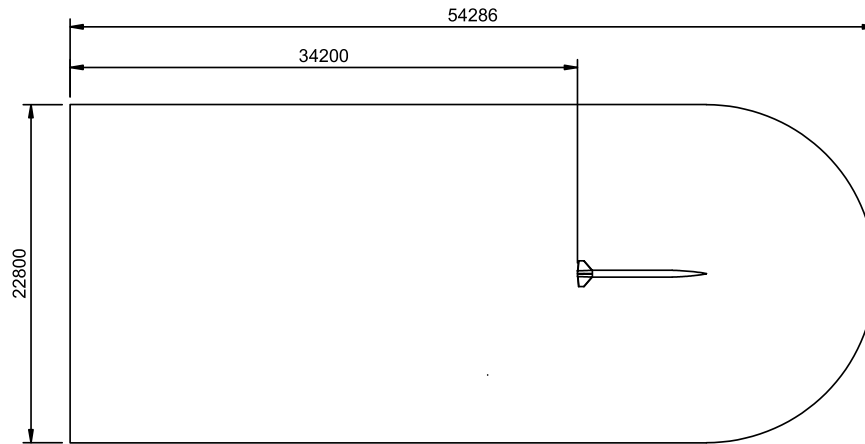


Figure 53. STeVe Fluid Domain Dimensions

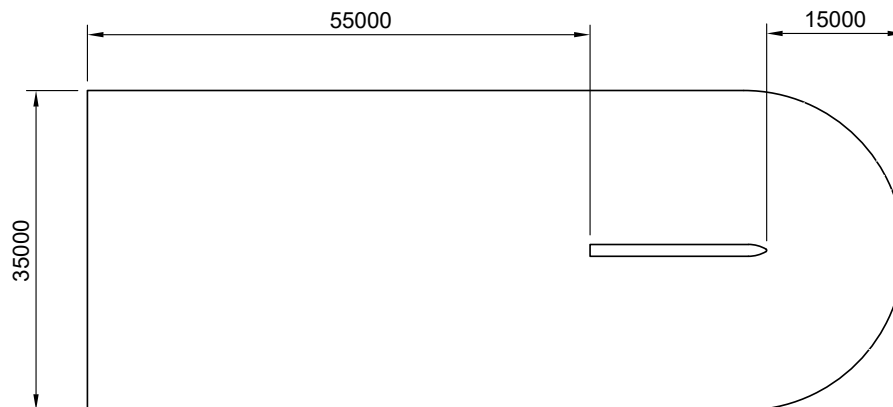


Figure 54. CLV Fluid Domain Dimensions

4.3 Meshing

When dealing with high-speed external flow over rockets, meshing can be broken down into three areas of importance: The boundary layer, the wake, and the free stream, as shown in Figure 55. Each of these areas contributes differently to solution accuracy and computational time; therefore, each area is meshed differently to optimise performance.

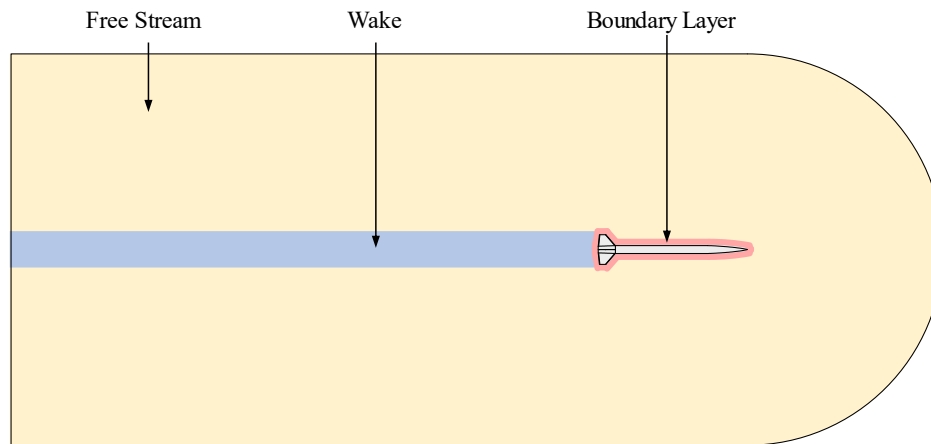


Figure 55. Meshing Areas

Boundary layer: The meshing technique used in the boundary layer is to create small, structured cells (prism layers or inflation layers) near the rocket surface to adequately capture the boundary layer formation. The process of correctly sizing the boundary layer is described in Chapter 2.1.4. A y^+ of 1 was targeted to adequately resolve the boundary layer without the use of non-linear wall functions. ‘Sizing the boundary layer’ involved determining the first cell height, the boundary layer thickness, the total prism layer thickness, and the growth rate. In the boundary layer, there are some areas where the prism layers need to be small, and some areas where prism layers do not need to be small, or might not be required at all. In the direction of the flow, the cells can be large; and perpendicular to the direction of flow, the cells must be small. This is because changes in flow properties are greater perpendicular to the direction of flow in the boundary layer. On the fuselage, small prism layers are crucial. However, in the areas of stagnation and recirculation (nose cone tip, fin leading and trailing edges, rocket base) there is not the same boundary layer profile that is seen on the fuselage body. Therefore, structured prism layers are not as important in the areas of stagnation and recirculation. In these areas, structured prism layers often create poor quality cells due to the sharp rocket geometry. By removing the prism layers in those areas, the main cause of many meshing and convergence problems are also removed, and small polyhedral cells can be used instead.

Wake: The wake resolution has a significant impact on the upstream flow. To capture the wake effectively, the wake refinement was split into three regions: The plume, wake 1, and wake 2, as shown in Figure 56. The plume aimed to just capture the plume and recirculation at the base of the rocket, and was the finest region. Wake 1 captured the wake from the plume to roughly halfway to the domain outlet, and was coarser. Wake 2 captured the wake to the domain outlet, and was even coarser.

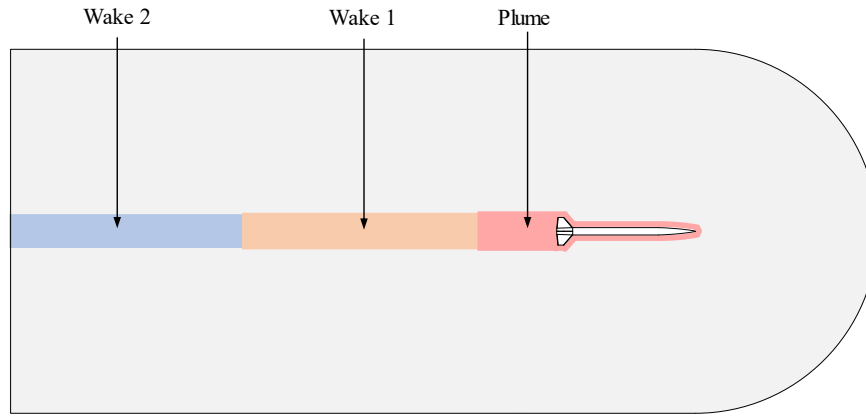


Figure 56. Wake Meshing Areas

Free stream: The free stream area is the coarsest as there is usually no change in flow properties between adjacent cells, so small cells are not required. This is not the case when there is shock wave formation, as the shock waves propagate into the free stream and there are large changes in flow properties across the shock waves. Shock wave refinement was dealt with in the following manner. As part of the general mesh refinement, the free stream was refined, and therefore the shock waves were refined. At supersonic speeds, convergence was observed, as shown in the Chapter 5.1. This showed that a standard mesh refinement can be used to account for shock waves in the free stream.

4.4 Fluid Properties and Flow Modelling

Air was modelled as an ideal gas for the launch vehicle simulations, with the properties summarised in Table 18.

Table 18. Air Properties for STeVe and CLV CFD

Description	Variable	Value
Dynamic viscosity *	μ	8.1332×10^{-5} Pa.s
Temperature	T	293.15 K
Heat capacity at constant pressure	C_p	1003.62 J/kg.K
Molar mass	M	28.97 kg/kmol

* Sutherland's Law was used to calculate the viscosity, as described in Chapter 3.3

The flow was modelled as an ideal gas, in 3D, using a steady solver. A coupled flow solver was chosen. The $k - \omega$ SST, RANS, turbulent flow solver was chosen with all y^+ wall treatment. $k - \omega$ SST was used as the turbulence model, due to positive research for high-speed external flow. $k - \omega$ SST also provided accurate results when compared to experimental data, as highlighted in the validation cases, shown in Chapter 3. A transition zone model was not selected, as the air flow quickly reaches very high Reynolds numbers, so modelling the transition zone becomes less important for accuracy (as discussed in detail in Chapter 2.1.8).

4.5 Boundary Conditions

The same boundary conditions were used as the boundary conditions in the validation cases. The boundary conditions are shown and described in Figure 57 and Table 19.

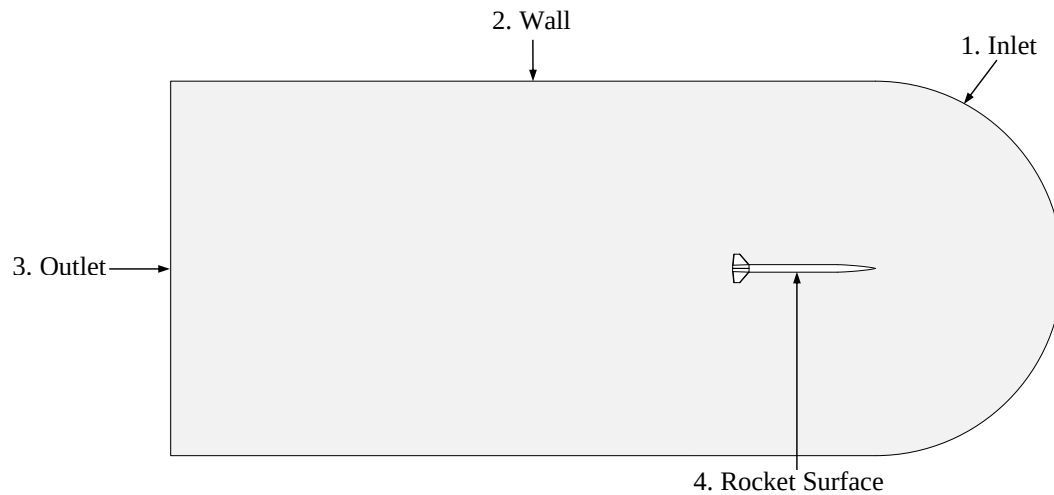


Figure 57. Boundary Conditions for Bullet Fluid Domain

Table 19. Boundary Conditions for STeVe and CLV

Number	Description	Boundary Condition	Variable	Value
1	Inlet	Free Stream	Temperature	293.15 K
2	Wall		Mach Number	x
3	Outlet		Static Pressure (Gauge)	0 Pa
4	Rocket Surface	Wall (No Slip)	-	-

Another method that was tested, was using a velocity inlet boundary at the inlet, free stream at the domain walls, and a pressure outlet at the outlet. This worked for some Mach numbers, however in the transonic region, it was difficult, and in some cases impossible, to achieve convergence. When a slip wall boundary condition was used for the outer walls, a noticeable shock reflection was observed. Therefore, free stream boundary conditions were used.

4.6 Semi-Empirical Methods

RASAero and Missile Datcom were used to generate comparative data points and as a means to assess confidence in the CFD results. In RASAero, the ‘All Turbulent Flow’ and the ‘Rogers Modified Barrowman’ options were selected, for reasons discussed in Chapter 2.5. A smooth surface finish (zero surface roughness) was used. It is not defined in the RASAero documentation whether the nose cone tip is measured from the tip before or after a round is applied. The nose cone length without the rounded tip was used as the length input into RASAero, as the results more closely resembled the CFD results and the RASAero graphics output matched the nose cone curve. The RASAero input data and geometry for STeVe and CLV are shown in Appendix B.

Missile Datcom was used as another data point to increase confidence in the CFD results. 20 discrete Mach numbers were used to characterise the flow for each angle of attack. A smooth surface finish (zero surface roughness) was used. Unlike RASAero, Missile Datcom has the ability to account for fin cant, which was done for STeVe. The Missile Datcom '*.mis*' input files and geometry for STeVe and CLV are shown in Appendix B.

5. Aerodynamic Characterisation Results

The aerodynamic characterisation results for both STeVe and CLV are shown in this chapter. This includes a section on mesh independence, static aerodynamic coefficients, and the flight trajectory.

5.1 Mesh Independence

A mesh independence study was conducted by varying the base cell size (keeping the prism layers unchanged) and obtaining a graph of drag coefficient vs. number of elements. A mesh independence study was conducted for CLV at Mach 0.6, 1.25, and 3, as shown in Figures 58 to 60.

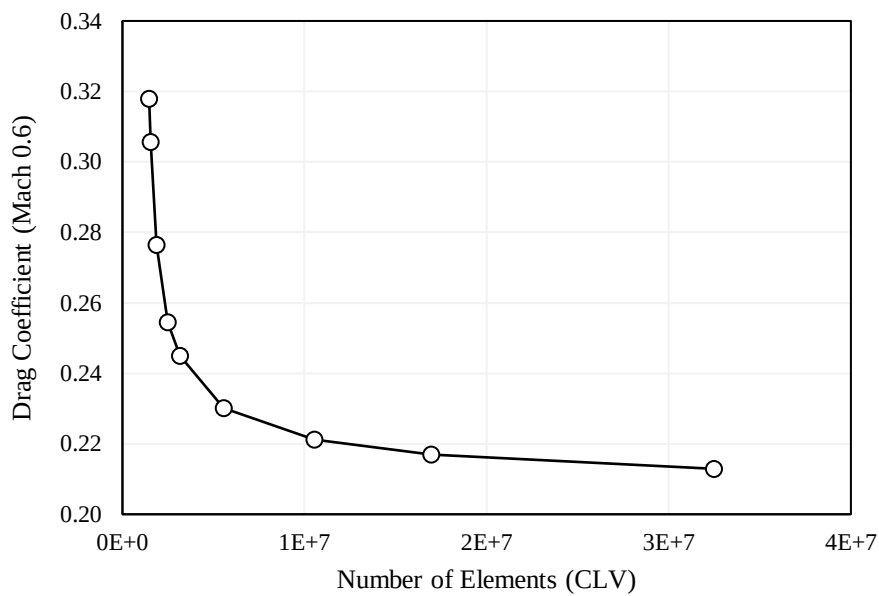


Figure 58. CLV Mesh Independence at Mach 0.6

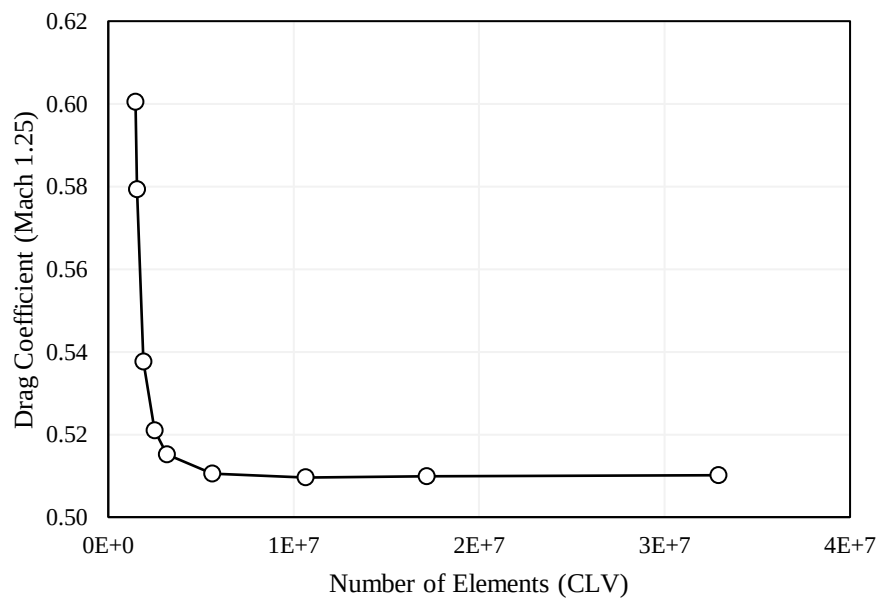


Figure 59. CLV Mesh Independence at Mach 1.25

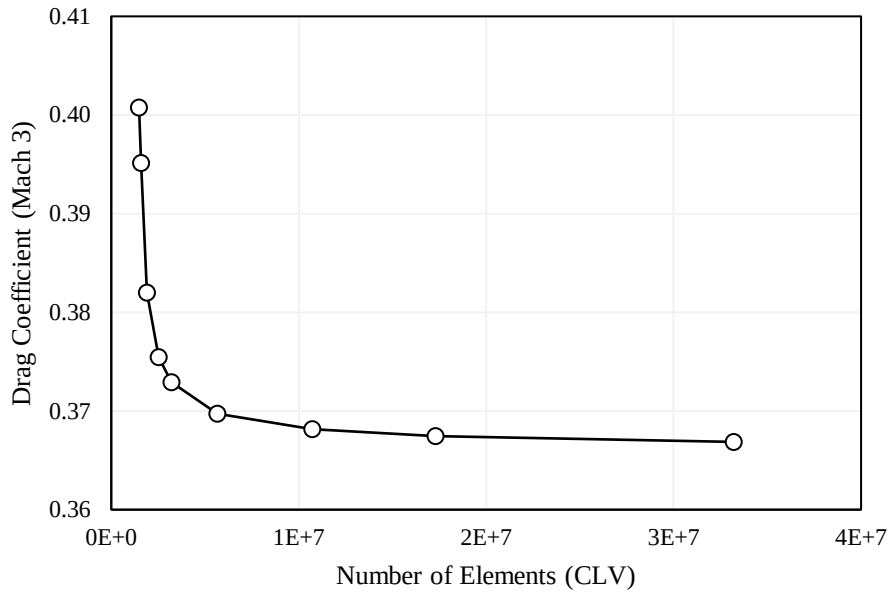


Figure 60. CLV Mesh Independence at Mach 3

To compare mesh independence between different Mach numbers, the percentage difference from the converged value was plotted, and shown in Figure 61.

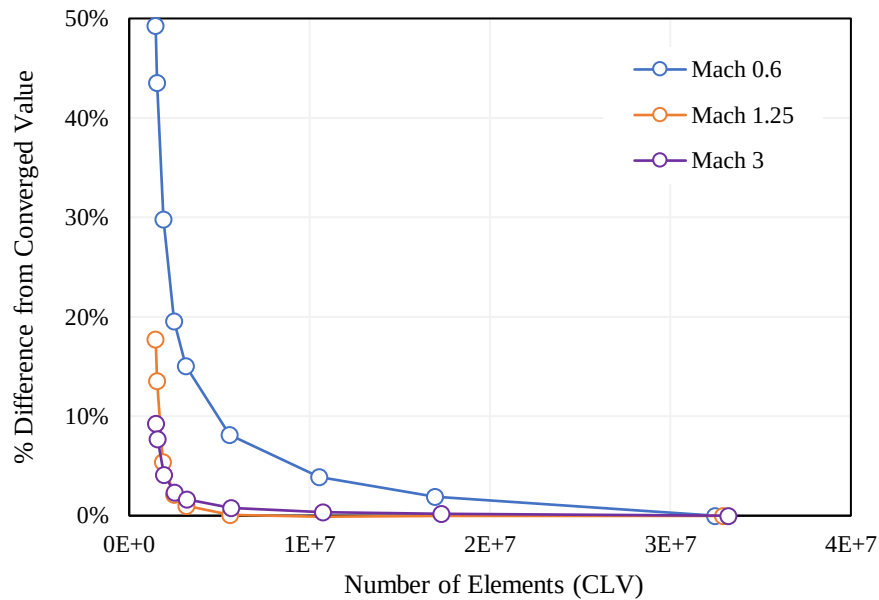


Figure 61. CLV Mesh Independence Summary

As shown above, the drag readings showed asymptotic convergence at all three Mach numbers for CLV. An interesting observation is that convergence occurred at a lower number of elements for the higher Mach numbers (1.25 and 3), while convergence was more difficult to achieve at Mach 0.6. This observation is shown in Figure 61. Table 20 shows the number of elements required to achieve convergence for the CLV simulation, further highlighting this observation.

Table 20. Elements Required to Achieve Convergence for CLV

Mach Number	Number of Elements to fall within 10% of Converged Value	Number of Elements to fall within 5% of Converged Value
0.6	4.6×10^6	8.7×10^6
1.25	1.7×10^6	1.9×10^6
3	1.4×10^6	1.8×10^6

A mesh independence study was conducted for STeVe at Mach 0.6 and Mach 1.25, as shown in Figures 62 and 63.

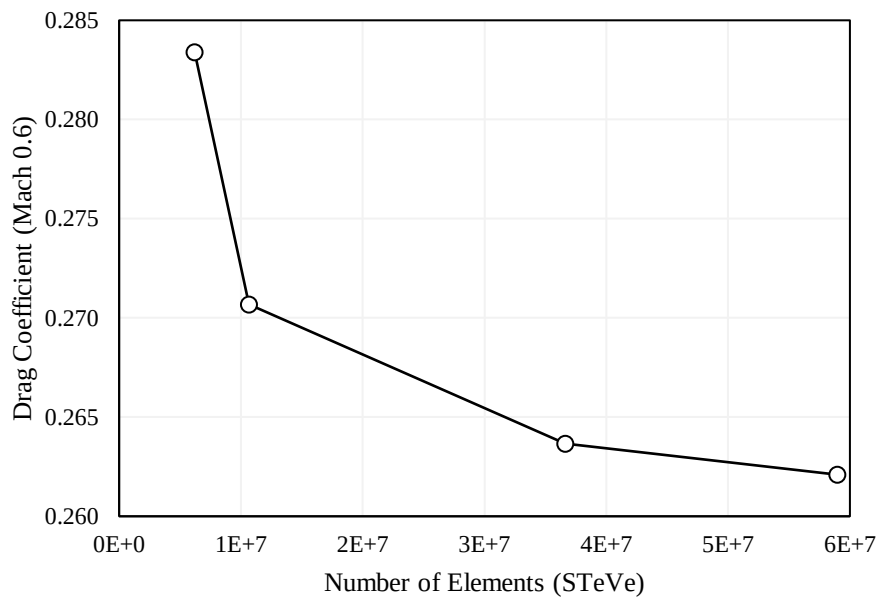


Figure 62. STeVe Mesh Independence at Mach 0.6

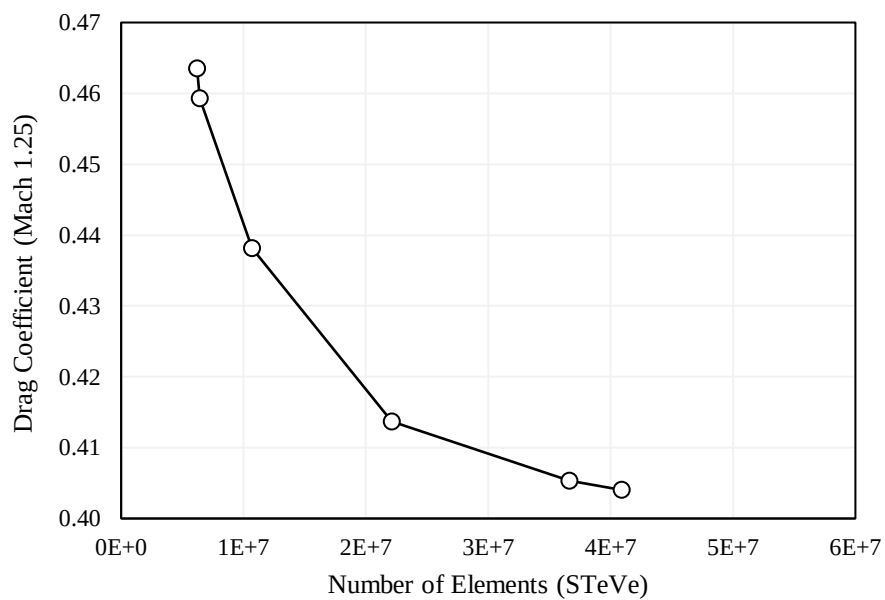


Figure 63. STeVe Mesh Independence at Mach 1.25

To compare mesh independence between different Mach numbers, the percentage difference from the converged value was plotted, and is shown in Figure 64.

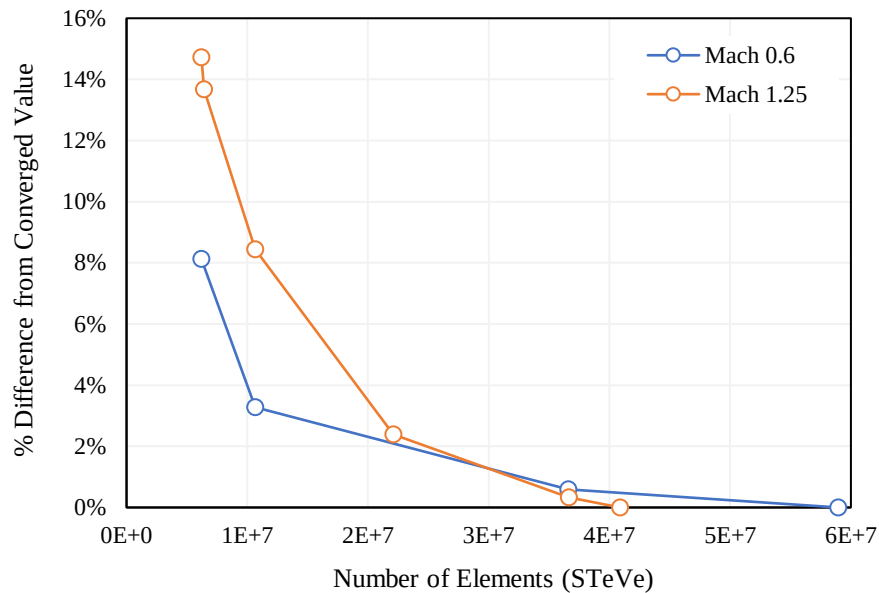


Figure 64. STeVe Mesh Independence Summary

Convergence was achieved, as shown by the asymptotic nature of the drag coefficient vs. number of elements plots. However, the convergence was not achieved as easily as compared to CLV. The convergence graphs for STeVe do not show as smooth of an asymptotic relationship as was shown for CLV. This could be a result of STeVe having a complex geometry, containing fins, which are difficult to mesh, and could have interfered with convergence.

Based on the mesh independence results, the factors in Table 21 were applied to the STeVe and CLV data points to account for the discretisation error, present in CFD.

Table 21. Discretisation Error Factor

Mach Number Range	STeVe ~10.7 million elements	CLV ~10.6 million elements
$M \leq 1$	0.967	0.981
$M > 1$	0.920	0.990

5.2 Static Aerodynamic Coefficients

The drag and lift coefficients were then obtained from CFD, taking into account the discretisation error. The drag coefficient vs. Mach number plot for STeVe is shown in Figure 65.

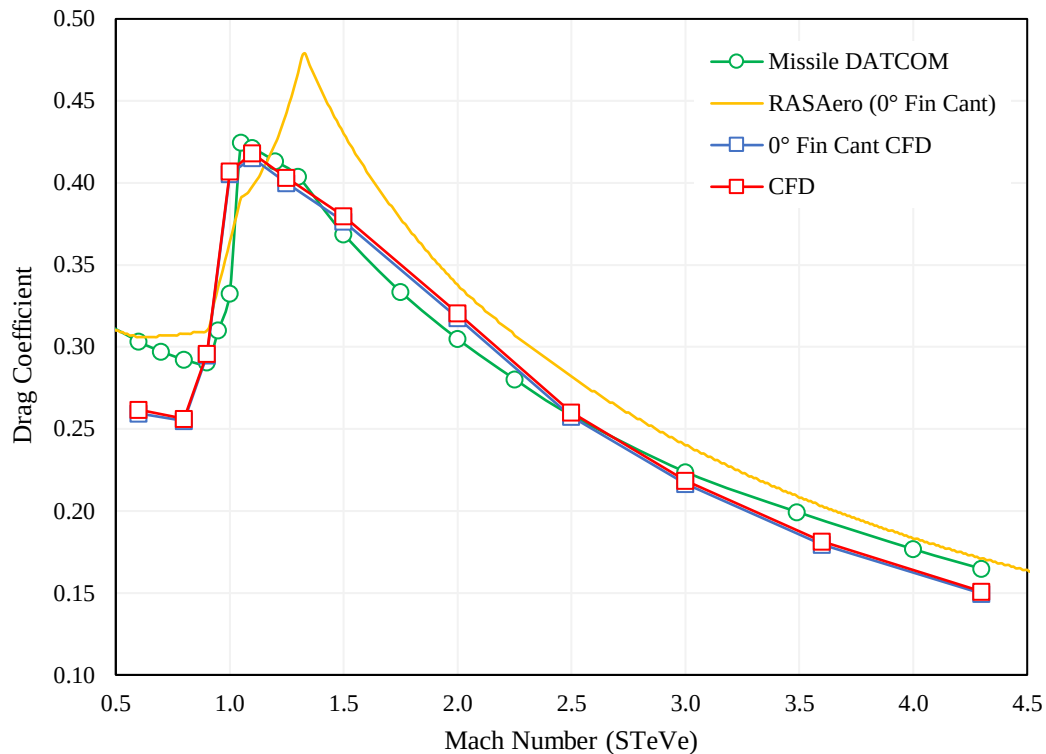


Figure 65. Drag Coefficient vs. Mach Number for STeVe

Due to the limitations of RASAero geometry creation, fin cant could not be added. To analyse if this would have a large effect on the results, a 0° and a 0.7° fin cant simulation were performed in CFD. As shown in the CFD results, there was less than 1% difference for all Mach numbers. It can be concluded that small fin cant angles, less than 0.7° , have a negligible effect on drag, and RASAero results can be used without any adjustment.

Both Missile Datcom and RASAero overpredicted the CFD results by roughly 15.6% in the subsonic region. RASAero overpredicted the peak drag coefficient by 14.6%, as well as shifting the Mach number at which peak drag occurred from Mach 1.1 to Mach 1.33. Missile Datcom overpredicted the peak drag by 1.6%. In the supersonic region, RASAero was within 8.5% of the CFD, while Missile Datcom was within 5.0% of the CFD. These results indicate that Missile Datcom is closer to the CFD results than RASAero.

The drag and lift coefficients were plotted as a function of angle of attack at Mach 4.3, as shown in Figures 66 and 67.

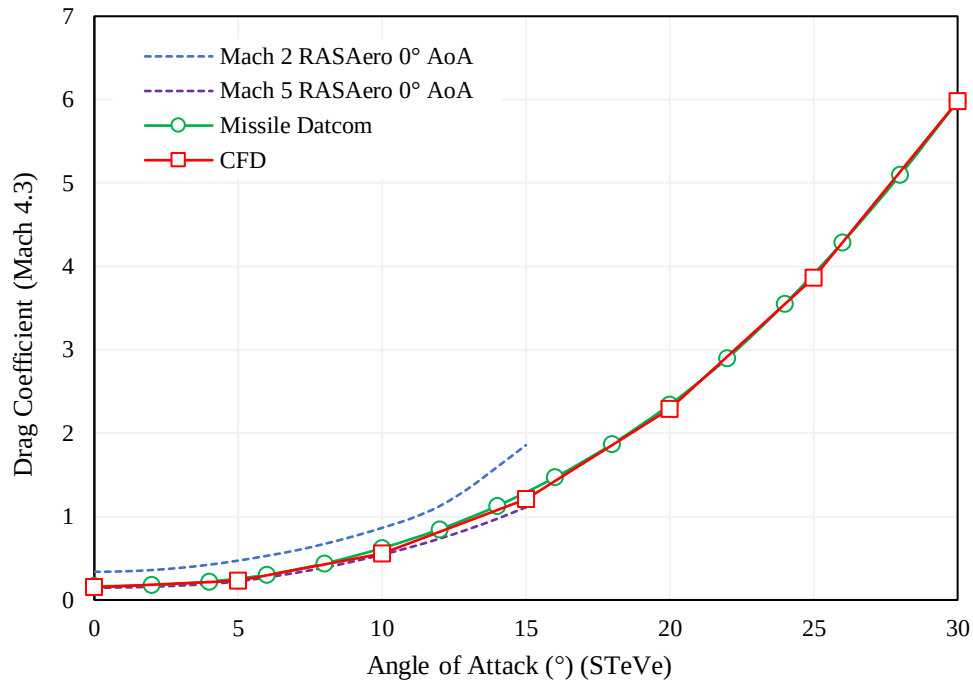


Figure 66. Drag Coefficient vs. Angle of Attack for STeVe

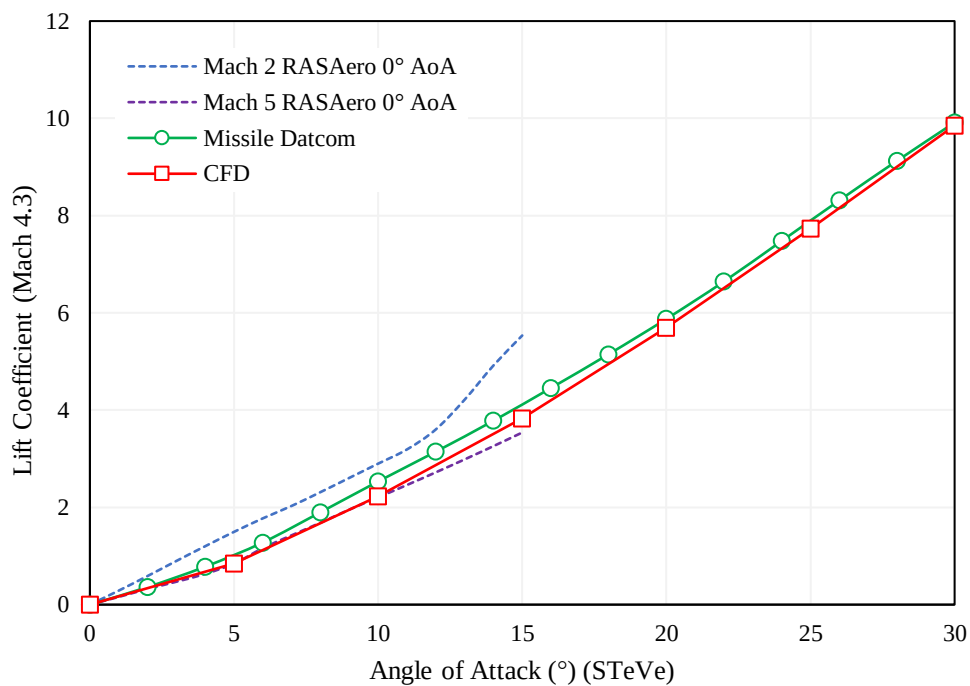


Figure 67. Lift Coefficient vs. Angle of Attack for STeVe

For the drag coefficient, the CFD and Missile Datcom results were within 11.6% up to 15° angle of attack and within 2.4% from 15° to 30° angle of attack. For the lift coefficient, the CFD and Missile Datcom results were within 13.6% up to 15° angle of attack and within 3.1% from 15° to 30° angle of attack. Due to reasons previously discussed in Chapter 2.5.1, RASAero is limited in varying angles of attack. In comparison to the limited RASAero data, the CFD data

were accurate. The CFD and semi-empirical methods are remarkably accurate for high angles of attack and Mach numbers, when simulating STeVe.

A breakdown of the total drag (using CFD and Missile Datcom) into pressure and friction drag for STeVe is shown in Figure 68.

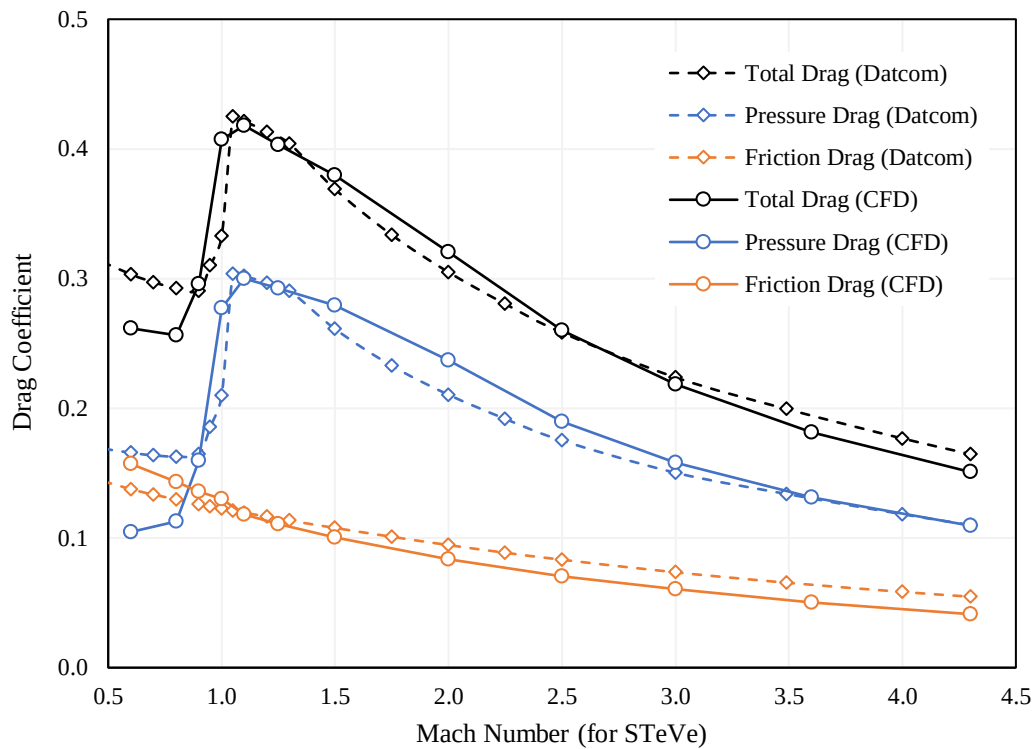


Figure 68. STeVe Drag Coefficient Summary

The friction drag from CFD and Missile Datcom showed a noticeable difference (33% at Mach 4.3). However, because the friction drag values are relatively small (compared to the pressure drag values), this difference did not have a considerable effect on the total drag. The friction drag shows a decreasing, largely linear relationship. The relationship is linear because the friction drag is unaffected by the substantial spike due to wave drag that is shown in the pressure drag graph. There was a smaller discrepancy in the pressure drag readings (11.2% at most) than the friction drag readings.

The grayscale Mach number plots for various Mach numbers and angle of attack over the vehicles are shown in Figures 69 to 72 below, to clearly see the shock wave formation. The higher Mach numbers are indicated by lighter shades.

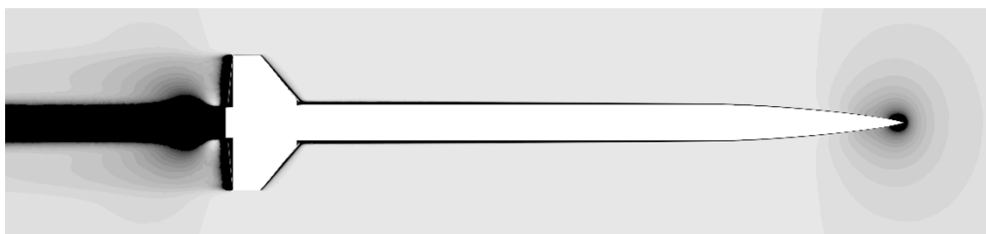


Figure 69. Mach Number Plot at Mach 0.6



Figure 70. STeVe Mach Number Plot at Mach 1.25

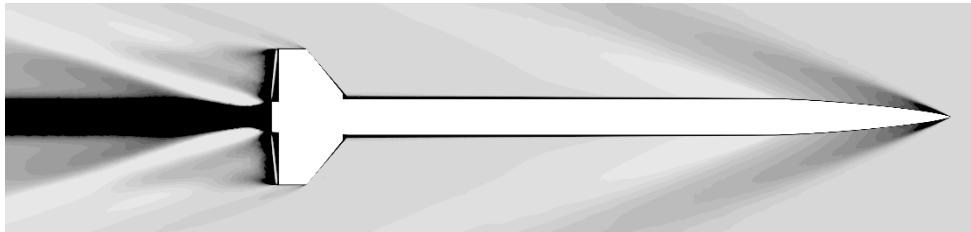


Figure 71. STeVe Mach Number Plot at Mach 3



Figure 72. STeVe Mach Number Plot at Mach 4.3 and 30° Angle of Attack

In subsonic flow, there are no shock waves, resulting in no wave drag and a lower pressure drag. There is a stagnation point at the nose tip, contributing to pressure drag. At Mach 1.25, there is clear shock wave formation at the nose tip, the fin leading edge, and the plume. At Mach 3, there are more shock waves forming at the fin tip. The shock angle is also more acute at Mach 3, than at Mach 1.25. For 0° angle of attack flow, the boundary layer growth can clearly be seen. The boundary layer (indicated by the dark zone) grows in thickness as the distance down the rocket body increases. For the flow at 30° angle of attack, there is not the classic shock wave formation seen in 0° angle of attack flow. At this extreme angle of attack, the base drag accounts for most of the pressure drag. The whole side of the rocket acts as a stagnation zone on the leading side, and a low-pressure zone on the trailing side. This results in substantial base drag.

A characterisation of the drag coefficient versus Mach number for CLV is given in Figure 73.

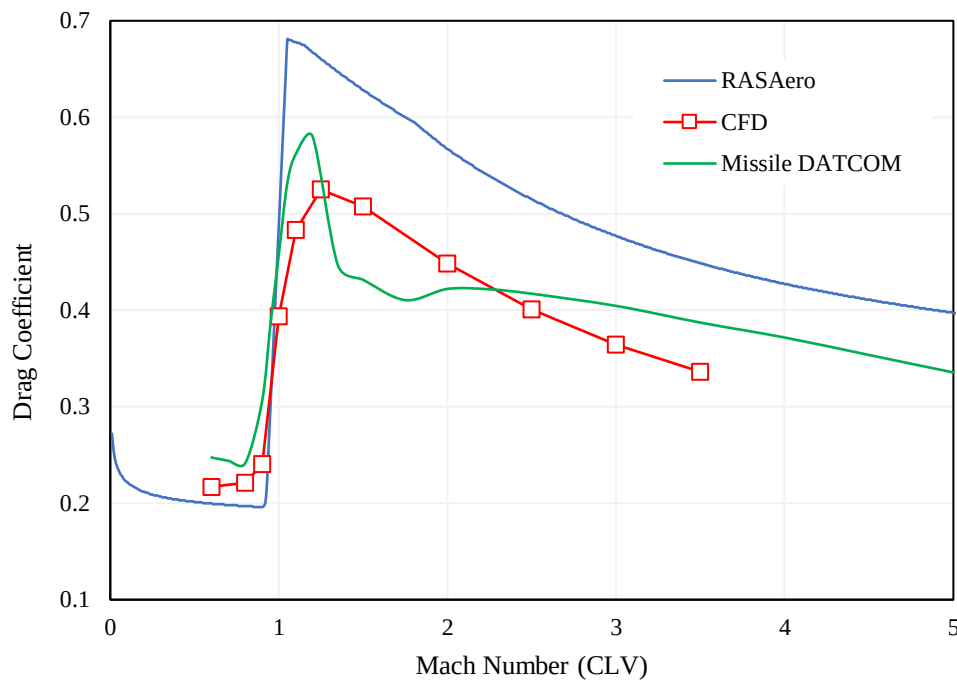


Figure 73. CLV Drag Coefficient vs. Mach Number

CLV was only simulated at a 0° angle of attack, monitoring the drag coefficient vs. Mach number. In the subsonic region, the CFD and semi-empirical data were all close to each other. Missile Datcom and RASAero overpredicted the peak drag coefficient by 10.5% and 29.7% respectively. This is quite a large overprediction, when compared to the STeVe simulation and other validation simulations. In the supersonic region (Mach 3.5), RASAero and Missile Datcom overpredict the drag coefficient by 23.3% and 6.4% respectively. This again, is quite a large overprediction. CLV is a simple rocket geometry, so it was expected to produce close results between the CFD and semi-empirical methods (as shown in the Dunn spinner validation case, Chapter 3.4.1). The only noticeable geometric difference in CLV is the large radius on the nose cone tip, which could be causing the differences seen in RASAero and Missile Datcom. For the Missile Datcom data between Mach 1.5 and Mach 2, there was a noticeable drop in drag, followed by an increase around Mach 2. This behaviour is unrealistic; however, these sharp changes are not uncommon to see when plotting drag from Missile Datcom. This is due to the Missile Datcom code switching from subsonic/transonic correlations to supersonic correlations around Mach 1.2.

Figure 74 shows a drag breakdown (using CFD and Missile Datcom) of the total drag into friction and pressure drag for CLV.

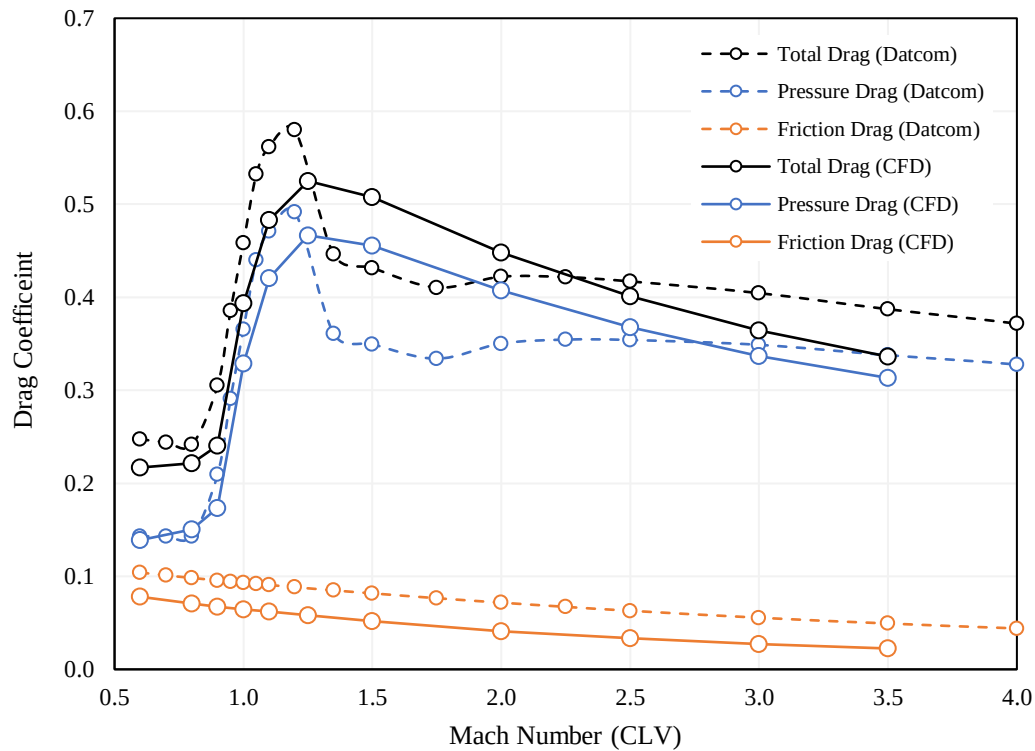


Figure 74. CLV Drag Breakdown

Missile Datcom consistently overpredicts the skin friction drag by 75% when compared to CFD. There was also a large discrepancy in the pressure drag, with a 23% error at Mach 1.5. Missile Datcom showed an unrealistic drop in pressure drag around Mach 1.5, followed by an increase in pressure drag around Mach 2.

The general behaviour of the drag coefficient vs. Mach number plot follows the same shape, as shown in the validation cases, as well as in Figures 65 and 73. The drag coefficient remains relatively constant until roughly Mach 0.8. At this point, there is a substantial increase until a peak is reached somewhere between Mach 1.05 to Mach 1.25. The point at which the drag coefficient rapidly increases is called drag divergence. For the region up until drag divergence, the skin friction drag, pressure drag, and total drag increase with Mach number at similar rates to the dynamic pressure. This is shown by the relatively constant C_d values up to drag divergence. At the drag divergence, there is shock wave formation, which greatly increases wave drag (a type of pressure drag). The pressure drag increases at much higher rates compared to the dynamic pressure, up until a peak C_d is reached. After the peak C_d , the dynamic pressure increases at slightly higher rates than the pressure drag, shown by slowly decreasing C_d values. The rate that the skin friction coefficient changes with Mach number stays relatively constant as it is not affected by the onset of wave drag.

Figure 75 shows a greyscale Mach number plot of CLV at Mach 3.5. Shock wave formation can be seen at the nose cone tip, as well as on the rocket plume.

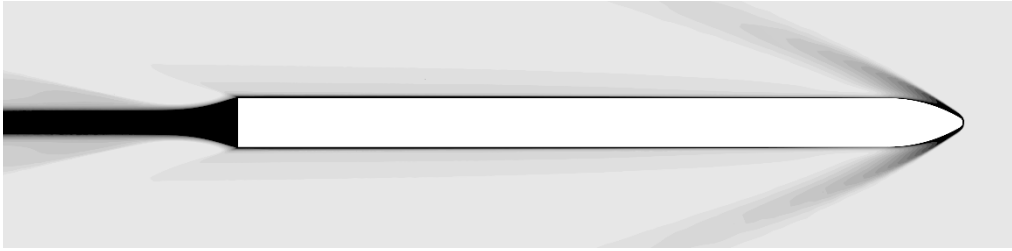


Figure 75. Mach Number Plot of CLV at Mach 3.5

5.3 Flight Trajectory

One of the main objectives of determining the aerodynamic coefficients is to accurately determine the flight trajectory and the aerodynamic forces acting on the rocket. Determining the change in Mach number with altitude allows for the calculation of the dynamic pressure and therefore, the drag force on the rocket. The drag force can then be used to perform stress analyses on the rocket structure.

The trajectory of CLV was determined using the velocity vs. time graph, shown in Figure 76. This approximate trajectory was generated using a two degree of freedom code with a simple gravity turn at 0° angle of attack (Wunderlin, 2024).

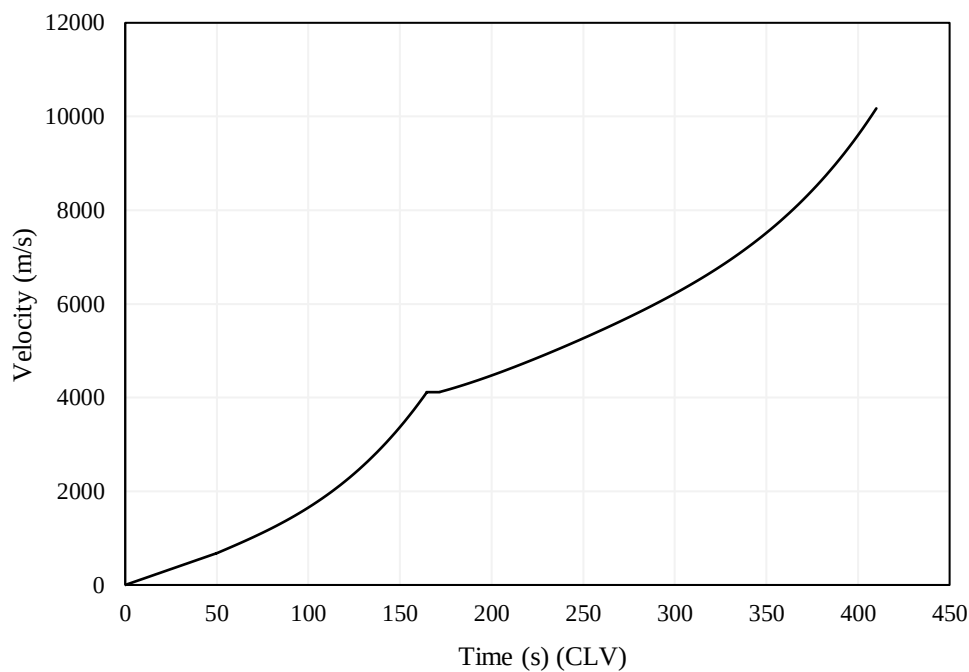


Figure 76. CLV Velocity vs. Time

The velocity vs. time graph was used to determine the altitude vs. time graph. The standard atmospheric temperature and static pressure tables were then used to determine the dynamic pressure and Mach number for the flight trajectory, as shown in Figure 77.

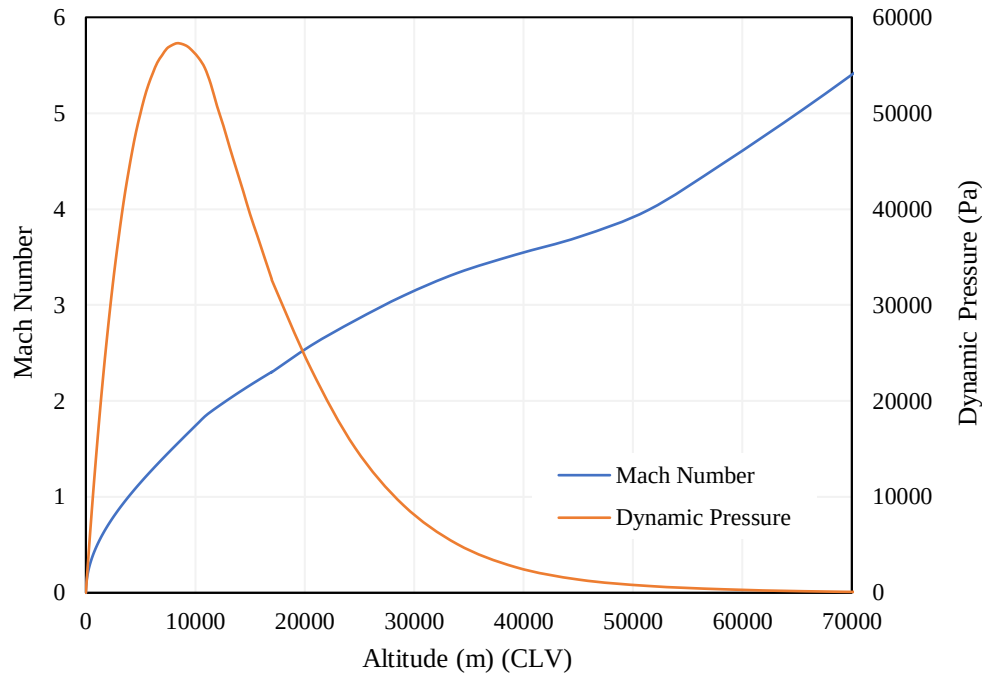


Figure 77. CLV Mach Number and Dynamic Pressure

The drag coefficient vs. Mach number plots (for CFD, Missile Datcom, and RASAero) can be used with the dynamic pressure plots to determine the drag force for the flight trajectory, as shown in Figure 78.

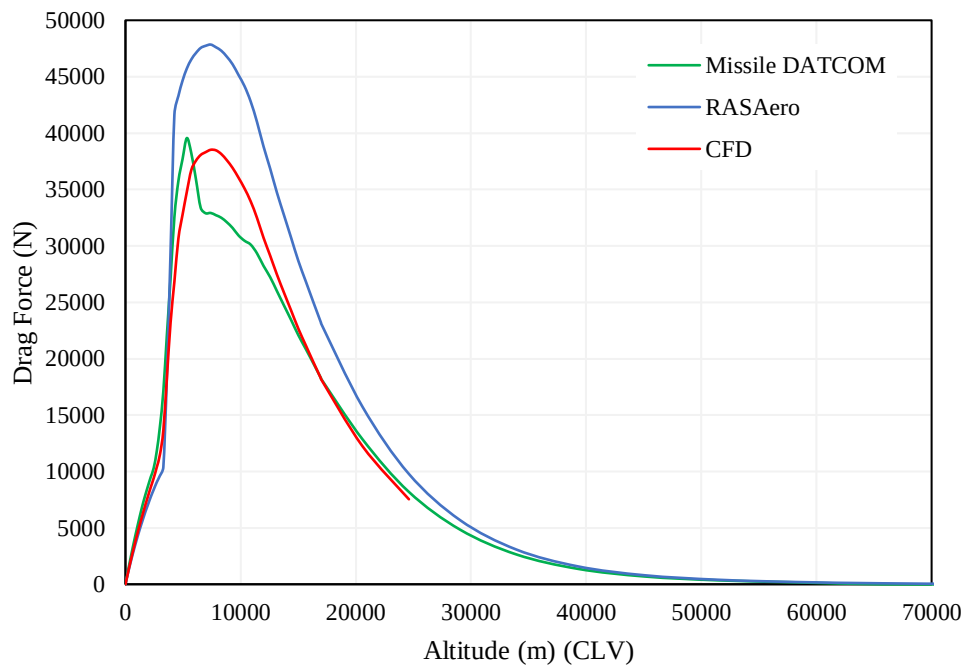


Figure 78. CLV Drag Force vs. Altitude

It is useful to see the effects that different aeroprediction tools have on the maximum drag force. The rocket must be designed using some maximum force, making the choice of software used to determine this force critical. The maximum force will determine the maximum stresses on the rocket, and therefore the minimum rocket wall thickness required to withstand those stresses. If the maximum drag force is overpredicted, then the rocket will be over-engineered,

reducing payload capacity. If the maximum drag force is underpredicted, then the rocket might catastrophically fail under the maximum drag force. Predicting the drag force accurately is also crucial as it directly affects the expected apogee, and it determines the thrust required to reach that apogee. During flight, rocket thrust can be modulated to manage structural loads. Since the dynamic pressure is directly proportional to the drag force, decreasing the thrust near the point of maximum dynamic pressure will decrease the forces acting on the structure.

For CLV, RASAero overpredicted the maximum drag force by 24.2% compared to the CFD, while Missile Datcom only overpredicted the maximum drag force by 2.7% when compared to the CFD. For altitudes less than 4.5 km and greater than 25 km, there was a negligible difference in the drag force between the various aeroprediction tools. RASAero predicted the shape of the drag force graph much better than Missile Datcom.

The same flight trajectory analysis was performed for STeVe, but instead using HyROPS (Hybrid Rocket Performance Simulator) to determine the flight trajectory. HyROPS (Chowdhury, 2012) is a 6 degree of freedom flight code, developed by ASRI, that determines the flight trajectory of a rocket with more accuracy than the CLV trajectory prediction. The velocity vs. time graph for STeVe is shown in Figure 79.

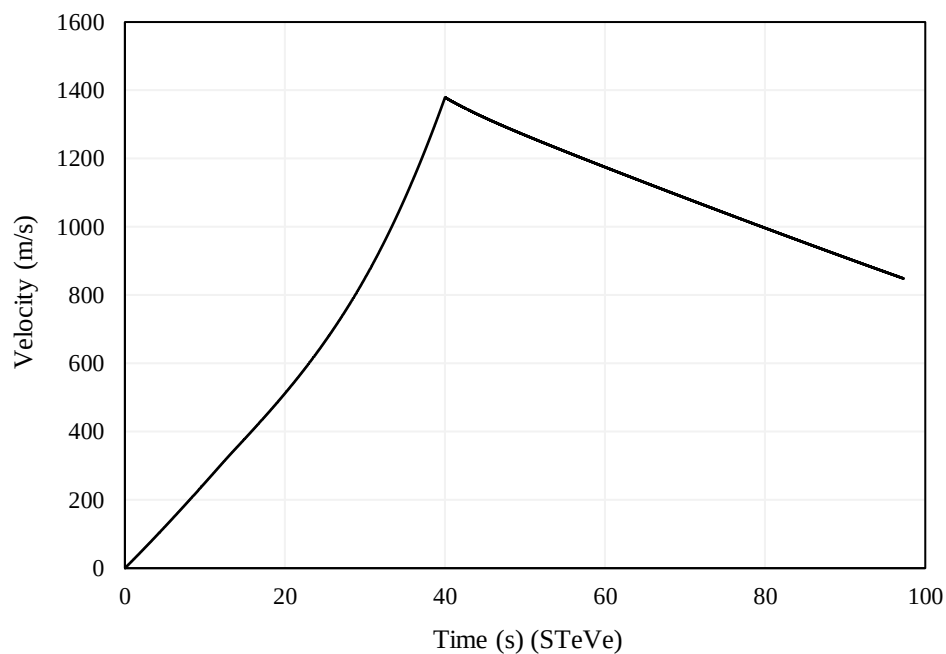


Figure 79. STeVe Velocity vs. Time

Unlike the data from CLV, HyROPS contained the dynamic pressure and Mach number in the output, so no further calculations were required. The dynamic pressure and Mach number graph is shown in Figure 80.

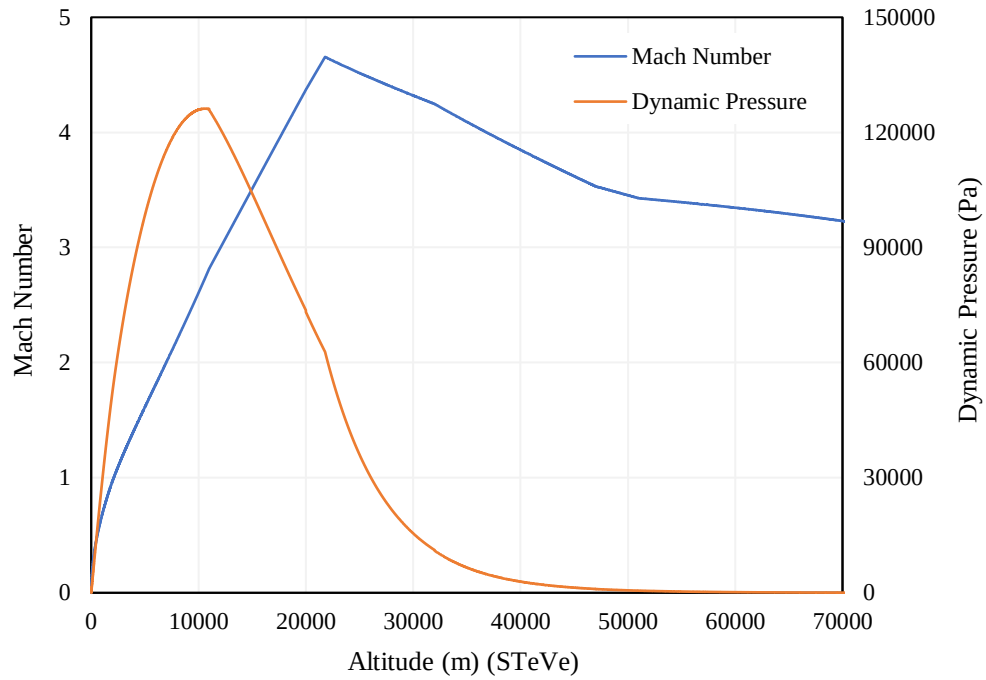


Figure 80. STeVe Mach Number and Dynamic Pressure

The drag coefficient vs. Mach number plots (for CFD, Missile Datcom, and RASAero) were then used along with the dynamic pressure to determine the drag force for the flight trajectory, as shown in Figure 81.

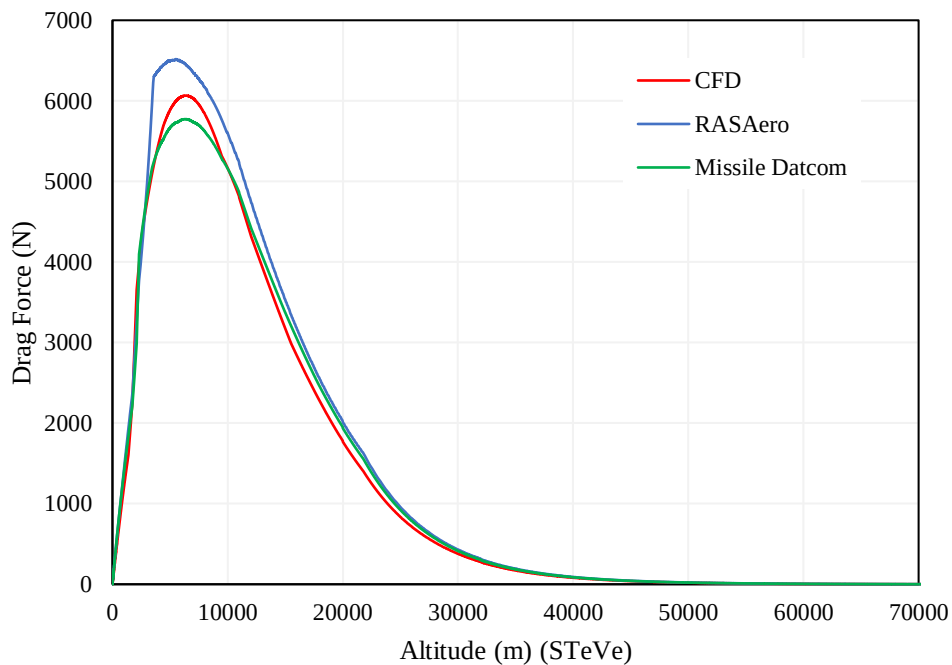


Figure 81. STeVe Drag Force vs. Altitude

RASAero overpredicted the CFD maximum drag force by 7.5%, and Missile Datcom underpredicted the maximum drag force by 4.8%. For altitudes less than 3.2 km and greater than 13.5 km, there were negligible differences in the drag force, between various aeroprediction tools.

6. Aerodynamic Heating

This chapter addresses the process and results of the aerodynamic heating analysis for this study. The important theory regarding aerodynamic heating is found in Chapter 2.3.

6.1 Introduction

In this study, the heat transfer coefficient and surface temperature were analysed for the nose cone of STeVe. As part of previous work done at ASRI, Sabbagha (2024) analytically determined the heat transfer coefficient and temperature along STeVe's nose cone. He approximated the nose cone as a flat plate parallel to the oncoming flow, where forced convection heat transfer dominated. For the nose cone tip temperature, equations from Jerger (1960) were used. Sabbagha (2024) concluded that with a pointed nose cone, virtually no material could withstand the temperatures at the nose cone tip. Therefore, further analysis was required on using rounded nose cone tips to reduce the temperature. The aerodynamic heating analysis in this study had the following aims:

- 1) Compare the analytical heat transfer coefficient and surface temperature results with CFD results.
- 2) Determine the heat transfer coefficient and surface temperature for varying nose cone tip radii.

6.2 CFD Setup

An identical setup to the other STeVe simulations was used, with the process described in Chapter 4. The only change was inputting the thermal conductivity, k , to a value of 2.5695 W/m.K using Sutherland's Law.

Sutherland's Law can be used to define the thermal conductivity of pure and non-reacting gases as a function of temperature.

$$k = k_{ref} \left(\frac{T}{T_{ref}} \right)^{\frac{3}{2}} \cdot \frac{T_{ref} + S_k}{T + S_k} \quad 6.1$$

where k is the thermal conductivity, in W/m.K,
the Sutherland temperature, $S_k = 194$ K (for air),
the reference temperature, $T_{ref} = 273.15$ K, and
the reference thermal conductivity, $k_{ref} = 0.0241$ W/m.K.

The thermal boundary condition on the rocket surface was set to adiabatic. The 'specified y^+ heat transfer coefficient' method was used to determine the heat transfer coefficient. A user specified y^+ value of 100 was used to obtain the heat transfer coefficient, as recommended in the Star-CCM+ user's manual (Siemens, 2023b). All CFD simulations were run at sea-level atmospheric conditions. This means that the raw heat transfer coefficients and temperatures obtained from CFD could not be compared to analytical calculations, as the analytical calculations were based on HyROPS flight data. The following methods were used to obtain comparable results between CFD and the analytical results for the heat transfer coefficient and the surface temperature.

Heat Transfer Coefficient: Comparing the heat transfer coefficient between different methods is a futile exercise, as “there is an infinite number of heat transfer coefficient and reference temperature combinations that give the same surface heat flux upon convergence.” (Siemens, 2023b). Therefore, comparing the trend of heat transfer coefficient is more important than comparing the absolute values. The input conditions for the analytical calculator were changed to sea-level conditions to act as a fairer comparison with the CFD results.

Surface Temperature: To obtain the nose cone surface temperature at flight conditions using CFD, the following method based on theory from Chapter 2.3 was used.

- 1) The adiabatic wall temperature, T_{aw} , was obtained using CFD at sea level atmospheric conditions, at a desired Mach number.
- 2) The recovery factor, r , for the CFD result was obtained using

$$r = \frac{T_{aw} - T_{\infty}}{T_0 - T_{\infty}} \quad 6.2$$

where the stagnation temperature is

$$T_0 = T_{\infty} \left(1 + \frac{\gamma - 1}{2} \cdot M_{\infty}^2 \right) \quad 6.3$$

- 3) The adiabatic wall temperature at the desired flight condition was then obtained using the following equation,

$$T_{aw} = T_{\infty} \cdot \left(r \cdot \frac{\gamma - 1}{2} \cdot M_{\infty}^2 + 1 \right) \quad 6.4$$

where T_{∞} is the free stream temperature at the desired flight condition.

No literature was found regarding the accuracy of this method. So first, this method was compared with the analytical results.

6.3 Results

The aerodynamic heating results are presented and discussed in two parts. Firstly, the analytical calculations were compared to the CFD simulation results. Secondly, the performance of various nose cone geometries was then evaluated using CFD.

6.3.1 Analytical Calculation Comparison

Since no experimental temperature data were available, the CFD results were compared to the analytical results as an initial check of accuracy. The heat transfer coefficient was plotted against distance along the nose cone for three Mach numbers, as shown in Figures 82 to 84. As sea-level atmospheric conditions were used in the CFD, the same conditions were inputted into the analytical calculator.

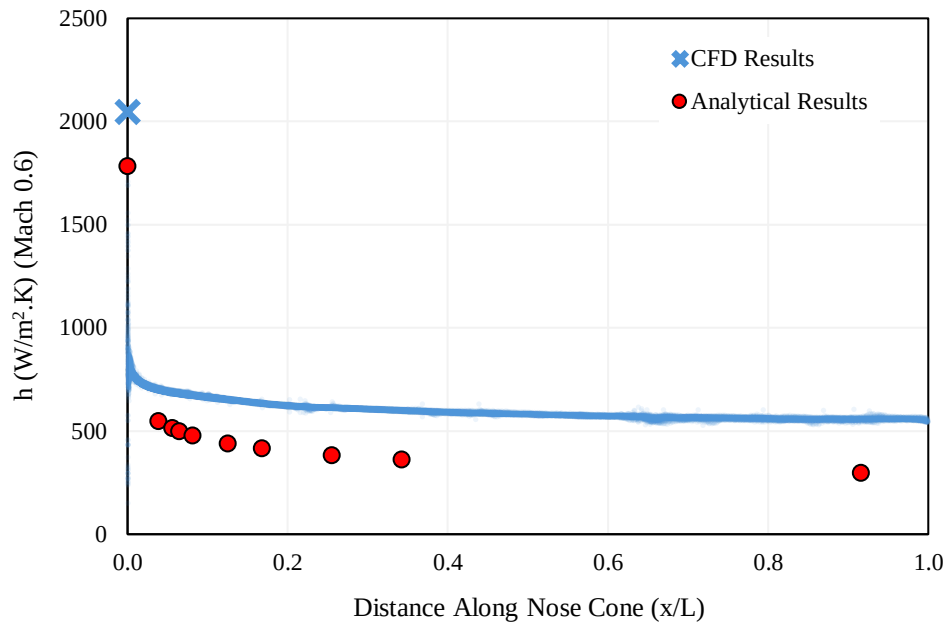


Figure 82. Heat Transfer Coefficient at Mach 0.6

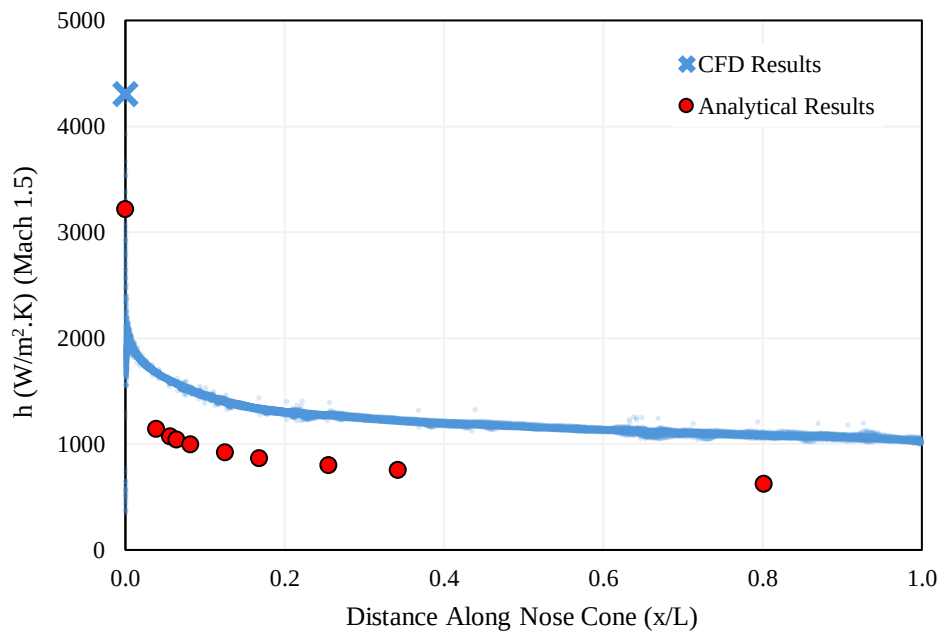


Figure 83. Heat Transfer Coefficient at Mach 1.5

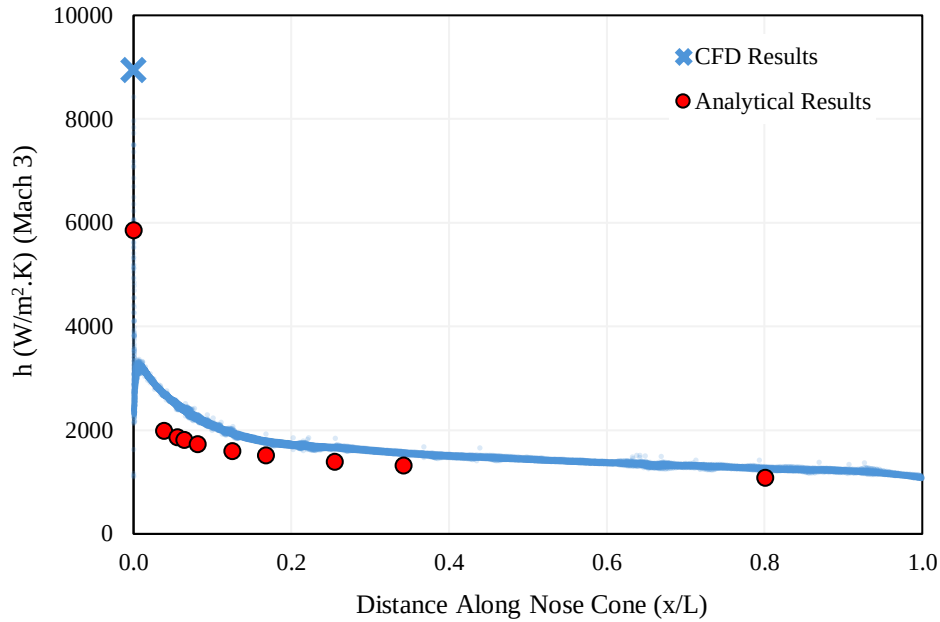


Figure 84. Heat Transfer Coefficient at Mach 3

CFD predicted a higher peak heat transfer coefficient by 14.6%, 33.7%, and 52.8% at Mach 0.6, 1.5, and 3 respectively. CFD predicted a higher heat transfer coefficient along the main nose cone surface by roughly 65.5%, 61.7%, and 18.4% at Mach 0.6, 1.5, and 3 respectively. These results show a noticeable difference between the CFD and analytical results for the heat transfer coefficient. However, this difference is expected as there is a pressure gradient present in the CFD. In the flat plate correlation, there is no pressure gradient, so the analytical results should be more inaccurate. If only the trend is analysed, the trends shown by the CFD and analytical results are similar.

The surface temperature, at flight conditions, was plotted against distance along the nose cone at Mach 0.6, 1.5 and 3, as shown in Figures 85 to 87.

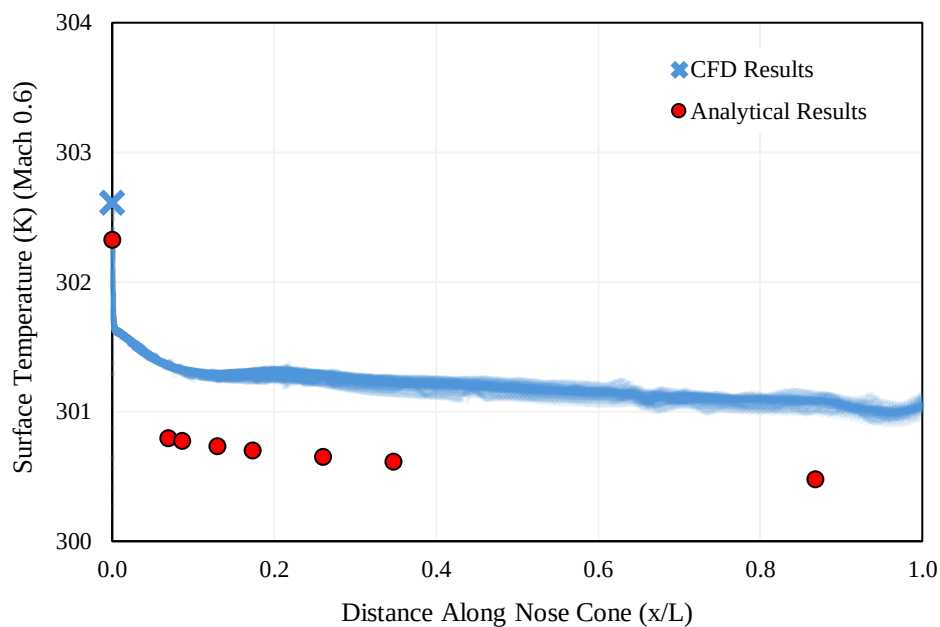


Figure 85. Surface Temperature at Mach 0.6

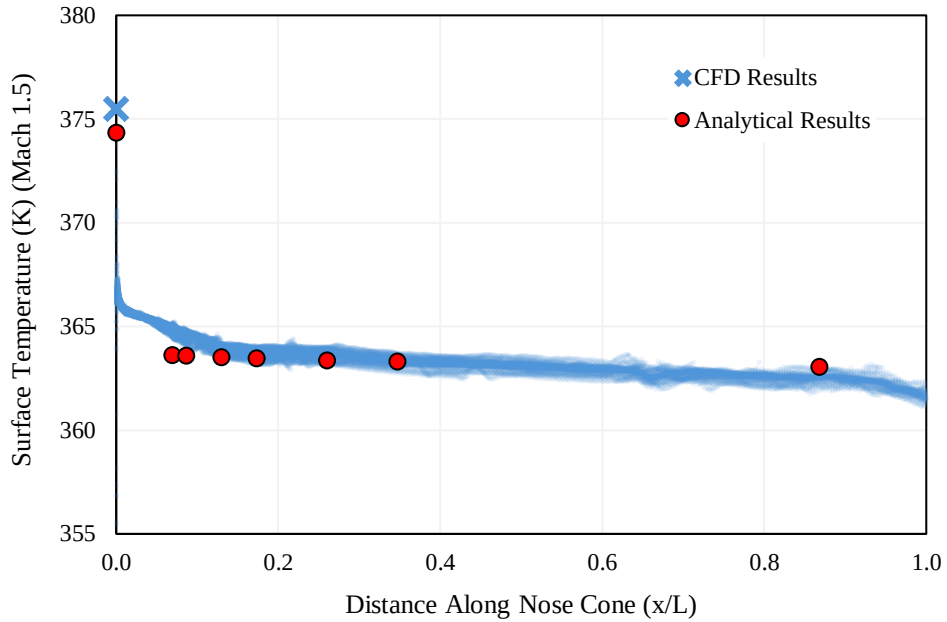


Figure 86. Surface Temperature at Mach 1.5

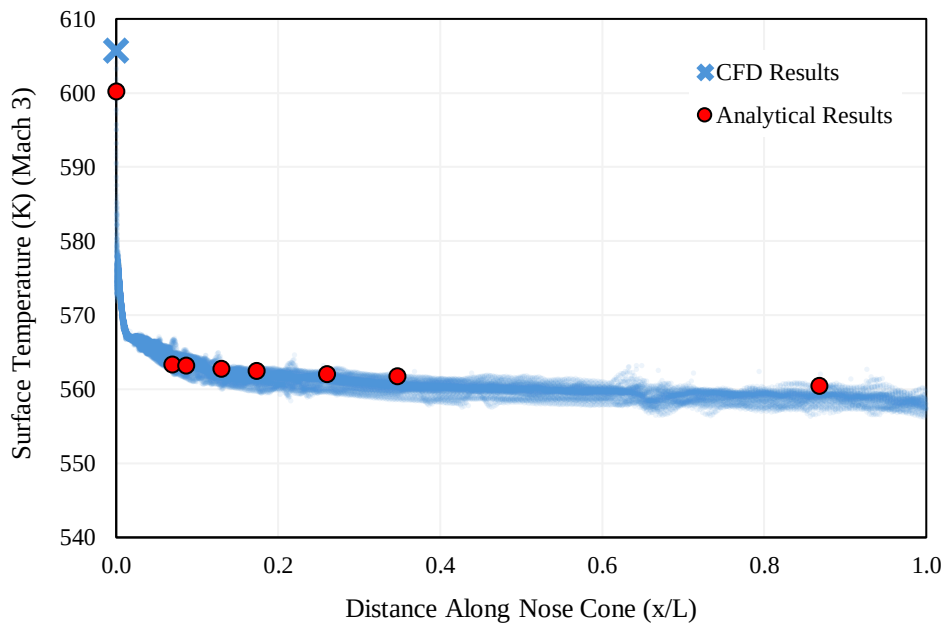


Figure 87. Surface Temperature at Mach 3

CFD predicted a higher peak surface temperature by 0.1%, 0.3%, and 0.9% at Mach 0.6, 1.5, and 3 respectively. Between 10% to 100% of the nose cone length, CFD varied by a maximum of 0.1% for all three Mach numbers. The CFD and analytical results for the heat transfer coefficient were close. Therefore, It would appear that using the recovery factor to convert adiabatic wall temperatures from CFD at sea level conditions to surface temperatures at flight conditions yields reasonable results.

6.3.2 Nose Cone Geometry Variation

STeVe was simulated with a Von Kármán nose cone with various tip radii, as shown in Figure 88.

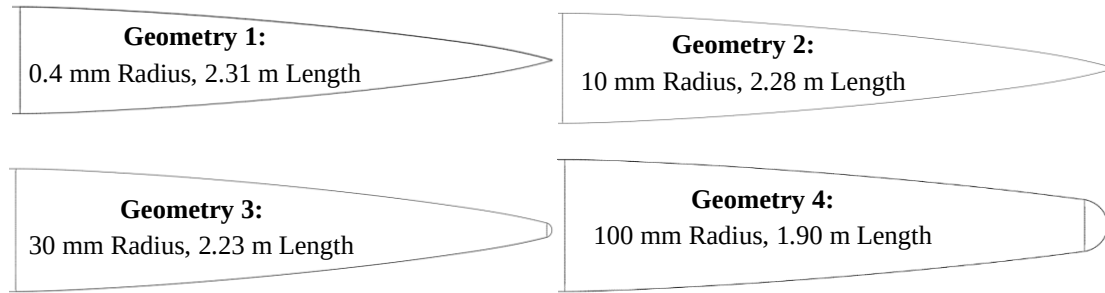


Figure 88. Von Kármán Nose Cone Variations

The heat transfer coefficient was plotted against distance along the nose cone for three Mach numbers, as shown in Figures 89 to 91.

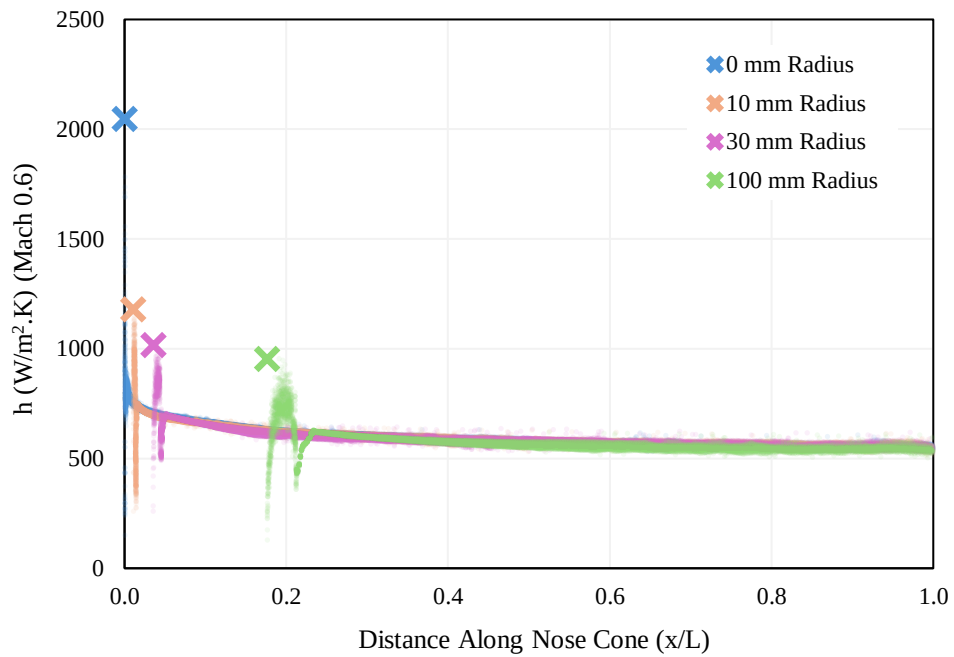


Figure 89. Heat Transfer Coefficient at Mach 0.6

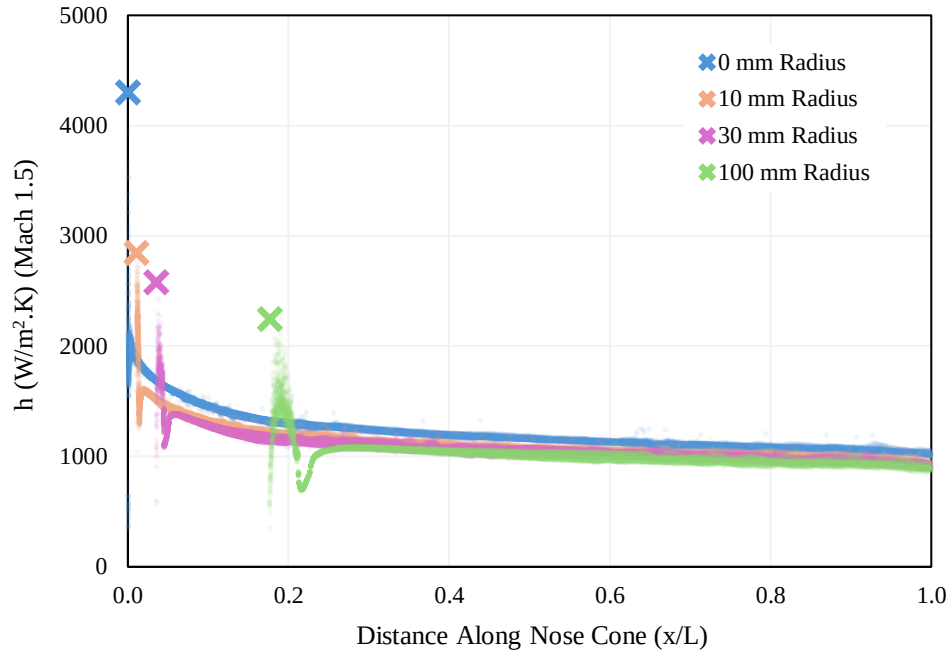


Figure 90. Heat Transfer Coefficient at Mach 1.5

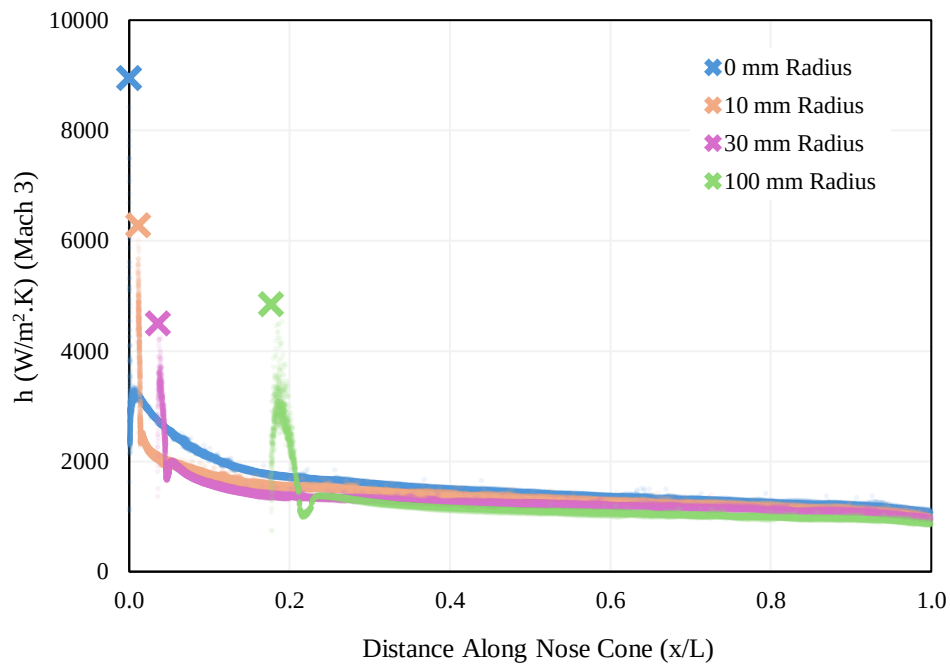


Figure 91. Heat Transfer Coefficient at Mach 3

It can be observed that the maximum heat transfer coefficient at the nose tip decreased as the nose tip radius increased. This was true for all Mach numbers, except that the maximum heat transfer coefficient at Mach 3 was larger for a 100 mm radius nose, than a 30 mm radius nose. There are clearly diminishing returns when increasing the nose cone tip. Increasing the tip radius from 0 mm to 10 mm, decreased the maximum heat transfer coefficient by 33% to 42% (depending on the Mach number). However, increasing the tip radius from 10 mm to 30 mm, only decreased the maximum heat transfer coefficient by 9 % to 28 % (depending on the Mach

number). Another takeaway is that the heat transfer coefficient remained largely similar along the main body of the nose cone, regardless of tip radius.

The adiabatic wall temperature (at flight conditions) was then plotted against distance along the nose cone for three Mach numbers, as shown in Figures 92 to 94.

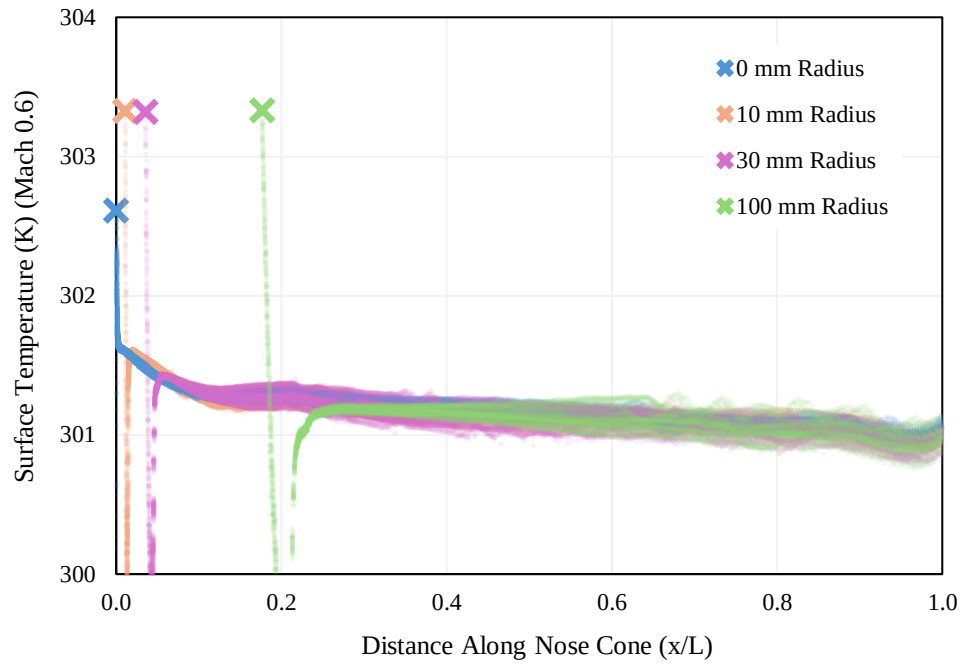


Figure 92. Surface Temperature at Mach 0.6

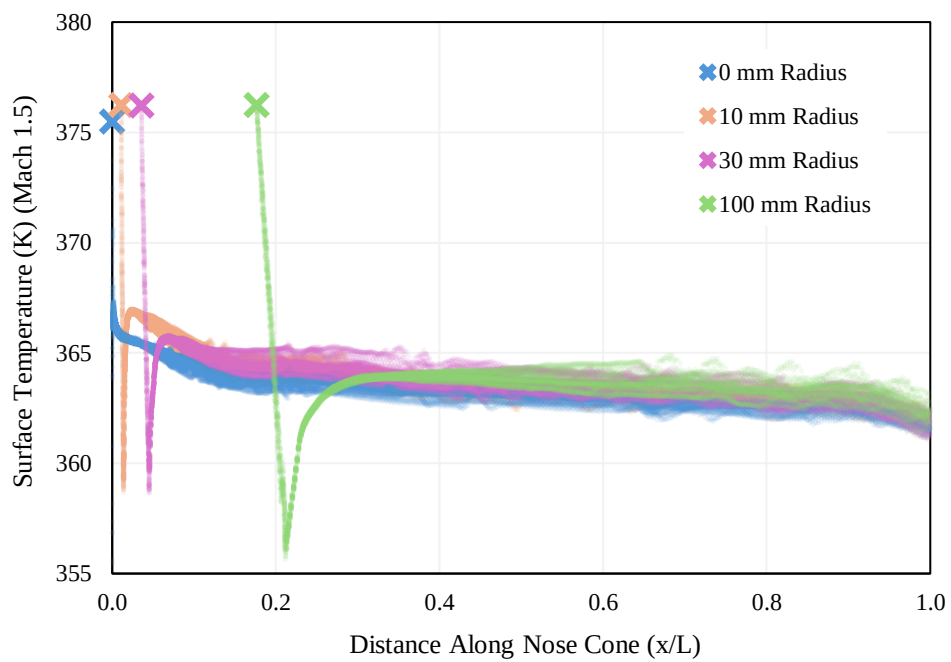


Figure 93. Surface Temperature at Mach 1.5

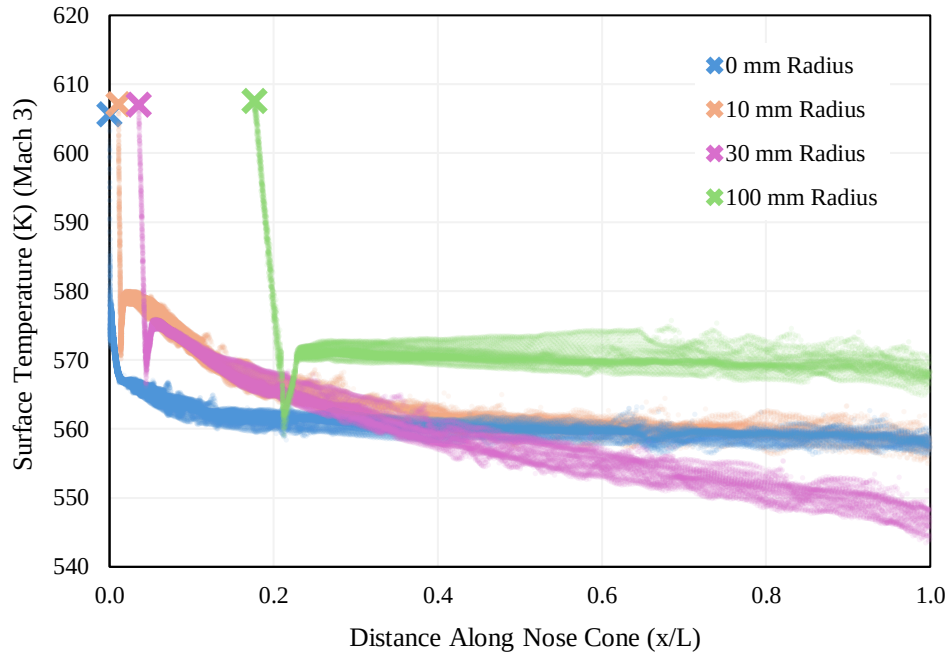


Figure 94. Surface Temperature at Mach 3

At all three Mach numbers, the maximum surface temperature at the different nose cone radii varied by 0.3% at most; indicating the tip radius did not affect the maximum surface temperature. At Mach 0.6 and 1.5, the surface temperature did not change along the length of the nose cone, as the tip radius was varied. At Mach 3, the 100 mm radius had the highest temperature along the length of the nose cone, while the 30 mm radius had the lowest temperature along the length of the nose cone. These results are inconclusive, as no clear trend is observed for surface temperature when varying the nose tip radius. Experimental data are required to confirm the accuracy of these plots. However, based on the surface temperature plots, a 30 mm tip radius would be best, as it results in the lowest surface temperatures at Mach 3.

In theory, the main drawback of increasing nose tip radius is that aerodynamic drag increases. To confirm this, the drag coefficient was plotted against Mach number for different nose cone radii, as shown in Figure 95.

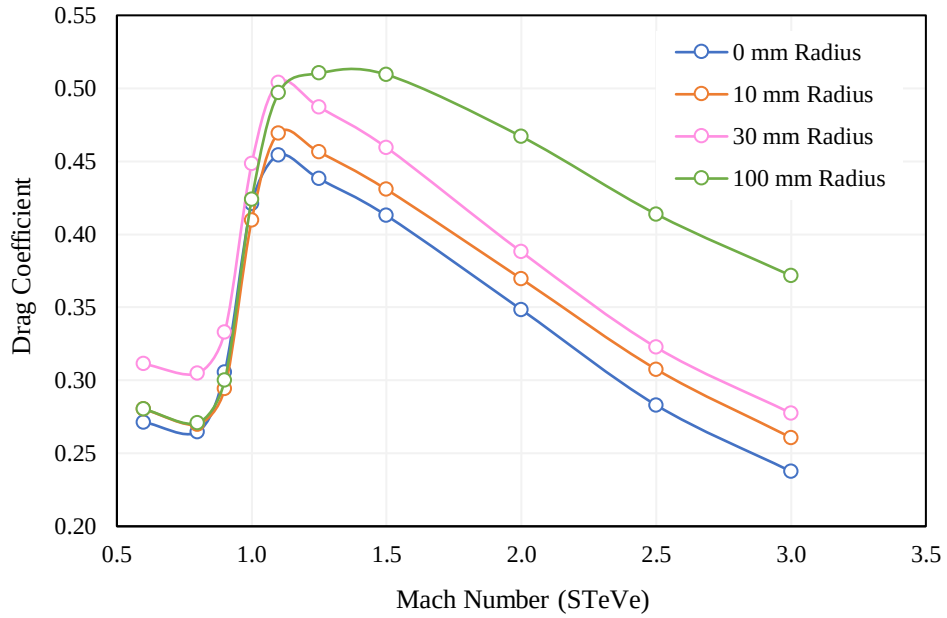


Figure 95. Drag Coefficient for Different Geometries

As expected, the drag coefficient increased with the nose tip radius. One exception is that the highest drag in the subsonic region was the 30 mm nose cone radius, not the 100 mm radius. However, in the supersonic region, the 100 mm radius had a higher drag than the 30 mm radius.

Mach number contour plots for the varying nose cone radii are shown at Mach 0.6, 1.25, and 3, in Figures 96 to 98.

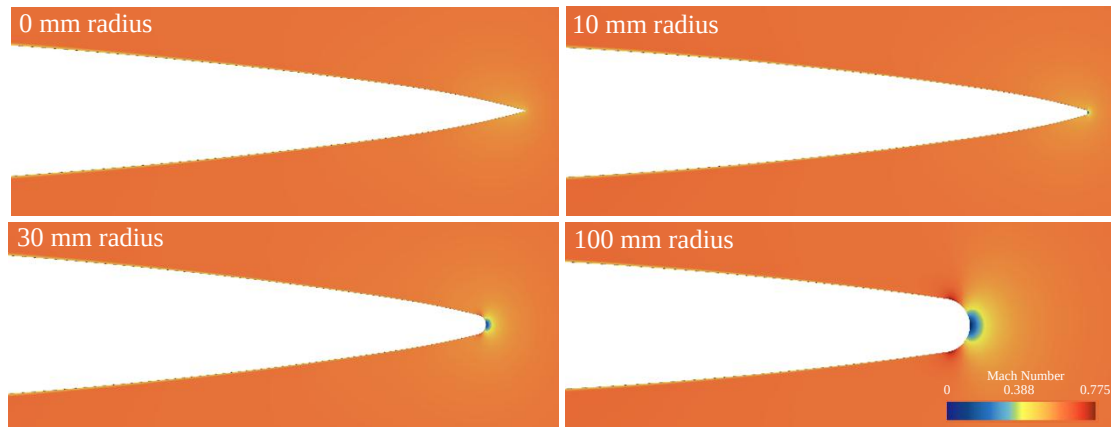


Figure 96. Mach Number Contour Plots at Mach 0.6

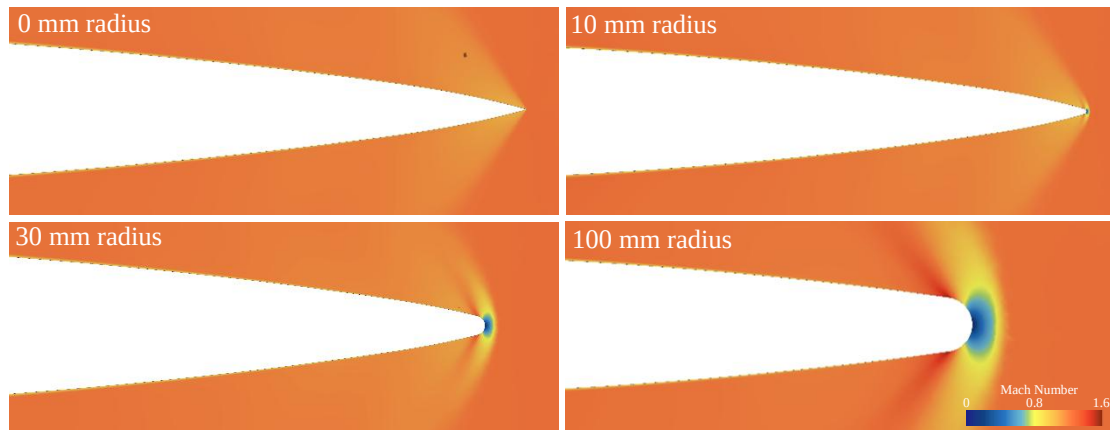


Figure 97. Mach Number Contour Plots at Mach 1.25

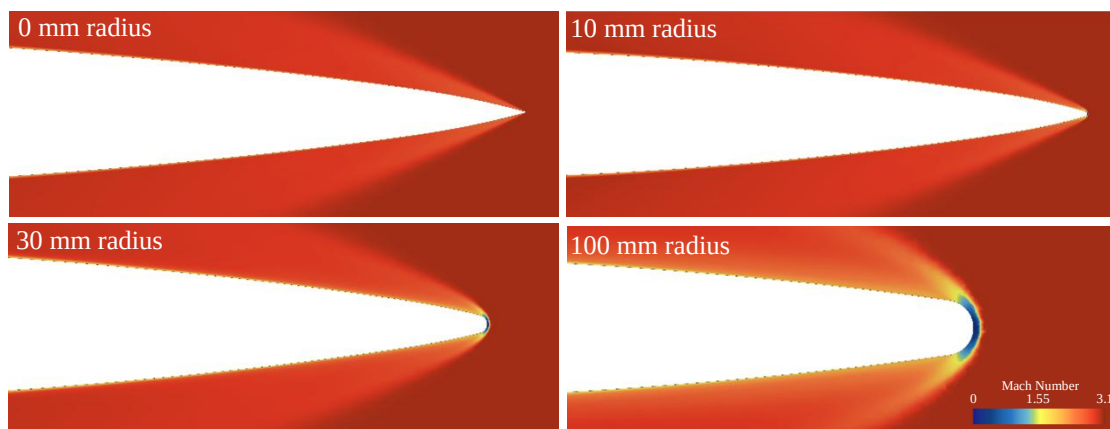


Figure 98. Mach Number Contour Plots at Mach 3

For all Mach numbers, the larger nose cone diameters resulted in larger stagnation zones at the nose cone tip. In the supersonic regime, the shock waves originated from the nose cone tip and are clearly shown in the figures above. As the nose cone tip radius was increased, the oblique shock wave transitioned to a bow shock wave. The distance from the bow shock wave to the nose cone tip was larger for Mach 1.25, than for Mach 3. At higher Mach numbers, the angle of the oblique shock waves were more acute. At Mach 3, there was a local area of lower Mach number flow present around the larger nose tip radii. This lower Mach number flow formed behind the bow shock. Temperature contour plots are shown in Figure 99, to observe the fluid temperature distribution at high Mach number flow.

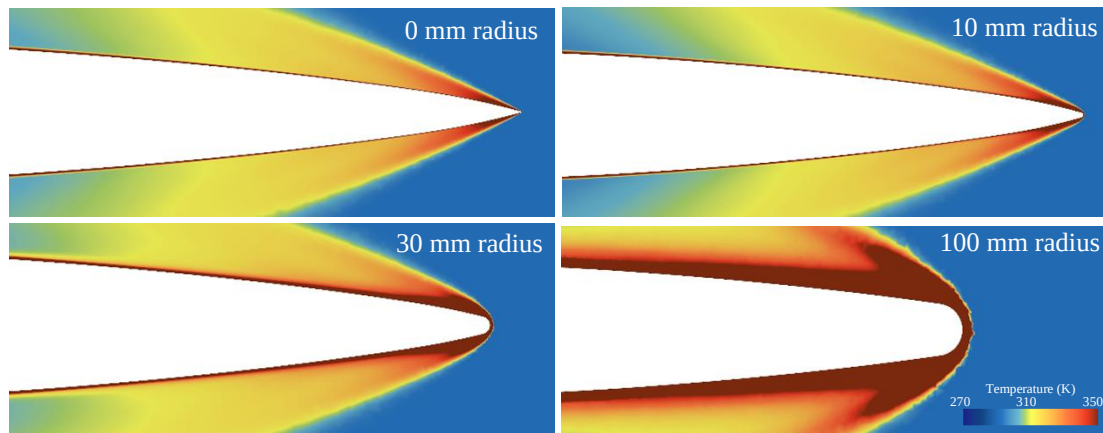


Figure 99. Temperature Contour Plots at Mach 3

The temperature contour plots show that increasing the nose tip radii, increases the fluid temperature surrounding the nose cone. This agrees with the theory on shock waves in Chapter 2.2.7. Downstream of a shock wave, the Mach number decreases and the temperature increases.

In summary, increasing the nose tip radius decreases the maximum heat transfer coefficient. However, increasing the nose tip radius increases the drag and the surrounding fluid temperature. All these factors must be considered when sizing the nose cone tip radius for STeVe and CLV.

7. Geometry Modification

This chapter analyses the effects of changing fuselage geometry. For both CLV and STeVe, three different analyses were performed:

- 1) Changing fuselage diameter, while keeping the fuselage length constant.
- 2) Changing fuselage length, while keeping the fuselage diameter constant.
- 3) Keeping the fuselage volume constant, while adjusting the fuselage diameter and length.

In this chapter, $C_d A$ is used to compare geometries, not C_d . This is because $C_d A$ gives a better representation of the drag force. When the rocket geometry is modified, the diameter is changed. Therefore, the reference area with which the drag coefficient is defined, is changed. Using $C_d A$ accounts for the diameter change and more appropriately represents the drag force. For a constant Mach number, the following equation shows that the drag force is dependent on the $C_d A$:

$$F_d = \overbrace{\left(\frac{1}{2}\rho u^2\right)}^{\text{Constant}} \times \overbrace{(C_d A)}^{\text{Varies with Geometry}} \quad 7.1$$

Missile Datcom was used to perform the simulations in this chapter, as many different geometries could be simulated in a short time, unlike with CFD simulations.

7.1 CLV Geometry Modification

Figure 100 and Table 22 show the effects on $C_d A$ for varying the diameter of CLV.

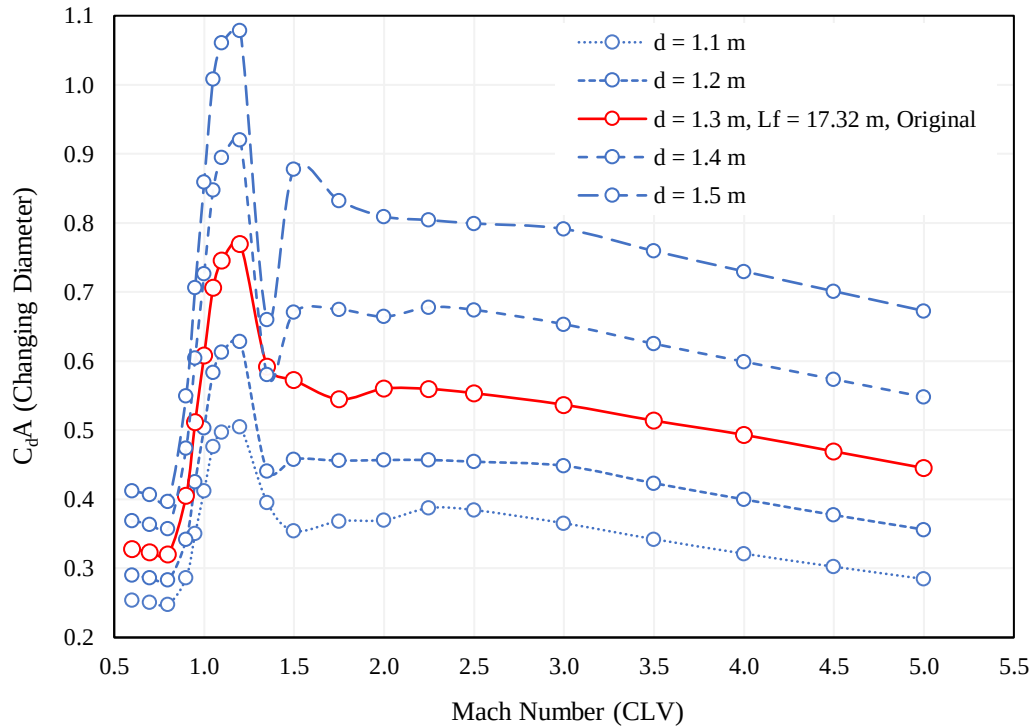


Figure 100. $C_d A$ Changing Diameter

Table 22. C_dA Comparison Changing Diameter

Diameter (m)	Area (%)	Mach 0.6 C_dA	Peak C_dA	Mach 3.5 C_dA
1.1 (15.4%)	-28.4%	-22.5%	-34.4%	-33.4%
1.2 (+7.7%)	-14.8%	-11.6%	-18.4%	-17.6%
1.3 (-)	-	-	-	-
1.4 (+7.7%)	+16.0%	+12.4%	+19.6%	+21.6%
1.5 (+15.4%)	+33.1%	+25.5%	+40.2%	+47.8%

Figure 101 and Table 23 show the effects on C_dA for varying the length of CLV.

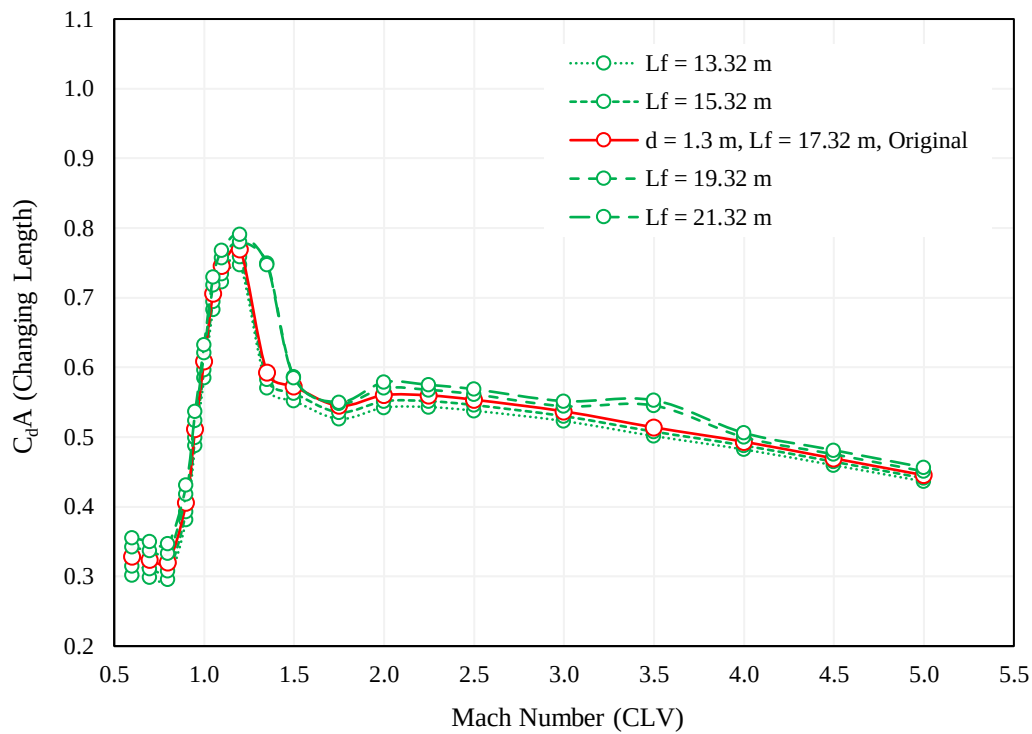


Figure 101. C_dA Changing Length

Table 23. C_dA Comparison Changing Length

Length (m)	Mach 0.6 C_dA	Peak C_dA	Mach 3.5 C_dA
13.32 (-23.1%)	-8.0%	-2.9%	-2.4%
15.32 (-11.5%)	-4.0%	-1.4%	-1.2%
17.32 (-)	-	-	-
19.32 (+11.5%)	+4.1%	+1.4%	+6.1%
21.32 (+23.1%)	+8.2%	+2.8%	+7.6%

Changing the rocket diameter had a substantial effect on C_dA , as shown in Figure 100. However, changing the rocket length had a negligible effect on C_dA , as shown in Figure 101. This is because pressure drag dominates over friction drag, especially at supersonic Mach numbers. Pressure drag is directly related to the cross-sectional area, as the cross-sectional area increases both base drag and wave drag, as shown above. When the cross-sectional area was

increased by 16.0%, the peak C_dA increased by 19.6%. However, when the rocket length was increased by 11.5%, the peak C_dA only increased by 1.4%.

Figure 102 and Table 24 show the effects on C_dA for varying the diameter and length of CLV.

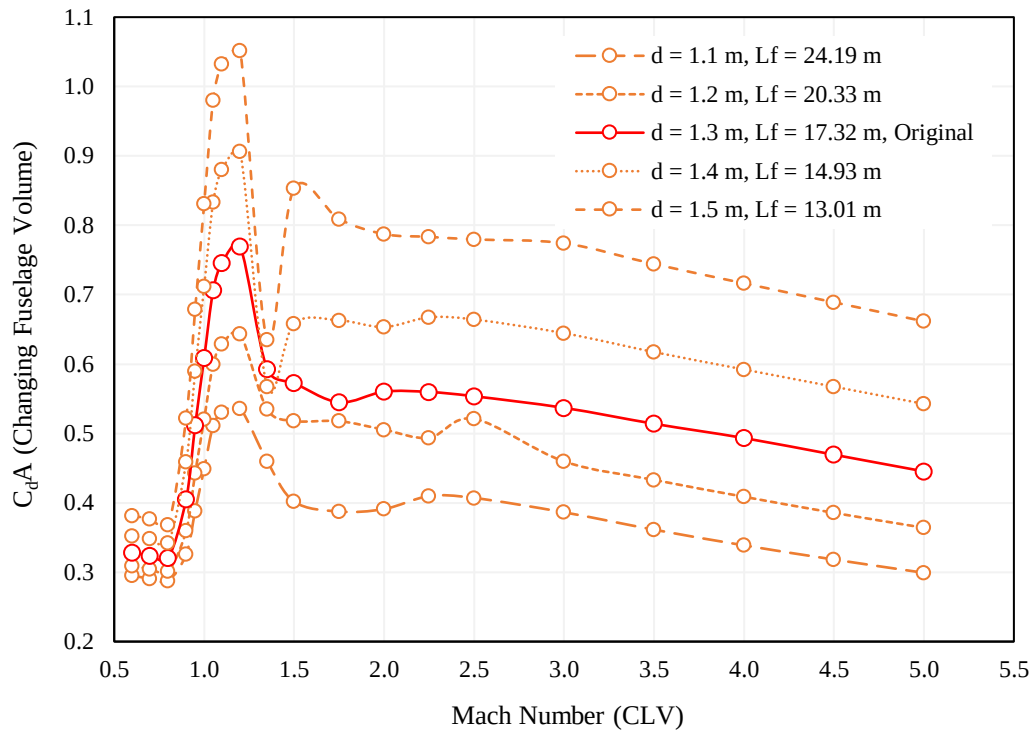


Figure 102. C_dA Constant Volume

Table 24. C_dA Comparison Constant Volume

Diameter (m)	Length (m)	Area (%)	Mach 0.6 C_dA	Peak C_dA	Mach 3.5 C_dA
1.1 (-15.4%)	24.19 (+39.7%)	-28.4%	-10.1%	-30.4%	-29.7%
1.2 (-7.7%)	20.33 (+17.4%)	-14.8%	-5.8%	-16.4%	-15.7%
1.3 (-)	17.32 (-)	-	-	-	-
1.4 (+7.7%)	14.93 (-13.8%)	+16.0%	+7.3%	+17.7%	+20.1%
1.5 (+15.4%)	13.01 (-24.9%)	+33.1%	+16.1%	+36.6%	+44.8%

When keeping the volume constant (increasing the diameter while decreasing the length), the peak C_dA increased by 17.7%, when the cross-sectional area was increased by 16.0%. Table 24 and Table 22 can be compared, to see the effects on keeping the fuselage volume constant. Keeping the fuselage volume constant when changing diameter, had a negligible effect on C_dA in the supersonic region. This is shown by the 'Peak C_dA ' and 'Mach 3.5 C_dA ' column being similar in Table 24 and Table 22. However, in the subsonic region, there was a noticeable effect. When only changing the rocket diameter, increasing the area by 16.0% increased the Mach 0.6 C_dA by 12.4%. When the rocket diameter was increased, keeping the volume constant, the area increased by 16.0% while the Mach 0.6 C_dA only increased by 7.3%. This shows that reducing rocket length, and therefore friction drag, plays more of an importance in the subsonic region.

Another point to note is that there are strange behaviours in the drag coefficient vs. Mach number graphs produced by Missile Datcom. One such behaviour is a large decrease and increase in drag around Mach 1.3, as seen in Figures 100 and 102. This unrealistic behaviour is common to see from Missile Datcom, based on the previous validation cases and Datcom results. This unrealistic behaviour is likely due to a change in solver correlation from the subsonic/transonic region to the supersonic region. In the Missile Datcom user's manual (Air Force Research Laboratory, 1998) it is stated that the body alone aerodynamic methodology switches from subsonic/transonic correlations to supersonic correlations at Mach 1.2. The fin alone aerodynamic methodology switches from transonic correlations to supersonic correlations at Mach 1.4. It therefore makes sense, that in some instances there might be large changes around this transition point between correlations.

The Datcom results were then used to determine the drag force for the flight trajectory. The effects of changing the rocket dimensions can then be visualised, as shown in Figure 103 and Table 25.

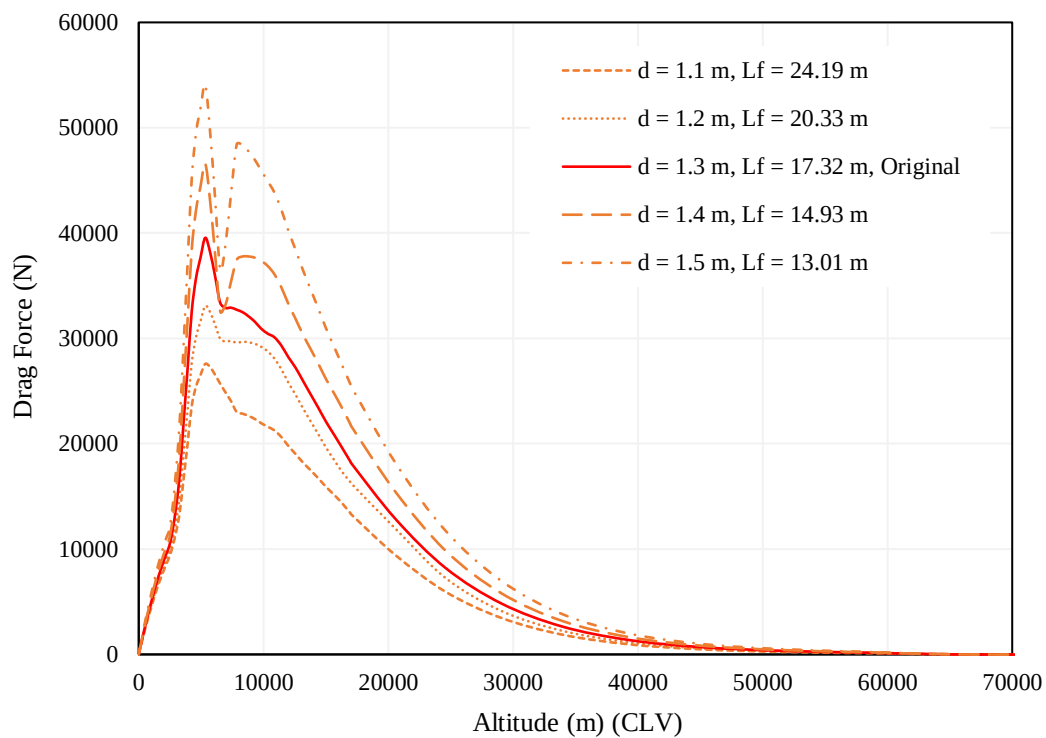


Figure 103. CLV Geometry Comparison

Table 25. Drag Force Comparison for a Constant Volume

Diameter (m)	Length (m)	Area (%)	Peak F_D (%)
1.1 (-15.4%)	24.19 (+39.7%)	-28.4%	-30.3%
1.2 (-7.7%)	20.33 (+17.4%)	-14.8%	-16.4%
1.3 (-)	17.32 (-)	-	-
1.4 (+7.7%)	14.93 (-13.8%)	+16.0%	+17.8%
1.5 (+15.4%)	13.01 (-24.9%)	+33.1%	+36.7%

Changing the diameter had a substantial impact on the maximum drag force experienced during flight. Increasing the area by 16.0%, increased the peak drag force by 17.8%. For altitudes below 3.5 km and above 35 km, there was a negligible difference in drag force. The sharp, unrealistic decrease and increase in drag force was due to the Missile Datcom solver switching from subsonic/transonic correlations to supersonic correlations around Mach 1.2.

7.2 STeVe Geometry Modification

Figure 104 and Table 26 show the effects on C_dA for varying the diameter of STeVe.

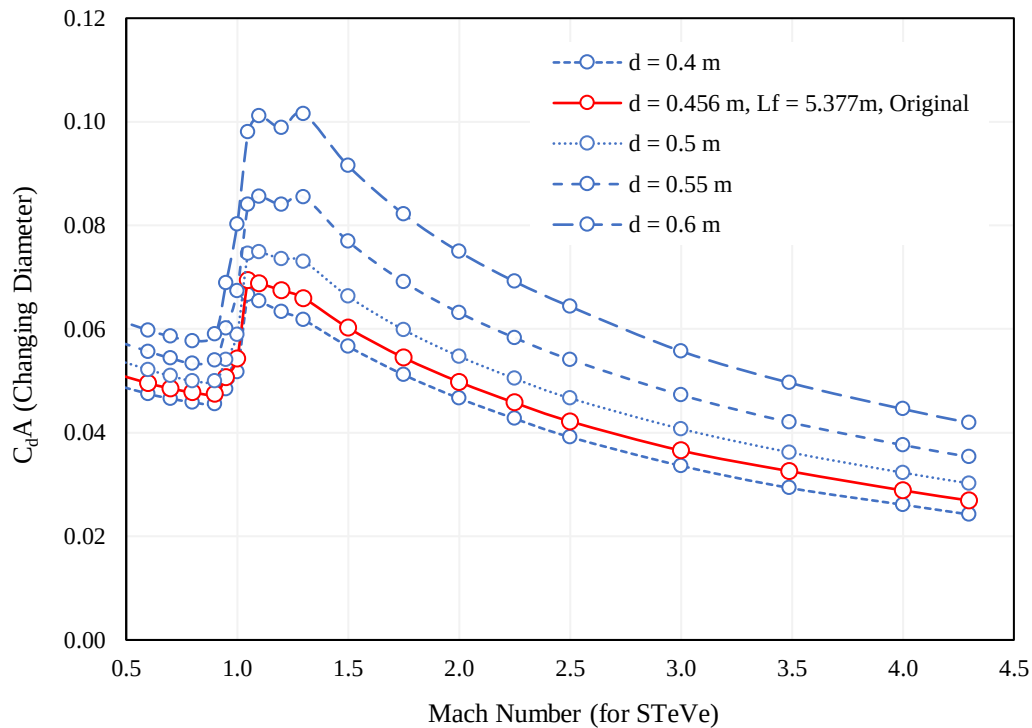


Figure 104. C_dA Changing Diameter

Table 26. C_dA Comparison Changing Diameter

Diameter (m)	Area (%)	Mach 0.6 C_dA	Peak C_dA	Mach 3.5 C_dA
0.4 (-12.3%)	-23.1%	-4.1%	-4.0%	-9.9%
0.456 (-)	-	-	-	-
0.5 (+9.6%)	+20.2%	+5.3%	+7.4%	+11.2%
0.55 (+20.6%)	+45.5%	+12.4%	+21.1%	+29.0%
0.6 (+31.6%)	+73.1%	+20.6%	+41.3%	+52.4%

Figure 105 and Table 27 show the effects on C_dA for varying the length of STeVe.

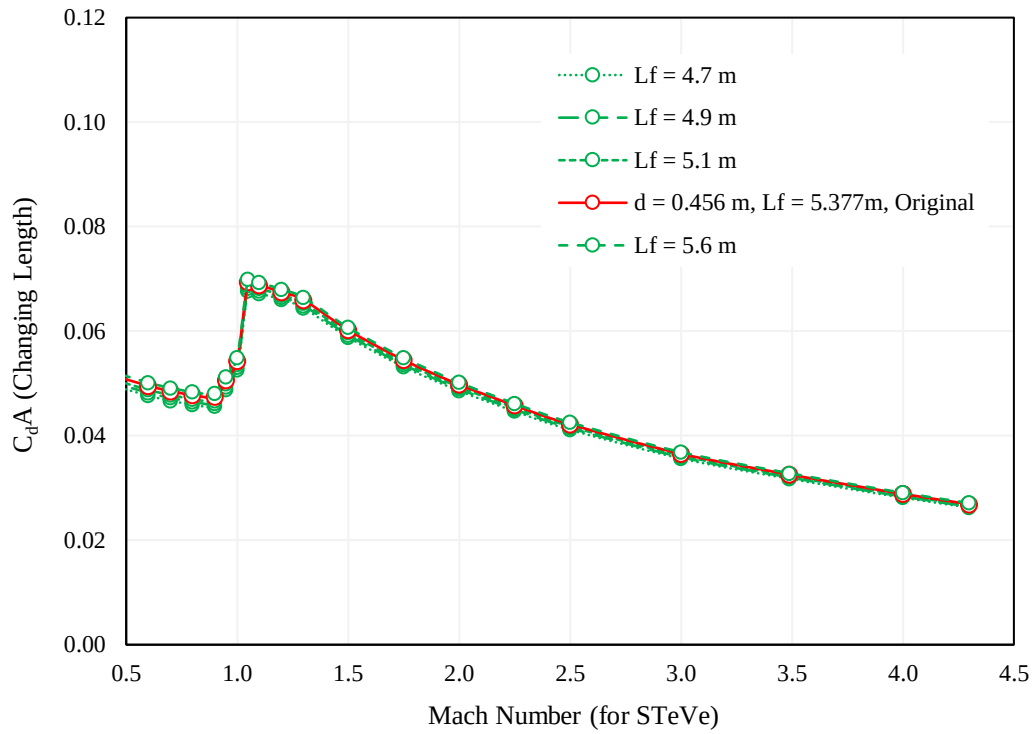


Figure 105. C_dA Changing Length

Table 27. C_dA Comparison Changing Length

Length (m)	Mach 0.6 C_dA	Peak C_dA	Mach 3.5 C_dA
4.7 (-12.6%)	-3.8%	-2.5%	-2.5%
4.9 (-8.9%)	-2.7%	-1.7%	-1.7%
5.1 (-5.2%)	-1.5%	-1.0%	-1.0%
5.377 (-)	-	-	-
5.6 (+4.1%)	+1.2%	+0.8%	+0.8%

Changing the rocket diameter did not have a substantial effect on C_dA , as shown in Figure 104. When the cross-sectional area was increased by 20.2%, the peak C_dA only increased by 7.4%. Changing the rocket length had an even smaller effect on C_dA , as shown in Figure 105. When the rocket length was increased by 4.1%, the peak C_dA increased by 0.8%. The effect on C_dA due to changing the fuselage dimensions is less for STeVe than for CLV. This could be because STeVe has four fins, which were not changed during the geometry comparison, and consistently contribute to the drag.

Figure 106 and Table 28 show the effects on $C_d A$ for varying the diameter and length of STeVe.

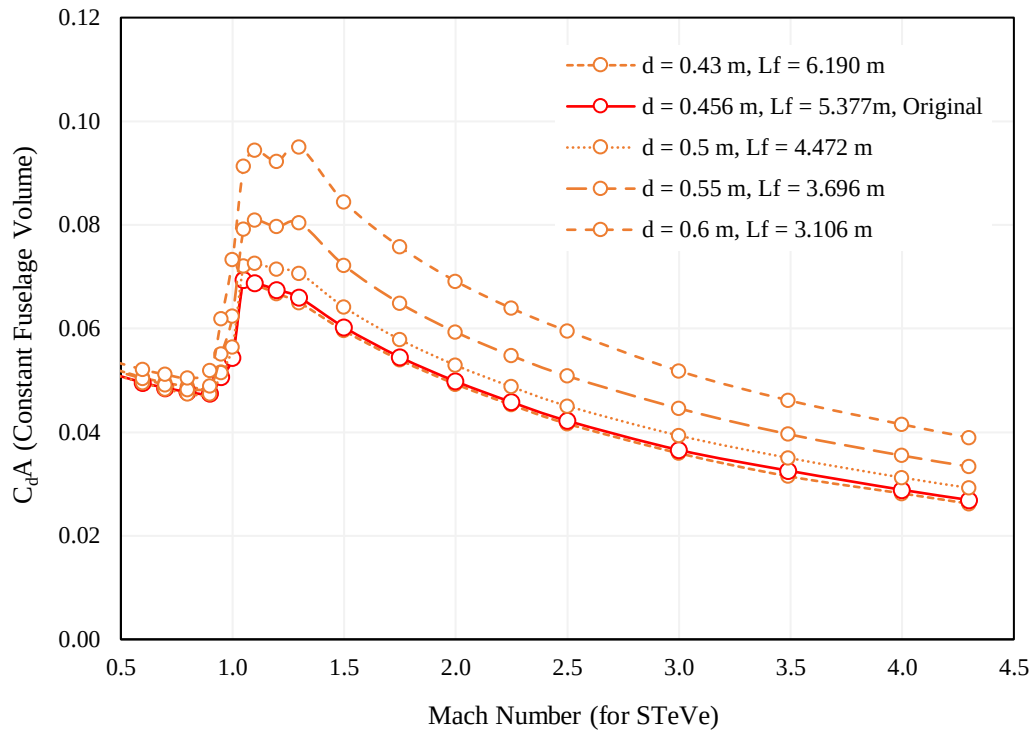


Figure 106. $C_d A$ Constant Fuselage Volume

Table 28. $C_d A$ Comparison for Constant volume

Diameter (m)	Length (m)	Area (%)	Mach 0.6 $C_d A$	Peak $C_d A$	Mach 3.5 $C_d A$
0.43 (-5.7%)	6.190 (+15.1%)	-23.1%	-2.0%	-0.4%	-3.3%
0.456 (-)	5.377 (-)	-	-	-	-
0.5 (+9.6%)	4.472 (-16.8%)	+20.2%	+0.1%	+3.9%	+7.6%
0.55 (+20.6%)	3.696 (-31.3%)	+45.5%	+1.5%	+14.1%	+21.6%
0.6 (+31.6%)	3.106 (-42.2%)	+73.1%	+5.1%	+31.6%	+41.7%

When keeping the volume constant (increasing the diameter while decreasing the length), the peak $C_d A$ only increased by 3.9%, when the cross-sectional area was increased by 20.2%. For STeVe, keeping the fuselage volume constant when changing diameter, had a noticeable effect on $C_d A$. For example, at Mach 0.6, the peak $C_d A$ only increased by 1.5% when the cross-sectional area was increased by 45.5%. This shows that there is the possibility to increase the diameter of STeVe without having a large increase in drag force.

The Datcom results were then used to determine the drag force for the flight trajectory. The effects of changing the rocket dimensions can then be visualised, as shown in Figure 107.

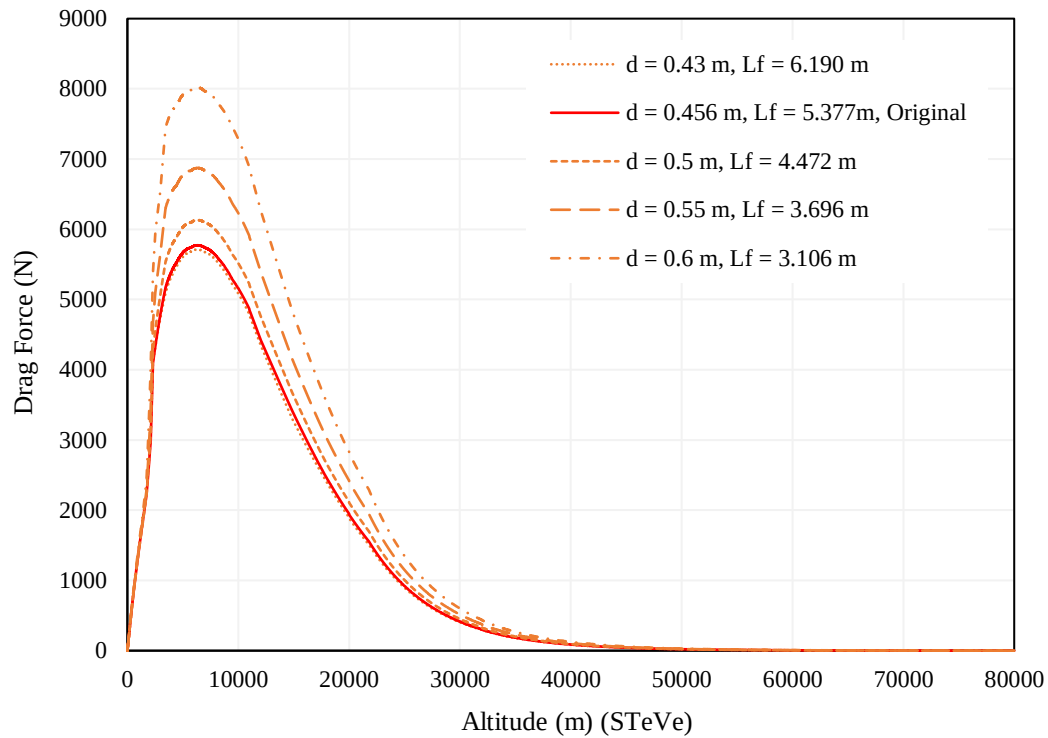


Figure 107. Drag Force Geometry Comparison

Table 29 shows the effects of changing the diameter and length on drag force. The results in Table 29 reinforce the previous points about STeVe, confirming that increasing STeVe's area does not increase the peak drag force in the same manner that was observed for CLV. Increasing the area by 20.2% only increased the peak drag force by 6.2%. This shows that the size of STeVe can be increased, without experiencing large drag penalties.

Table 29. Drag Force Comparison for a Constant Volume

Diameter (m)	Length (m)	Area (%)	Peak F_D (%)
0.43 (-5.7%)	6.190 (+15.1%)	-23.1%	-1.1%
0.456 (-)	5.377 (-)	-	-
0.5 (+9.6%)	4.472 (-16.8%)	+20.2%	+6.2%
0.55 (+20.6%)	3.696 (-31.3%)	+45.5%	+19.1%
0.6 (+31.6%)	3.106 (-42.2%)	+73.1%	+38.9%

8. Conclusion

The results from the previous chapters were discussed in detail at the end of the respective chapters. This conclusion chapter first describes how the aims were achieved, then summarises the main findings and recommendations. The possibility for future work related to this master's research is then discussed.

8.1 Achieving the Aims

- i) *Characterise the external flow over the STeVe sounding rocket and CLV rocket using CFD and semi-empirical methods.*

This aim was achieved, as outlined in Chapters 4 and 5. The drag coefficient was determined for STeVe up to Mach 4.3 using CFD and semi-empirical methods. The drag and normal force coefficient was determined up to 30° angle of attack at Mach 4.3 for STeVe using CFD and semi-empirical methods. The drag coefficient was determined for CLV up to Mach 3.5 using CFD and semi-empirical methods. The drag force was determined for a given flight trajectory, for both STeVe and CLV.

- ii) *Compare the accuracy of various semi-empirical methods in determining aerodynamic coefficients.*

This aim was achieved in Chapter 3. The validation cases were used to test the accuracy of various semi-empirical methods against experimental wind tunnel data. Both RASAero and Missile Datcom were within an acceptable level of accuracy versus experimental data, especially in the transonic and supersonic regions. Missile Datcom was sometimes found to be inaccurate in the transitional areas between flow regimes.

- iii) *Investigate the effects of aerodynamic heating on STeVe.*

This aim was achieved in Chapter 6. A method of using the adiabatic wall temperature and the recovery ratio to obtain the surface temperature at flight conditions was shown to be close to the analytical calculations.

- iv) *Alter the geometry of the launch vehicles to reduce drag and heat transfer.*

This aim was achieved in Chapter 7. Missile Datcom was used to change the fuselage length and diameter of STeVe and CLV. The C_dA values were plotted and the drag for a given trajectory was determined. In Chapter 6, CFD was used to compare the aerodynamic heating on STeVe's nose cone for different nose cone tip radii.

8.2 General Findings

- 1) Semi-empirical methods:
 - i) The RASAero and Missile Datcom semi-empirical codes have been shown to provide results of reasonable accuracy for drag and lift, up to Mach 5 and 50° angle of attack. In general, there was no tendency for one code to be more accurate than another.
 - ii) Missile Datcom sometimes produces unrealistic peaks or troughs in the drag coefficient vs. Mach number plots, when transitioning between flow regimes.

- iii) RASAero (0° fin cant) can be used to obtain results for geometries with small fin cants (less than 0.7°) with negligible errors.
 - iv) The subsonic (Mach < 0.8) drag predictions vary substantially for both RASAero and Missile Datcom.
 - v) A power law nose cone can approximate a secant ogive nose cone to an acceptable degree of accuracy.
- 2) CFD:
- i) Discretisation error is more prevalent at lower Mach numbers.
 - ii) Using the $k - \omega$ SST turbulence model, and aiming for a y^+ of 1, produces acceptable results for high-speed external flow.
 - iii) A bullet fluid domain with free stream boundary conditions was found to produce accurate results.
- 3) Characterisation:
- i) Pressure and skin friction drag are similar in magnitude in the subsonic region. However, pressure drag dominates in the supersonic region, where it accounts for 75% to 95% of the total drag.
 - ii) Dynamic pressure dominates over the drag coefficient in the calculation of drag force.
 - iii) The semi-empirical methods were closer to the CFD results for characterising STeVe, compared to CLV.
- 4) Aerodynamic heating:
- i) The adiabatic wall temperature and the recovery factor from CFD were used to approximate the surface temperature at a given free stream temperature. This method yielded results close to the analytical results.
 - ii) Increasing the nose tip radius decreased the maximum heat transfer coefficient, but had a negligible effect on the heat transfer coefficient along the nose cone body.
 - iii) Increasing the nose tip radius had a negligible effect on the surface temperature. However, the temperature contour plots showed an increase in surrounding fluid temperature with an increase in nose tip radius.
- 5) Geometric changes:
- i) In general, changing the rocket diameter had an impact on $C_d A$, while changing the rocket length had a negligible impact on $C_d A$.
 - ii) STeVe and CLV responded differently to fuselage geometry changes. CLV was more susceptible to changes in diameter than STeVe. For STeVe, a 9.6% increase in diameter, only increased the peak $C_d A$ by 3.9%. For CLV, a 7.7% increase in diameter, increased the $C_d A$ by 17.7%

8.3 Recommendations

- 1) Missile Datcom results must be manually checked to ensure there are no unrealistic changes in the drag plot.
- 2) RASAero and Missile Datcom codes should be used in tandem. If there is a substantial discrepancy in results, CFD should be used as the benchmark to determine which code is more likely to be accurate.
- 3) Semi-empirical codes should be used for geometry optimisation.
- 4) CFD results should be used as input into PyROPS, once the rocket design is finalised.

8.4 Future Work

- 1) Conduct wind tunnel testing.
 - Determining the static and dynamic aerodynamic coefficients would be beneficial for flight prediction, determining the drag forces (designing the rocket structure), and designing roll control systems.
 - The CFD model can then be fully validated.
- 2) Perform an unsteady CFD simulation.
 - The dynamic coefficients could then be determined and compared with data from Missile Datcom.
- 3) Use the static aerodynamic characterisation results from CFD as input into PyROPS.
 - Develop a method for obtaining the full range of data points required for PyROPS to run. This would require interpolating a discrete number of data points for varying Mach numbers and angles of attack, based on several CFD simulations.
- 4) Perform a geometry optimisation.
 - Use a feedback method of geometry optimisation that would aim to optimise a chosen variable by automatically adjusting the geometry based on a set of constraints. Optimisation would involve the nose cone, fins, fuselage diameter and length. One possible software, called HEEDS (Hierarchical Evolutionary Engineering Design System), can perform such tasks and is integrated with Star-CCM+.
- 5) Further investigate the effects of aerodynamic heating.
 - Perform a conjugate heat transfer CFD simulation, simulating both the fluid and rocket body.
 - Obtain temperature data to validate CFD simulations and the method used to calculate the surface temperature. This could be done with wind tunnel tests or real rocket flights.

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Appendix A: Drawings

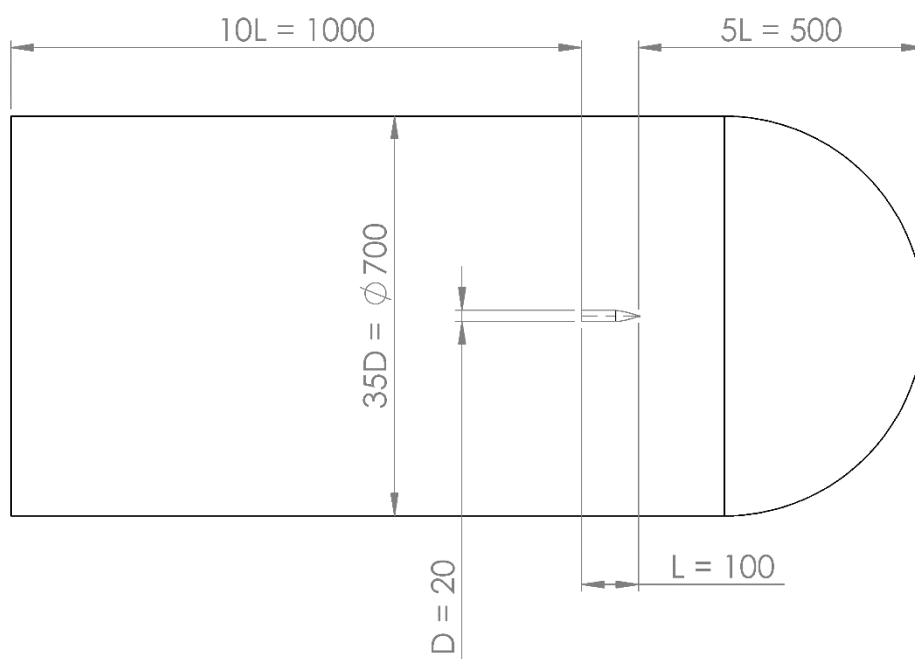
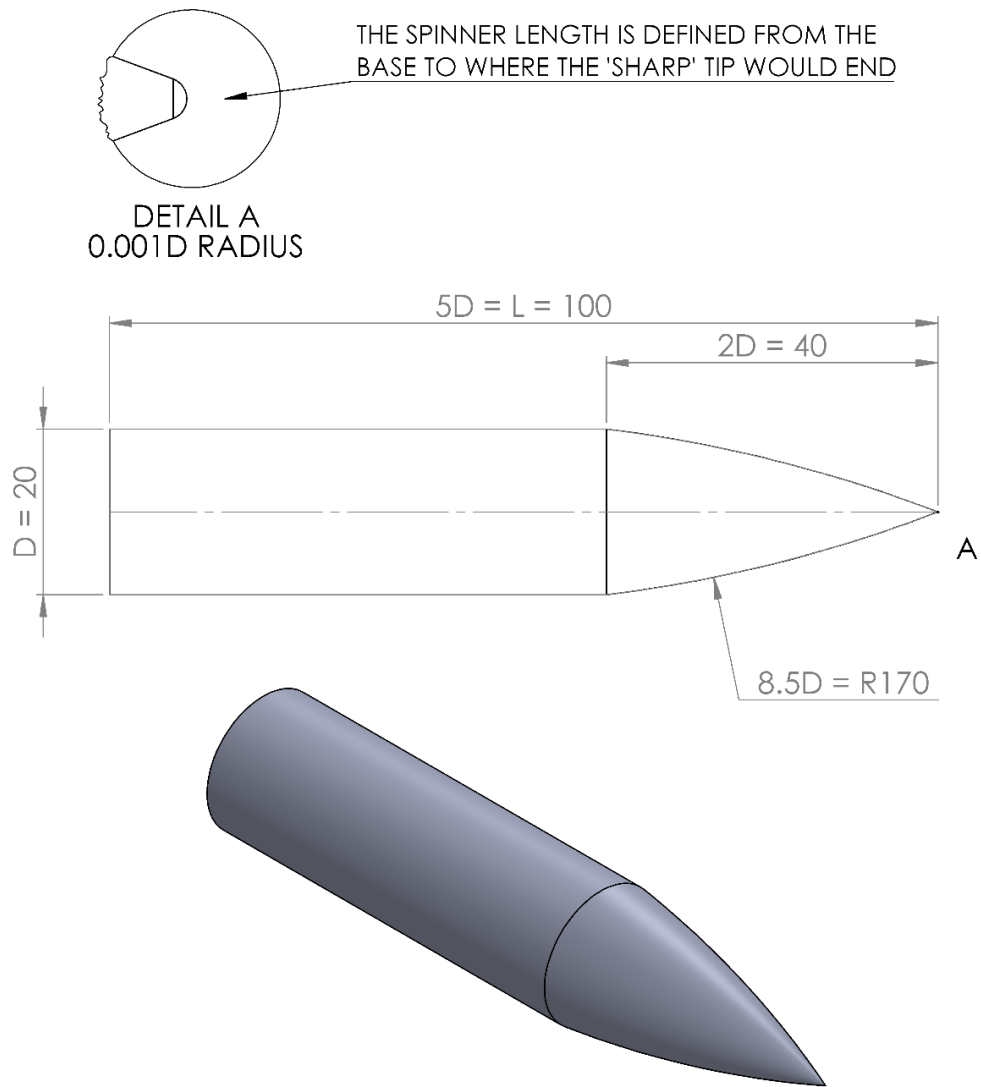
Validation Cases:

- Dunn spinner
- Dunn finner
- Dunn modified finner
- Shantz finner
- Hammargren Maxus

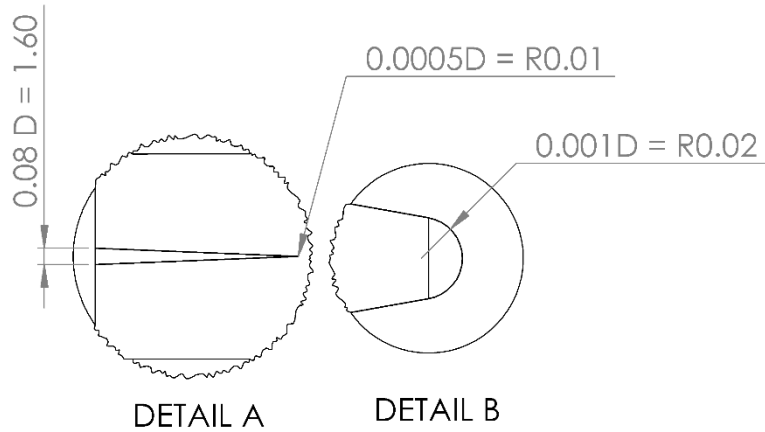
ASRI Rockets:

- STeVe
- CLV

Dunn Spinner

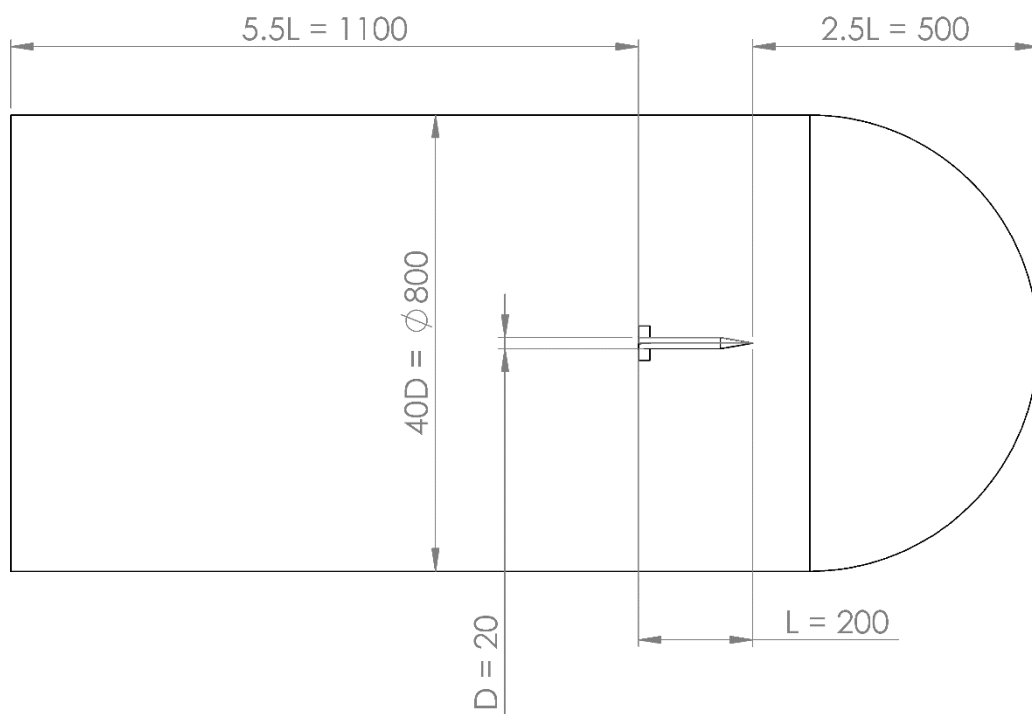
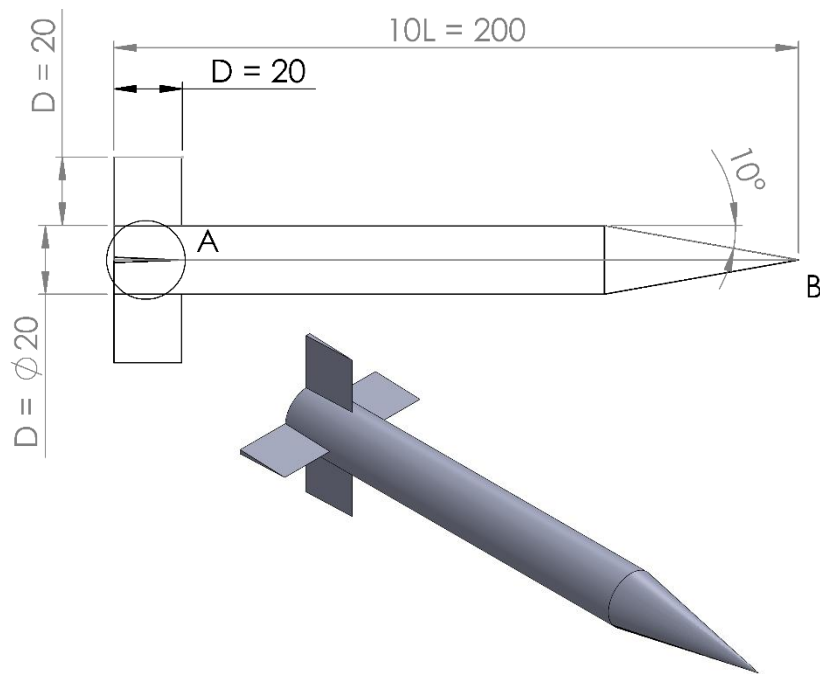


Dunn Finner

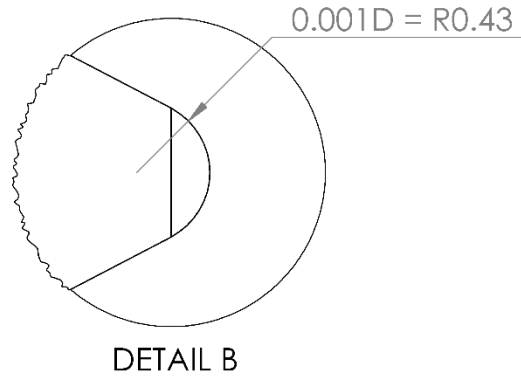
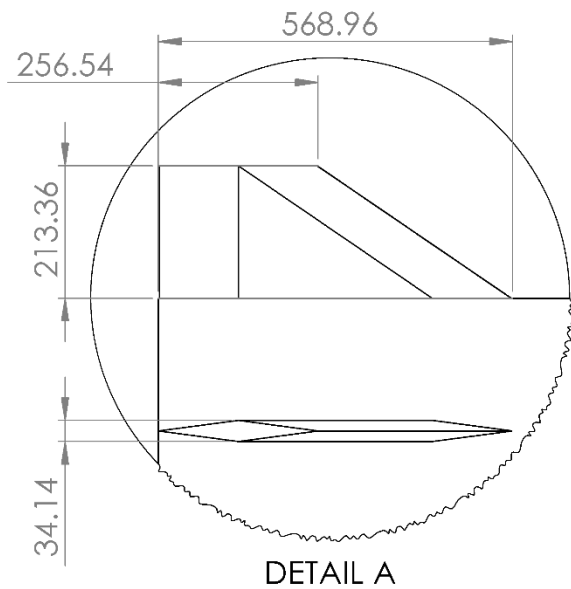


A: THE FIN CHORD LENGTH IS DEFINED FROM THE TRAILING EDGE TO WHERE THE 'SHARP' LEADING EDGE WOULD END.

B: THE TOTAL FINNER LENGTH IS DEFINED FROM THE BASE TO WHERE THE 'SHARP' TIP WOULD END.

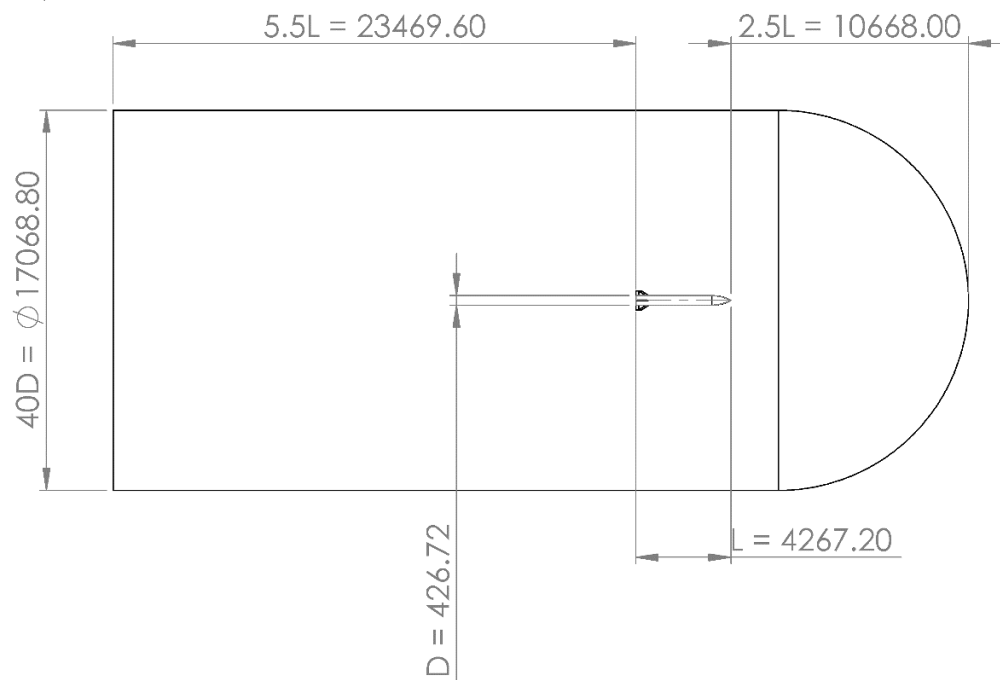
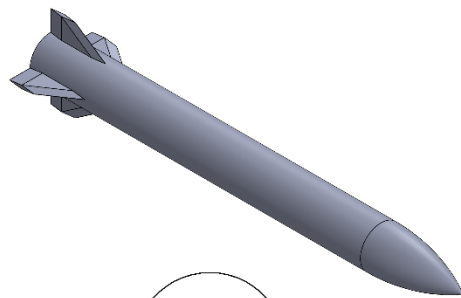


Dunn Modified Finner

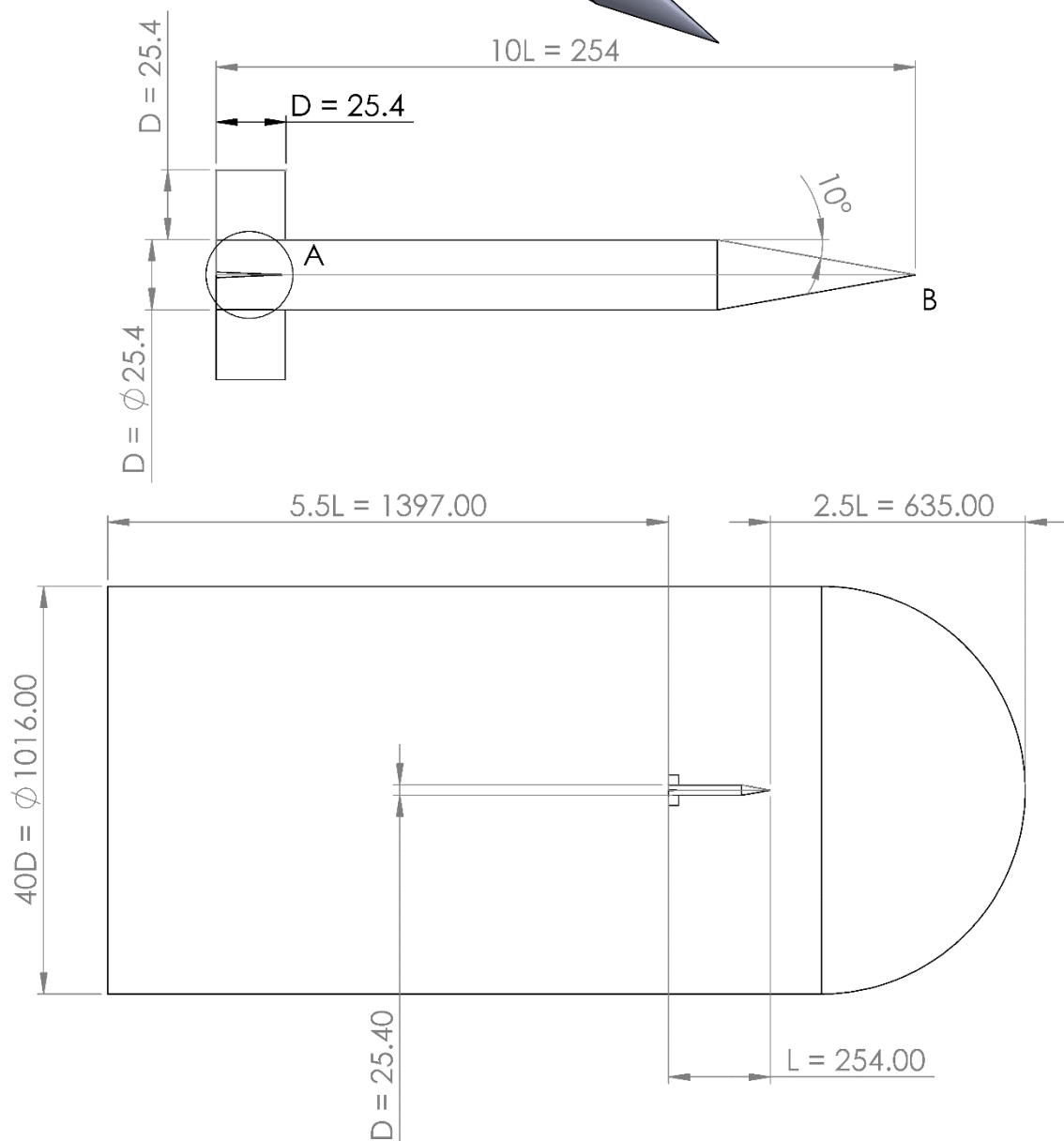
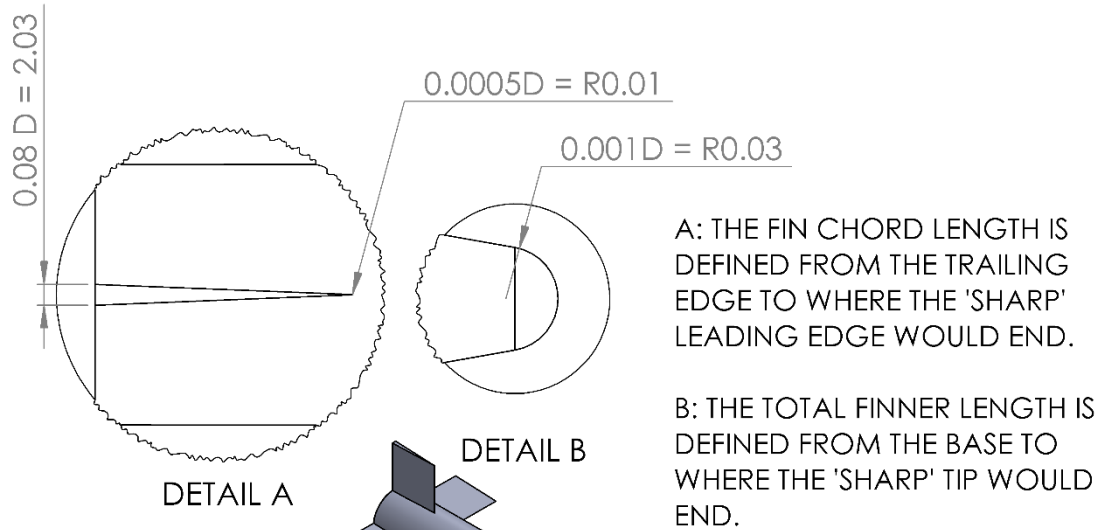


A: THE FIN CHORD LENGTH IS DEFINED FROM THE WHERE THE 'SHARP' TRAILING EDGE WOULD BE TO WHERE THE 'SHARP' LEADING EDGE WOULD END. THE LEADING AND TRAILING EDGE RADII WERE $0.0005D = 0.21$ mm.

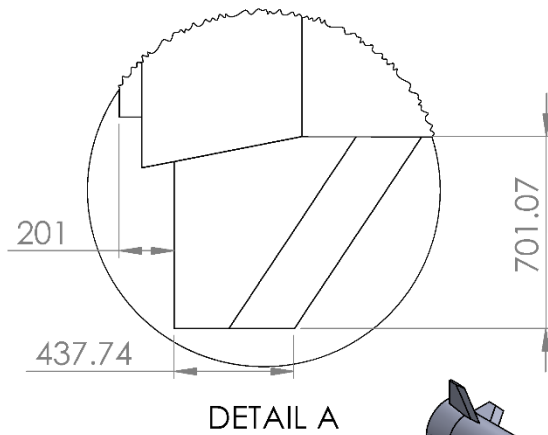
B: THE TOTAL FINNER LENGTH IS DEFINED FROM THE BASE TO WHERE THE 'SHARP' TIP WOULD END.



Shantz Finner

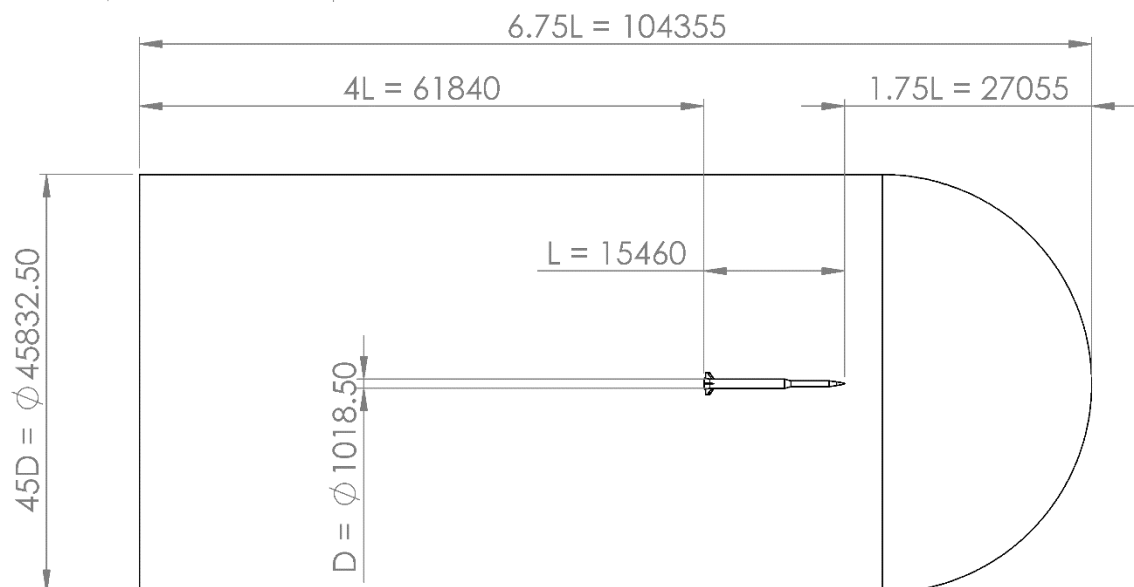
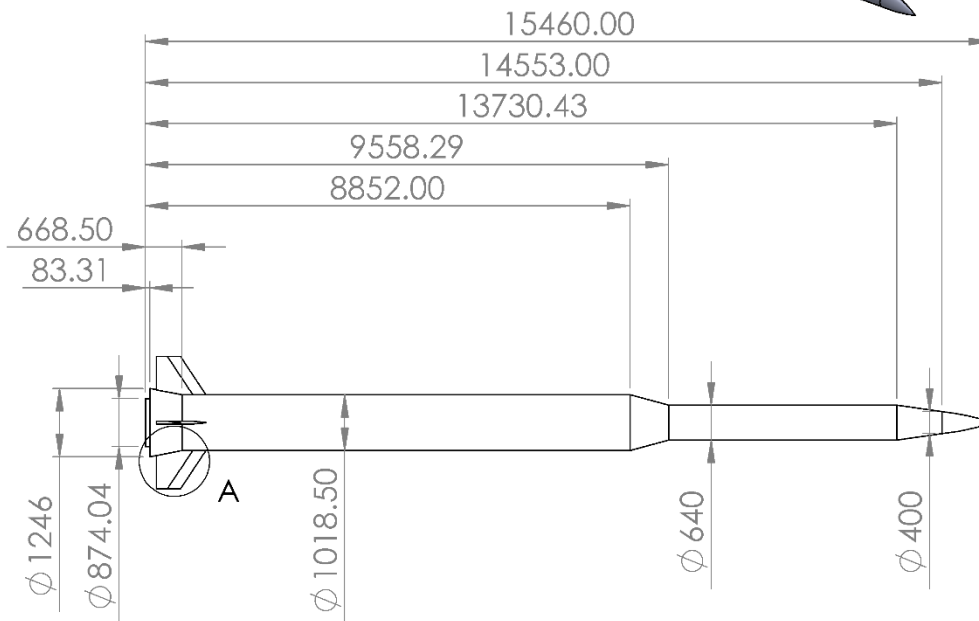
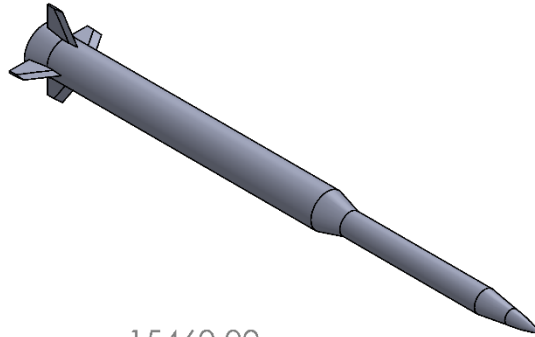


Hammargren Maxus

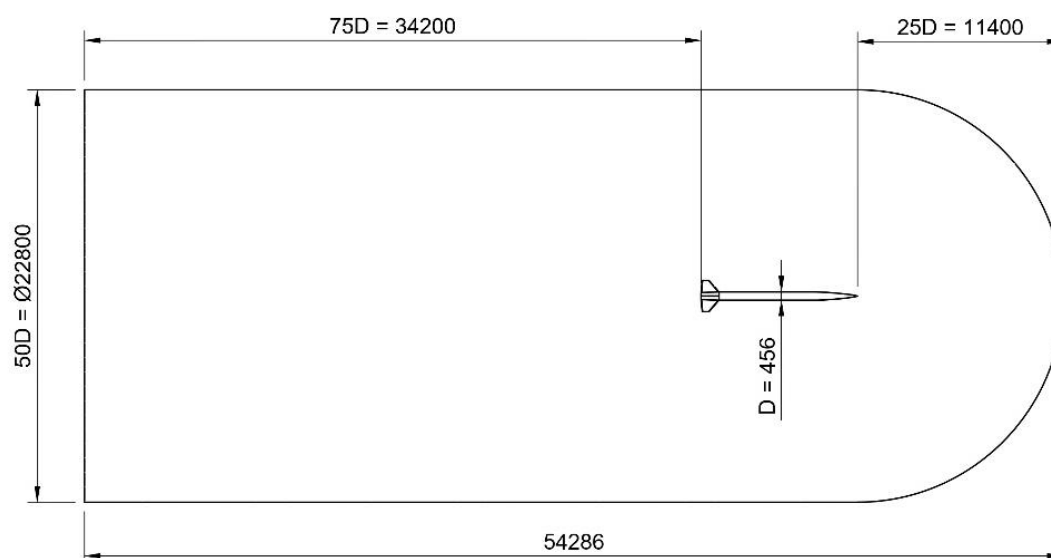
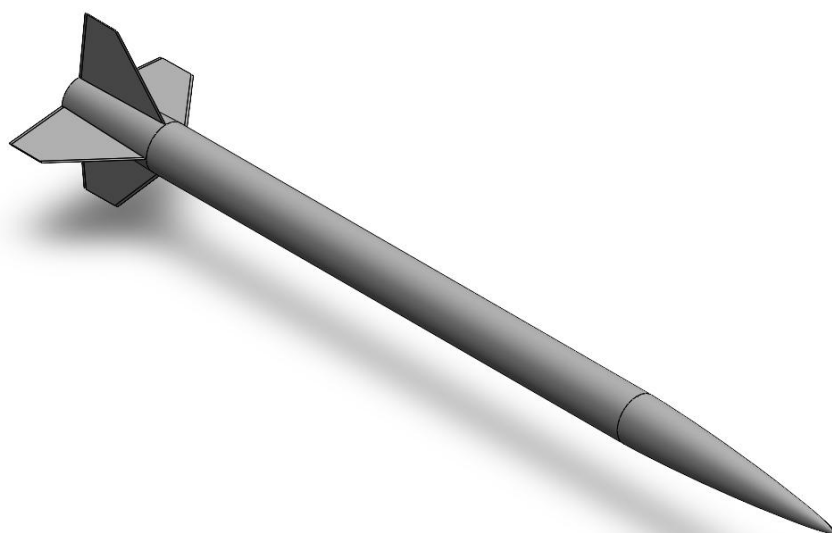
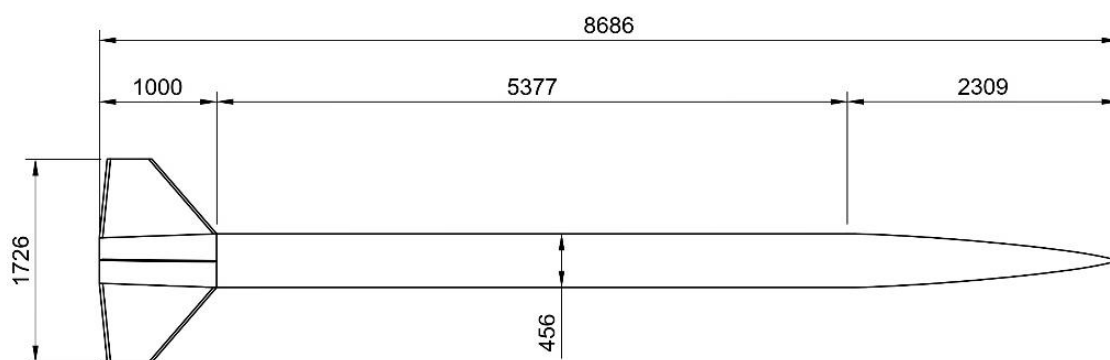


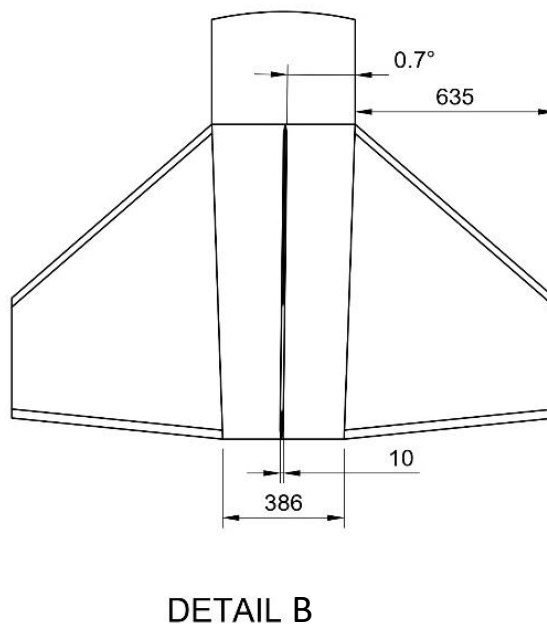
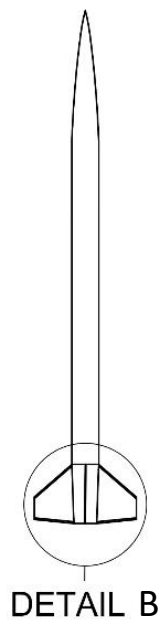
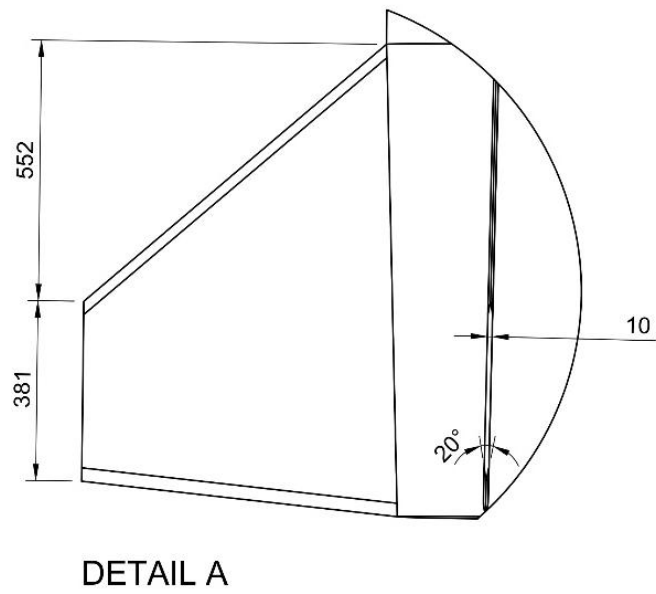
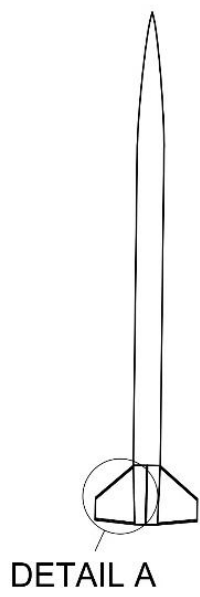
A: THE FIN CHORD LENGTH IS DEFINED FROM THE TRAILING EDGE TO WHERE THE 'SHARP' LEADING EDGE WOULD END.

B: THE TOTAL LENGTH IS DEFINED FROM THE BASE TO WHERE THE 'SHARP' TIP WOULD END.

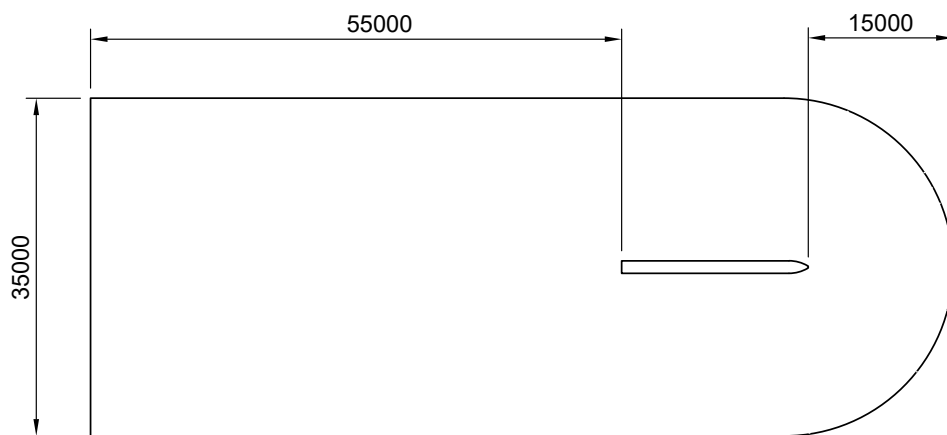
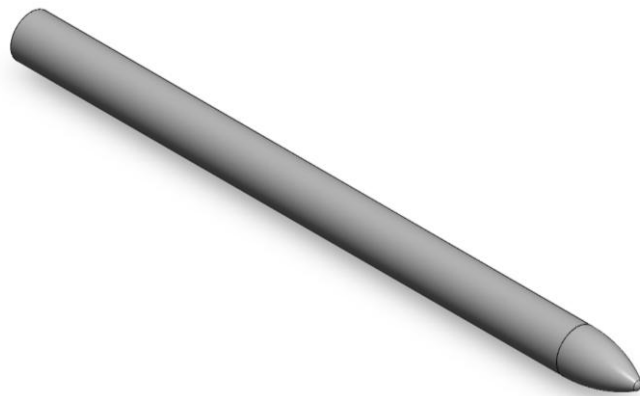
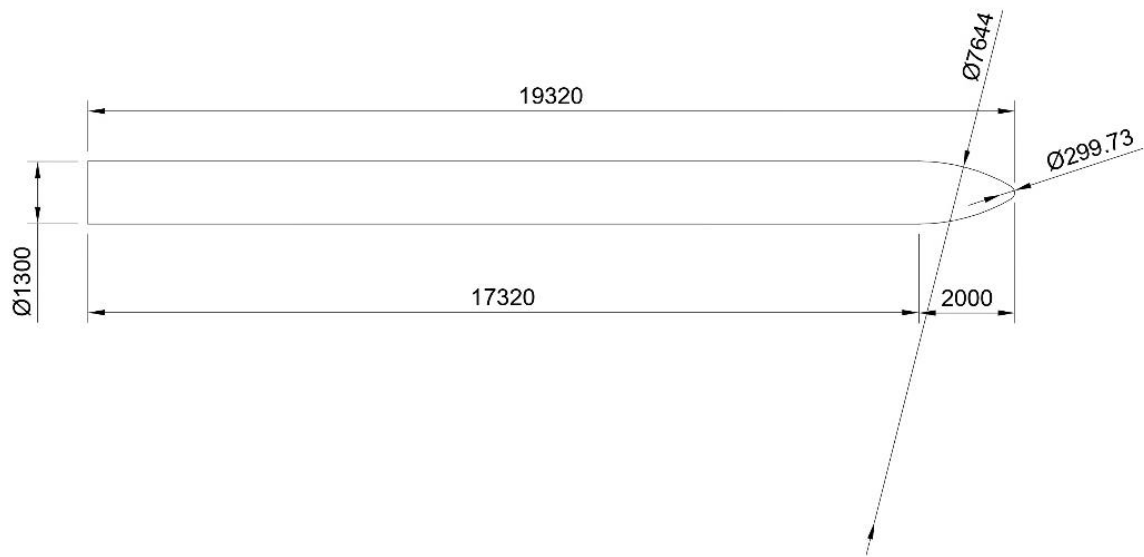


STeVe





CLV



Appendix B: Semi-Empirical Input Data

Validation Cases:

- Dunn spinner
- Dunn finner
- Dunn modified finner
- Shantz finner
- Hammargren Maxus

ASRI Rockets:

- STeVe
- CLV

Dunn Spinner: The RASAero geometry inputs are shown below:

Table 30. Dunn Spinner RASAero Nose Cone Geometry

Description	Value
Type	Power Law
Diameter	0.7874 in
Length	1.5748 in
Nose Tip Radius	0.0008
Power Law	0.7500

Table 31. Dunn Spinner RASAero Body Tube Geometry

Description	Value
Length	2.3622 in

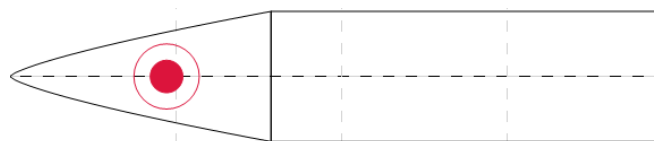


Figure 108. Dunn Spinner RASAero STeVe Geometry

Dunn Spinner: The Missile Datcom inputs are shown below:

```

CASEID Dunn Spinner
DIM M
DERIV RAD
$REFQ
  RHR=0.0,
  XCG=0.04,
  LREF=0.02,
  SREF=0.00031416,$
$FLTCON
  NMACH=20.0,
  MACH=0.2,0.4,0.6,0.7,0.8,0.9,0.95,1.0,1.05,1.1,1.2,1.3,1.4,1.5,1.75,
    2.0,2.5,3.0,3.5,4.0,
  ALT=0.0,
  NALPHA=0.,
  ALPHA=0.0,
  BETA=0.,$

```

```

$AXIBOD
  TNOSE=POWER,
  POWER=0.75,
  LNOSE=0.04,
  DNOSE=0.02,
    BNOSE=0.001,
  LCENTR=0.06,
  DCENTR=0.02,
  DEXIT=0.0,$
SAVE
PLOT
PART
NEXT CASE

```

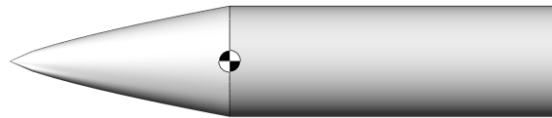


Figure 109. Missile Datcom Dunn Spinner Geometry

Dunn Finner: The RASAero geometry inputs are shown below:

Table 32. Dunn Finner RASAero Nose Cone Geometry

Description	Value
Type	Conical
Diameter	0.7874 in
Length	2.2328 in
Nose Tip Radius	0.0008 in

Table 33. Dunn Finner RASAero Body Tube Geometry

Description	Value
Length	5.6412 in

Table 34. Dunn Finner RASAero Fin Geometry

Description	Value
Fin Type	Single Wedge
Fin Count	4
Root Chord	0.7874
Span	0.7874
Tip Chord	0.7874
Sweep Distance	0.0000
Fin Thickness	0.0630
Distance from the base of the tube	0.7874
Fin LE Radius	0.0008

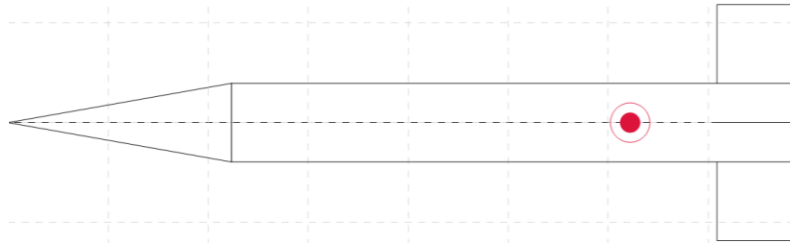


Figure 110. Dunn Finner RASAero STeVe Geometry

Dunn Finner: The Missile Datcom inputs are shown below:

```

CASEID Dunn Finner
DIM M
DERIV RAD
$REFQ
    RHR=0.0,
    XCG=0.04,
    LREF=0.02,
    SREF=0.00031416,$
$FLTCON
    NMACH=20.0,
    MACH=0.2,0.4,0.6,0.8,0.9,0.95,1.0,1.05,1.1,1.15,
        1.2,1.3,1.4,1.5,1.75,2.0,2.25,2.5,2.75,3.0,
    ALT=0.0,
    NALPHA=0.,
    ALPHA=0.0,
    BETA=0.,$
$AXIBOD
    TNOSE=CONICAL,
    LNOSE=0.05671282,
    DNOSE=0.02,
        BNOSE=0.00002,
    LCENTR=0.1432872,
    DCENTR=0.02,
    DEXIT=0.0,$
$FINSET1
    SECTYP=USER,
    XCORD=0.0,0.1,0.2,0.3,0.4,0.5,0.6,0.7,0.8,0.9,1.0,
    MEAN=0.0,0.0,0.0,0.0,0.0,0.0,0.0,0.0,0.0,0.0,0.0,
    THICK=0.0,0.008,0.016,0.024,0.032,0.04,0.048,0.056,0.064,0.072,0.08,
    SSPAN=0.0,0.02,
    CHORD=0.02,0.02,
    XLE=0.18,0.18,
    SWEEP=0.0,
    NPANEL=4.0,
    PHIF=0.0,90.0,180.0,270.0,$
SAVE
PLOT
PART
NEXT CASE

```



Figure 111. Missile Datcom Dunn Finner Geometry

Dunn Modified Finner: The RASAero geometry inputs are shown below:

Table 35. Dunn Modified Finner RASAero Nose Cone Geometry

Description	Value
Type	Conical
Diameter	16.8000 in
Length	33.6000 in
Nose Tip Radius	0.0168 in

Table 36. Dunn Modified Finner RASAero Body Tube Geometry

Description	Value
Length	134.4000

Table 37. Dunn Modified Finner RASAero Fin Geometry

Description	Value
Fin Type	Hexagonal
Fin Count	4
Root Chord	22.3500 in
Span	8.4000 in
Tip Chord	10.0500 in
Sweep Distance	12.3000 in
Fin Thickness	1.3441 in
Distance from the base of the tube	22.3500 in
LE Diamond Airfoil Length	5.0000 in
TE Diamond Airfoil Length	5.0500 in
Fin LE Radius	0.0083 in

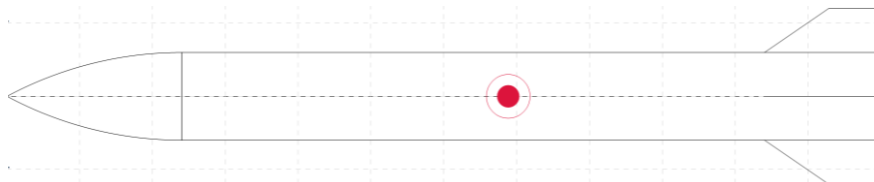


Figure 112. Dunn Modified Finner RASAero STeVe Geometry

Dunn Modified Finner: The Missile Datcom inputs are shown below:

```

CASEID Dunn Modified Finner
DIM M
DERIV RAD
$REFQ
  RHR=0.0,
  XCG=2.0,
  LREF=0.42672,
  SREF=0.14301,$
$FLTCON
  NMACH=1.0,
  MACH=1.3,
  ALT=0.0,
  NALPHA=11.0,

```

```

ALPHA=0.0,5.0,10.0,15.0,20.0,25.0,30.0,35.0,
40.0,45.0,50.0,
BETA=0.,$
$AXIBOD
TNOSE=OGIVE,
LNOSE=0.85344,
DNOSE=0.42672,
BNOSE=0.00043,
LCENTR=3.41376,
DCENTR=0.42672,
DEXIT=0.0,$
$FINSET1
SECTYP=HEX,
SSPAN=0.0,0.21336,
CHORD=0.56896,0.25654,
XLE=3.69824,4.01066,
ZUPPER=0.030002,0.066539,
LMAXU=0.225,0.5,
LFLATU=0.55,0.0,
NPANEL=4.0,
PHIF=0.0,90.0,180.0,270.0,$
SAVE
PLOT
PART
NEXT CASE

```

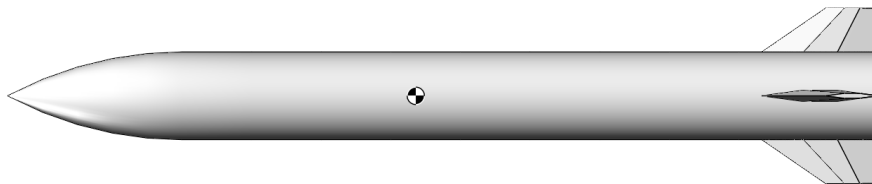


Figure 113. Missile Datcom Dunn Modified Finner Geometry

Shantz Finner: The RASAero geometry inputs are shown below:

Table 38. Shantz Finner RASAero Nose Cone Geometry

Description	Value
Type	Conical
Diameter	1.0000 in
Length	2.8356 in
Nose Tip Radius	0.0010 in

Table 39. Shantz Finner RASAero Body Tube Geometry

Description	Value
Length	7.1644 in

Table 40. Shantz Finner RASAero Fin Geometry

Description	Value
Fin Type	Single Wedge
Fin Count	4
Root Chord	1.0000 in
Span	1.0000 in
Tip Chord	1.0000 in
Sweep Distance	0.0000 in
Fin Thickness	0.0800 in
Distance from the base of the tube	1.0000 in
Fin LE Radius	0.0010 in

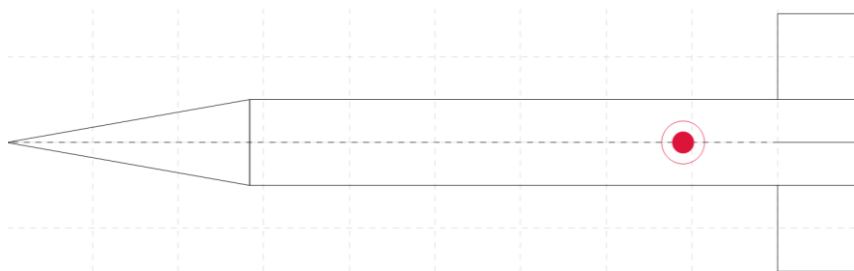


Figure 114. Shantz Finner RASAero STeVe Geometry

Shantz Finner: The Missile Datcom inputs are shown below:

```

CASEID Shantz Finner
DIM M
DERIV RAD
$REFQ
  RHR=0.0,
  XCG=0.04,
  LREF=0.0254,
  SREF=0.00050671,$
$FLTCON
  NMACH=2.0,
  MACH=1.58,3.86,
  ALT=0.0,
  NALPHA=12.0,
  ALPHA=0.0,2.0,4.0,6.0,8.0,10.0,12.0,14.0,
    16.0,18.0,20.0,22.0,
  BETA=0.,$
$AXIBOD
  TNOSE=CONE,
  LNOSE=0.07202528,
  DNOSE=0.0254,
  BNOSE=0.0000254,
  LCENTR=0.1819747,
  DCENTR=0.0254,
  DEXIT=0.0,$
$FINSET1
  SECTYP=USER,
  XCORD=0.0,0.1,0.2,0.3,0.4,0.5,0.6,0.7,0.8,0.9,1.0,
  MEAN=0.0,0.0,0.0,0.0,0.0,0.0,0.0,0.0,0.0,0.0,0.0,
  THICK=0.0,0.007992,0.015984,0.023976,0.031969,
    0.039961,0.047953,0.055945,0.063937,0.071929,0.079921,
  SSPAN=0.0,0.0254,
  CHORD=0.0254,0.0254,
  XLE=0.2286,0.2286,
  SWEEP=0.0,

```

```

NPANEL=4.0,
PHIF=0.0,90.0,180.0,270.0,$
SAVE
PLOT
PART
NEXT CASE

```



Figure 115. Missile Datcom Shantz Finner Geometry

Hammargren Maxus: The RASAero geometry inputs are shown below:

Table 41. Hammargren Maxus RASAero Nose Cone Geometry

Description	Value
Type	Power Law
Diameter	15.7480 in
Length	35.7087 in
Nose Tip Radius	0.0402 in
Power Law	0.7300

Table 42. Hammargren Maxus RASAero Transition 1 Geometry

Description	Value
Length	32.3846 in
Rear Diameter	25.1969 in

Table 43. Hammargren Maxus RASAero Body Tube 1 Geometry

Description	Value
Length	164.2575 in

Table 44. Hammargren Maxus RASAero Transition 2 Geometry

Description	Value
Length	27.8067 in
Rear Diameter	40.0984 in

Table 45. Hammargren Maxus RASAero Body Tube 2 Geometry

Description	Value
Length	32.3846 in
Rear Diameter	25.1969 in

Table 46. Hammargren Maxus RASAero Transition 3 Geometry

Description	Value
Length	322.1850 in

Table 47. Hammargren Maxus RASAero Fin Geometry

Description	Value
Fin Type	Hexagonal Blunt Base
Fin Count	4
Root Chord	35.6039 in
Span	27.5831 in
Tip Chord	17.3228 in
Sweep Distance	18.3795 in
Fin Thickness	1.9685 in
Distance from the base of the tube	17.1984 in
LE Diamond Airfoil Length	9.4488 in
Fin LE Radius	0.0984 in

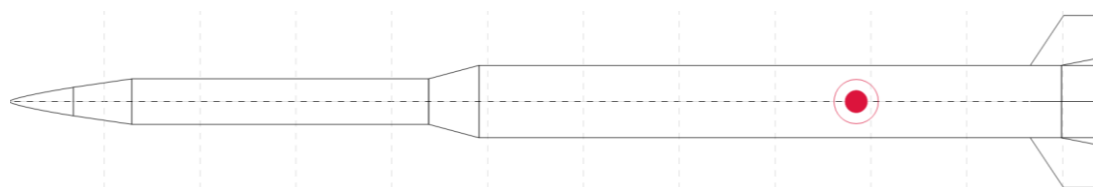


Figure 116. Hammargren Maxus RASAero STeVe Geometry

Hammargren Maxus: The Missile Datcom inputs are shown below:

```

CASEID Hammargren Maxus
DIM M
DERIV RAD
$REFQ
  RHR=0.0,
  XCG=5.0,
  LREF=1.0185,
  SREF=1.219,$
$FLTCON
  NMACH=20.0,
  MACH=0.3,0.6,0.7,0.8,0.9,0.95,1.0,1.05,1.1,1.2,1.3,1.5,1.75,
    2.0,2.5,3.0,4.0,6.0,8.0,10.0,
  ALT=0.0,
  NALPHA=0.,
  ALPHA=0.0,
  BETA=0.,$
$AXIBOD
  X0=0.0,
  NX=50.0,
  X=0.0,0.02060,0.0412,0.0618,0.0825,0.1031,0.1237,0.1443,0.1649,0.1855,
X(11)=0.2061,0.2268,0.2474,0.2680,0.2886,0.3092,0.3298,0.3504,0.3710,0.3917,
X(21)=0.4123,0.4329,0.4535,0.4741,0.4947,0.5153,0.5360,0.5566,0.5772,0.5978,
X(31)=0.6184,0.6390,0.6596,0.6803,0.7009,0.7215,0.7421,0.7627,0.7833,0.8039,
X(41)=0.8245,0.8452,0.8658,0.8864,0.9070,1.7296,5.9017,6.6080,14.7915,15.376
7,
  R=0.0,0.0071,0.0140,0.0209,0.0276,0.0342,0.0406,0.0470,0.0532,0.0593,
R(11)=0.0653,0.0711,0.0769,0.0825,0.0880,0.0934,0.0986,0.1038,0.1088,0.1137,
R(21)=0.1185,0.1232,0.1278,0.1322,0.1366,0.1408,0.1449,0.1489,0.1528,0.1566,

```

```

R(31)=0.1603,0.1638,0.1673,0.1706,0.1738,0.1769,0.1799,0.1828,0.1856,0.1883,
R(41)=0.1908,0.1933,0.1956,0.1979,0.2000,0.3200,0.3200,0.5093,0.5093,0.6230,
  DEXIT=0.0,$
$FINSET1
  SSPAN=0.0,0.70107,
  CHORD=0.959,0.43774,
  XLE=14.3,14.82126
  NPANEL=4.0,
  SECTYP=HEX,
  ZUPPER=0.026069,0.057112,
  LMAXU=0.25,0.5,
  LFLATU=0.75,0.5,
  PHIF=0.0,90.0,180.0,270.0,$
SAVE
PLOT
PART
NEXT CASE

```

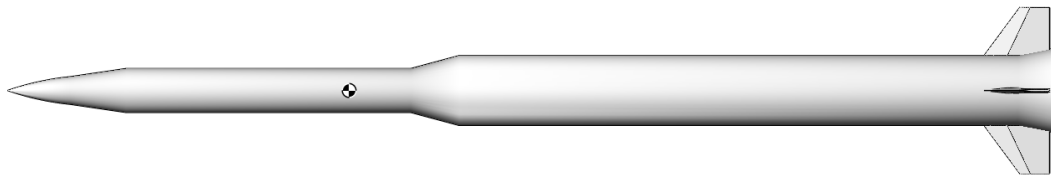


Figure 117. Missile Datcom Hammargren Maxus Geometry

STeVe: The RASAero geometry inputs are shown below:

Table 48. STeVe RASAero Nose Cone Geometry

Description	Value
Type	Von Karman
Diameter	17.9528 in
Length	90.9187 in
Nose Tip Radius	0.0000 in

Table 49. STeVe RASAero Body Tube Geometry

Description	Value
Length	211.7049 in

Table 50. STeVe RAS Aero Boat Tail and Fin Geometry

Description	Value
Length	39.3763 in
Diameter	15.2002 in
Fin Type	Hexagonal Airfoil
Fin Count	4
Root Chord	39.3763 in
Span	25.0008 in
Tip Chord	15.0063 in
Sweep Distance	21.7321 in
Fin Thickness	0.3937 in
Distance from the base of the tube	39.3763 in
LE Diamond Airfoil Length	1.1336 in
TE Diamond Airfoil Length	1.1336 in
Fin LE Radius	0.0000 in

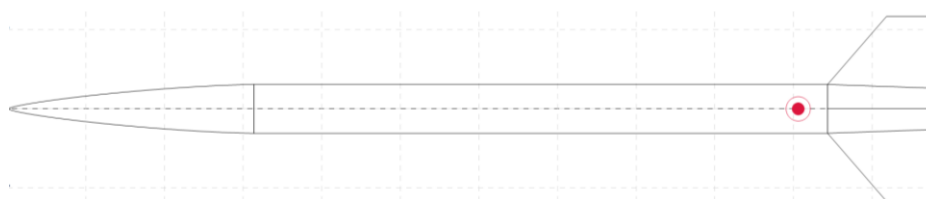


Figure 118. STeVe RAS Aero STeVe Geometry

STeVe: The Missile Datcom inputs are shown below:

```

CASEID STeVe Original Configuration
DIM M
DERIV RAD
$REFQ
    RHR=0.0,
    XCG=4,
    LREF=0.456,
    SREF=0.16331,$
$FLTCON
    NMACH=20.0,
    MACH=0.4,0.6,0.7,0.8,0.9,0.95,1.0,1.05,1.1,1.2,
    1.3,1.5,1.75,2.0,2.25,2.5,3.0,3.49,4.0,4.3,
    ALT=0.0,
    NALPHA=1.0,
    ALPHA=0.0,
    BETA=0.,$
$AXIBOD
    TNOSE=KARMAN,
    LNOSE=2.309,
    DNOSE=0.456,
    BNOSE=0.0,
    LCENTR=5.377,
    DCENTR=0.456,
    TAFT=CONICAL,

```

```

LAF T=1.0,
DAFT=0.386,
DEXIT=0.0,$
$FINSET1
SECTYP=HEX,
SSPAN=0.193,0.828,
CHORD=1.0,0.381,
XLE=7.686,8.238,
ZUPPER=0.005,0.0131,
LMAXU=0.05715,0.15,
LFLATU=0.8857,0.7,
NPANEL=4.0,
PHIF=0.0,90.0,180.0,270.0,$
$DEFLCT
DELTA1=-0.7,-0.7,-0.7,-0.7,$
SAVE
PLOT
PART
NEXT CASE

```

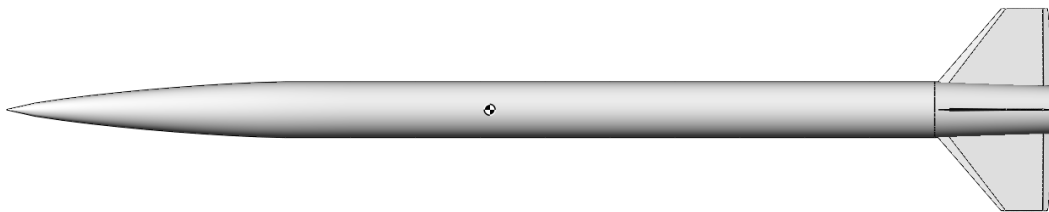


Figure 119. Missile Datcom STeVe Geometry

CLV: The RASAero geometry inputs are shown below:

Table 51. CLV RASAero Nose Cone Geometry

Description	Value
Type	Ogive
Diameter	51.1811 in
Length	83.9433 in
Nose Tip Radius	5.9002 in

Table 52. CLV RASAero Body Tube Geometry

Description	Value
Length	681.8898 in

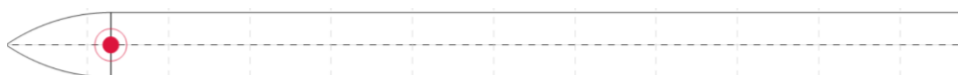


Figure 120. RASAero CLV Geometry

CLV: The Missile Datcom inputs are shown below:

```

CASEID CLV Original Configuration
DIM M
DERIV RAD
$REFQ

```



```

RHR=0.0,
XCG=4,
LREF=1.3,
SREF=1.327323,$
$FLTCON
NMACH=20.0,
MACH=0.6,0.7,0.8,0.9,0.95,1.0,1.05,1.1,1.2,1.35,
    1.5,1.75,2.0,2.25,2.5,3.0,3.5,4.0,4.5,5.0,
ALT=0.0,
NALPHA=1.0,
ALPHA=0.0,
BETA=0.,$
$AXIBOD
TNOSE=OGIVE,
LNOSE=2.0,
DNOSE=1.3,
BNOSE=0.149865,
LCENTR=17.32,
DCENTR=1.3,
DEXIT=0.0,$
SAVE
PLOT
PART
NEXT CASE

```



Figure 121. Missile Datcom CLV Geometry